



IALA S-201 Product Specification

Data Classification and Encoding Guide

Edition 2.0.0 – May 2025

IALA AtoN Product Specification

ANNEX A

Data Classification Encoding Guide (DCEG)

1 Overview

1.1 Preface

The “Data Classification and Encoding Guide” has been developed to provide consistent, standardized instructions for encoding S-100 compliant AtoN data. The purpose of the Data Classification and Encoding Guide is to facilitate S-201 encoding. The document describes how to encode information that the AtoN service provider considers relevant to an AtoN. A “AtoN service provider” is a AtoN authority or an organization authorized by a government or other relevant government institution.

1.2 S-201 Appendix A; Data Classification and Encoding Guide - Metadata

Note: This information uniquely identifies this Appendix to the Product Specification and provides information about its creation and maintenance.

Title: The International Association of Lighthouse Authorities Aids To Navigation Product Specification, Appendix A – Data Classification and Encoding Guide

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Maintenance: Changes to S-201 Appendix A; Data Classification and Encoding Guide are coordinated by the IALA ARM Committee WG2

1.3 Terms, definitions and abbreviations

1.3.1 Terms and definitions

aggregation

special form of association that specifies a whole-part relationship between the aggregate (whole) and a component

attribute

named property of an entity

NOTE: Describes the geometrical, topological, thematic, or other characteristic of an entity

feature

Abstraction of real world phenomena

NOTE: A feature may occur as a type or an instance. The terms “feature type” or “feature instance” should be used when only one is meant

EXAMPLE: The feature instance named “Eiffel Tower” may be classified with other phenomena into a feature type “tower”

enumeration

A fixed list of valid identifiers of named literal values. Attributes of an enumerated type may only take values from this list (source: ISO 19136:XX, *Geographic information — Geography Markup Language (GML)*)

geometric primitive

geometric object representing a single, connected, homogeneous element of geometry

NOTE: Geometric primitives are non-decomposed objects that present information about geometric configuration. They include **points**, **curves**, surfaces and solids

1.3.2 Abbreviations

AtoN	Aids To Navigation
IALA	International Association of Lighthouse Authorities
IHO	International Hydrographic Organization

1.4 Use of language

Within this document:

“Must” indicates a mandatory requirement;

“Should” indicates an optional requirement, that is the recommended process to be followed, but is not mandatory;

“May” means “allowed to” or “could possibly”, and is not mandatory.

1.5 Maintenance

Changes to the Data Classification and Encoding Guide must occur in accordance with the S-201 AtoN Product Specification clause 1.4.1.

2 General

The S-201 Data Classification and Encoding Guide describes how data describing the real world should be captured using the types defined in the S-201 Feature Catalogue. It provides the encoding rules and guidance required to create S-201 AtoNs. This standard is specifically concerned with those entities in the real world that are of relevance to Aids to Navigation. This concept is considered to be geo-spatial. As a result, the model defines real world entities as a combination of descriptive and spatial characteristics. Within the model these sets of characteristics are defined in terms of feature, spatial. A type is defined as a stereotype of class that is used to specify a domain of instances (objects) together with the operations applicable to the objects. A type may have attributes and may be related to other types.

2.1 Feature types

Feature types contain descriptive attributes and do not contain any geometry (i.e. information about the shape and position of a real world entity).

Features have two aspects – feature type and feature instance. A feature type is a class and is defined in a Feature Catalogue. A feature instance is a single occurrence of the feature type and represented as an object in a data set. A feature instance is located by a relationship to one or more spatial instances. A feature instance may exist without referencing a spatial instance.

2.2 Spatial primitives

The allowable spatial primitive for each feature type is defined in the Feature Catalogue. Within this document, allowable primitives are included in the tables containing a description of each feature type. Allowable spatial primitives are point, curve and surface.

Each spatial value must be referenced by a feature instance.

2.3 Information types

An information type is an identifiable object that can be associated with features in order to carry information particular to the associated features. Information types can also be associated with other information types. This can be done where there is further supplementary information that is relevant to the information type. Information types carry attributes but not geometry.

2.4 Attributes

Attributes may be simple type or complex type. Complex (C) attributes are aggregates of other attributes that can be simple type or complex type. Simple attribute types consist of Text (TE), Date (DA), Time (TI), DateTime (DT), Enumeration (EN), Boolean (BO), Integer (IN) and Real (RE).

The binding of attributes to feature types, the binding of attributes to attributes to construct complex attributes, and attribute multiplicity is defined in the Feature Catalogue. Within this document, the allowable attributes is included in the description of each feature type, as well as the allowable values for enumeration type attributes.

Real or integer attribute values must not be padded by non-significant zeroes. For example, for a signal period of 2.5 seconds, the value populated for the attribute **signal period** must be 2.5 and not 02.50.

2.4.1 Multiplicity

In order to control the number of allowed attribute values or sub attribute instances within a complex attribute, S-100 uses the concept of multiplicity. This defines lower and upper limits for the number of values, whether the order of the instances has meaning and if an attribute is mandatory or not. Common examples are shown in the table below:

Format : MinOccurs, MaxOccurs (if * Infinite) ordered – sequential

Multiplicity	Explanation
0,1	An instance is not required; there can be only one instance.
1,1	An instance is required and there must only be one instance.
0,*	An instance is not required and there can be an infinite number of instances.
1,*	An instance is required and there can be an infinite number of instances.
1,* ordered	An instance is required and there can be an infinite number of instances, the order of which has a specific meaning.
2,2	Two instances are required and no more than two.

Example: A red and white tower is encoded with attribute **colour** = 3 (red) and **colour** = 1 (white).

2.4.2 Simple attribute types

Each simple attribute is assigned to one of 10 types:

- EN Enumeration: A fixed list of valid identifiers of named literal values. Attributes of an enumerated type may only take values from this list.
- BO Boolean: A value representing binary logic. The value can be either *True* or *False*. The default state for Boolean type attributes (i.e. where the attribute is not populated for the feature) is *False*.
- RE Real: A signed Real (floating point) number consisting of a mantissa and an exponent. The representation of a real is encapsulation and usage dependent.
Examples: 23.501, -0.0001234, -23.0, 3.141296
- IN Integer: A signed integer number. The representation of an integer is encapsulation and usage dependent.
Examples: 29, -65547
- TE Free text: A `CharacterString`, that is an arbitrary-length sequence of characters including accents and special characters from a repertoire of one of the adopted character sets.
- DA Date: A date provides values for year, month and day according to the Gregorian Calendar. Character encoding of a date is a string which must follow the calendar date format (complete representation, basic format) for date specified by ISO 8601:1988.
Example: 19980918 (YYYYMMDD)
- TI Time: A time is given by an hour, minute and second. Character encoding of a time is a string that follows the local time (complete representation, basic format) format defined in ISO 8601:1988.
Time zone according to UTC is optional.
Example: 183059 or 183059+0100 or 183059Z
The complete representation of the time of 27 minutes and 46 seconds past 15 hours locally in Geneva (in winter one hour ahead of UTC), and in New York (in winter five hours behind UTC), together with the indication of the difference between the time scale of local time and UTC, are used as examples.
Geneva: 1527460100
New York: 1527460500
- DT Date and Time: A `DateTime` is a combination of a date and a time type. Character encoding of a `DateTime` shall follow ISO 8601:1988.
Example: 19850412T101530
- CL(OE) Codelist (open enumeration): A fixed list of valid identifiers of named literal values. Attributes of an enumerated type may take values from this list or the pattern "other: ..." i.e., the word 'other' followed by a colon and a single space character, followed by one or more alphanumeric strings separated by single spaces.
Example: other: loxodromic
- UI URI: Uniform Resource Identifier (URI) is a unique sequence of characters that identifies an abstract or physical resource.

EXAMPLE <http://registry.iho.int>

2.4.3 Mandatory and conditional attributes

Some attributes are mandatory and must be populated for a given feature type. There are some reasons why attribute values may be considered mandatory:

- They are required to support correct portrayal;
- Certain features make no logical sense without specific attributes;
- Some attributes are required for safety of navigation.

Where a value of a mandatory attribute is not known, the attribute should be populated with an empty (null) value. Where the value of a non-mandatory attribute is not known, the attribute should not be populated (i.e. not included in the dataset). Within this document, mandatory attributes (multiplicity 1,1; 1,n (n>1); or 1,*) are identified in the description of each feature type. Missing attribute values

Where a value of a mandatory attribute is not known, the attribute should be populated with an empty (null) value.

Where the value of a non-mandatory attribute is not known, the attribute should not be included in the dataset.

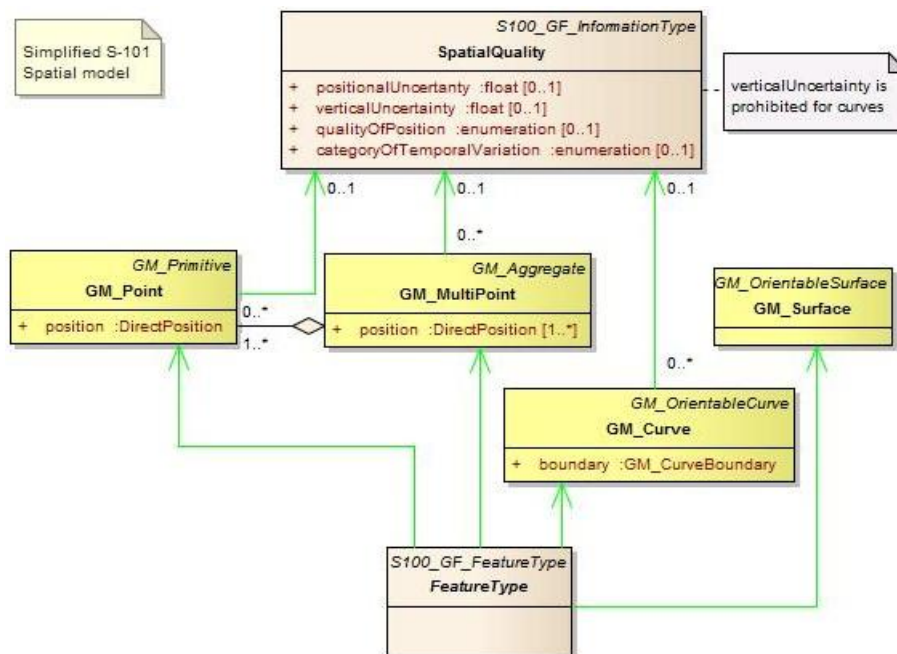
In a base dataset (EN application profile), when an attribute code is present but the attribute value is missing, it means that the producer wishes to indicate that this attribute value is unknown.

In an Update dataset (ER application profile), when an attribute code is present but the attribute value is missing it means:

- that the value of this attribute is to be replaced by an empty (null) value if it was present in the original dataset, or
- that an empty (null) value is to be inserted if the attribute was not present in the original dataset.

2.4.4 Spatial attribute types

Spatial attribute types must contain a referenced geometry and may be associated with spatial quality attributes. Each spatial attribute instance must be referenced by a feature instance or another spatial attribute instance.



Spatial quality attributes are carried in an information class called **Spatial quality**. Only points, multipoints and curves can be associated with Spatial quality. Currently no use case for associating surfaces with spatial quality attributes is known, therefore this is prohibited. Vertical uncertainty is prohibited for curves as this dimension is not supported by curves.

2.4.5 Dates

When encoding dates using the attributes **dredged date**, **fixed date range**, **periodic date range**, **reported date**, **survey date range** and **swept date**, and no specific year, month or day is required, the following values must apply in conformance to ISO 8601:1988.

- No specific year required, same day each year: --MMDD
- No specific year required, same month each year: --MM
- No specific day required: CCYYMM
- No specific month required: CCYY

Notes: CCYY = calendar year; MM = month; DD = day.
In the first two values, the dashes (--) must be included.

2.4.5.1 Seasonal features

If it is required to show seasonality of features, it must be done using the attribute **Status** = 5 (periodic/intermittent). If it is required to encode the start and/or end dates of the season, this must be done using the complex attribute **periodic date range**.

2.4.6 Times

If it is required to show the beginning and end of the active period of a feature, it must be encoded using the complex attribute **time range**, sub-attributes **time end** and **time start**. When using these sub-attributes, all times must be encoded as Coordinated Universal Time (UTC). The attribute descriptions for **time end** and **time start** states that the mandatory format is CCYYMMDDThhmmss, where T is the separator, and this format must be used.

2.4.7 Colours and colour patterns

If it is required to encode multiple colours on a feature, they must be encoded using the attributes **colour pattern** and **colour** as follows:

- For horizontal stripes (**colour pattern** = 1), the first colour in the list must be the top-most, and subsequent colours follow sequentially from top to bottom. For example, **colour** = 3,1 to encode a red stripe above a white stripe.
- For vertical stripes (**colour pattern** = 2), the first colour in the list must be the left-most, and subsequent colours follow sequentially from left to right. For example, **colour** = 3,1,3 to encode red, white, red vertical stripes
- For diagonal stripes (**colour pattern** = 3), the first colour in the list must be the top-left-most, and subsequent colours follow sequentially from top left to bottom right. For example, **colour** = 1,3,1,3,1 to encode white, red, white, red, white diagonal stripes.
- For squares (**colour pattern** = 4), the first colour in the list must be the top-left-most square. Subsequent colours follow sequentially from left to right along the top row then repeated for subsequent rows until the bottom right-most square is reached. For example, **colour** = 1,3,3,1 to encode white, red squares on the top row and red, white squares on the bottom row.
- For border stripes (**colour pattern** = 6), the first colour in the list must be the border stripe, the second colour must be that of the background. For example, **colour** = 3,1 to encode a red border stripe on a white background. Where a border stripe is combined with other patterns, the border stripe colour must be the first colour in the list, and subsequent colours must be interpreted in accordance with the rules defined for the additional patterns. Therefore, if a pattern contains a border stripe as well as other patterns, the border stripe must be the first value in the list of **colour pattern**.

Note that the attribute **colour pattern** is mandatory for any feature (except **Light**) that has more than one colour.

2.4.8 Radar conspicuous features

The attribute **radar conspicuous** is used to encode whether or not a feature is radar conspicuous.

Remarks:

- If it is required to encode a feature which has no radar reflector, but is radar conspicuous, it must be indicated using attribute **radar conspicuous**.

- If it is required to encode a surface or point feature which is radar conspicuous because it is fitted with a radar reflector, it must be indicated using **radar conspicuous** on the feature.
- If it is required to encode radar reflectors on curve features (e.g. overhead cables), this must be done using the feature **Radar Reflector**.

2.5 Datasets

A Dataset is a grouping of features, attributes, geometry and metadata which comprises a specific coverage.

2.6 Description of table format for S-201 meta and geo features

X.X Clause heading

<u>IHO Definition:</u> FEATURE: Definition. (Authority for definition).				
S-201 Geo Feature: Feature S-201 feature				
Primitives: Point, Curve, Surface Allowable geometric primitive(s)				
<i>Real World</i> Example if real world instance(s) of the Feature.	<i>Paper Chart Symbol</i> Example(s) of paper chart equivalent symbology for the Feature.	<i>ECDIS Symbol</i> Example(s) of ECDIS symbology for the Feature.		
S-201 Attribute	Allowable Encoding Value *		Type	Multiplicity
Category of beer	1 : ale 2 : lager 3 : porter 4 : stout 5 : pilsener		EN	1,1
This section lists the full list of allowable attributes for the S-201 feature. Attributes are listed in alphabetical order. Sub-attributes (Type prefix (S)) of complex (Type C) attributes are listed in alphabetical order and indented directly under the entry for the complex attribute	This section lists the allowable encoding values for S-201 (for enumerate (E) Type attributes only). Further information about the attribute is available.		Attribute type	Multiplicity describes the “cardinality” of the attribute in regard to the feature.
Fixed date range			C	0,1
Date end			(S) DA	0,1
Date start			(S) DA	0,1
X Sub-clause heading(s) Introductory remarks. Includes information regarding the real world entity/situation requiring the encoding of the Feature, and where required nautical cartographic principles relevant to the Feature to aid the compiler in determining encoding requirements. Specific instructions to encode the feature. <u>Remarks:</u> • Additional encoding guidance relevant to the feature. X.X Sub-sub-clause heading(s) Clauses related to specific encoding scenarios for the Feature. (Not required for all Features). <u>Remarks:</u> • Additional encoding guidance relevant to the scenario (only if required). <u>Distinction:</u> List of features in the Product Specification distinct from the Feature.				

* For (EN) type attributes, the enumerates listed are only those allowable for the particular occurrence of the attribute relevant to the feature. Allowable values may vary for the attribute depending on the feature to which the attribute is bound. Such bindings are defined in the S-201 Feature Catalogue. The full list of enumerates that may be assigned to an attribute in S-201 can be found in Section **XX** – Attribute and Enumerate Descriptions – of this document.

3 Aids to Navigation - Overview

3.1 Geo features forming parts of navigational aids

Aids to navigation are composed of fixed or floating structure features carrying equipment features.

The most common structure features are: **Cardinal Beacon, Isolated Danger Beacon, Lateral Beacon, Safe Water Beacon, Special Purpose/General Beacon, Cardinal Buoy, Installation Buoy, Isolated Danger Buoy, Lateral Buoy, Safe Water Buoy, Special Purpose/General Buoy, Daymark, Light Float, Light Vessel, Landmark, Offshore Platform, Pile, SiloTank**

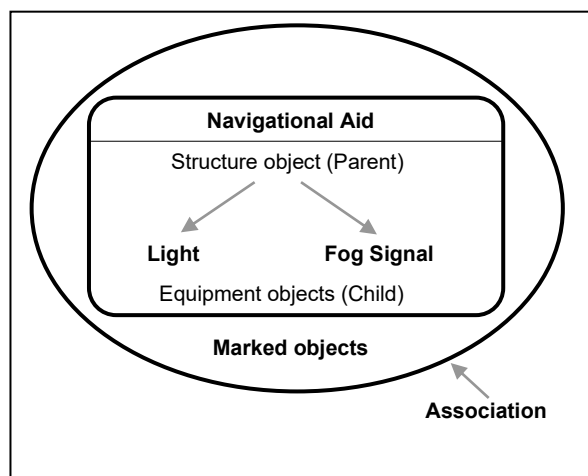
Equipment features consist of: **Daymark, Fog Signal, Light, Environment Observation Equipment, Retroreflector, Radar Transponder Beacon, Topmark.**

Where the source indicates the existence of a navigation aid equipment feature, but there is no indication as to the type or shape of the structure feature, a **Special Purpose Beacon** feature, with the attribute **beacon shape** populated with an empty (null) value, must be encoded as the master feature.

Radar reflectors must not be encoded as separate features when attached to navigational aids. If it is required to encode their existence, it must be done using the attribute **radar conspicuous**.

3.2 Relationships

A parent to child relationship must be created in order to relate the different features comprising a navigational aid. Where a parent to child relationship is created, there must be only one parent (structure) feature related to one or more child (equipment) features. A child feature must not be related to more than one parent feature, and a feature must not be both a parent and a child feature.

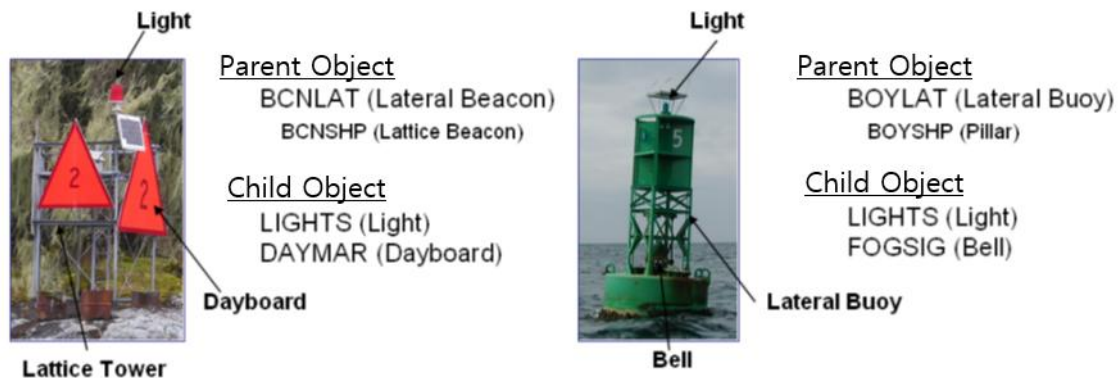


Navigational Aids – Parent / Child Relationship

When the navigational aid contains a structure feature, this feature must be the parent feature, and the equipment features must be the child. Note that **Daymark** may be a parent feature or a child feature; where a navigational aid contains a **Daymark** and there is no other base structure (which can serve as the parent feature) indicated on the source, the **Daymark** feature should be encoded as the parent feature.

When the nature of the base structure on land is unknown or there is no structure feature, one of the equipment features may be chosen as the parent feature, giving priority to a **Light** feature, if one exists. Alternatively, a **Pile** feature of type point or a **Special Purpose/General Beacon** feature may be encoded as the structure feature at the same position as the equipment features. When the nature of the base structure in the water is unknown, an ECDIS Base Display feature, e.g. **Pile** feature of type point or a **Special Purpose/General Beacon** feature, must be encoded as the structure feature at the same position as the equipment features.

Real World Examples



Navigational Aids – Parent / Child Relationship: Real World Examples

In the above real world examples, the parent and child features that make up the navigational aids are point spatial types, and they must share the same geographic point spatial type.

If it is required to encode the name of the navigational aid, it must be done using the complex attribute **feature name** on the parent feature. The name must not be repeated for the child features. If the name is painted on the structure, it must be encoded with the same spelling in the complex attribute **feature name** (sub-attribute **name**, no value populated for sub-attribute **language**) if it is based on the Latin alphabet. If the name is not based on the Latin alphabet, it must be encoded using **feature name (name)**, with an appropriate value populated for the sub-attribute **language**, and transliterated for encoding on an iteration of **feature name (name)** with no value populated for **language**.

All point features comprising a navigational aid must share the same geographic point spatial instance.

The navigational aid may be associated with the features which it marks using the collection feature **Association**. Several navigational aids and several marked features may be associated in the same relationship.

3.3 Buoyage systems and direction of buoyage

Systems of buoyage are described as lateral, cardinal, or a combination of lateral and cardinal. Lateral systems depend on a direction of buoyage being defined. The cardinal system depends solely on the main points of the compass. Special purpose buoys often mark the limits or centre of an area (e.g. an exercise area, a dumping ground) and do not necessarily have lateral or cardinal system characteristics.

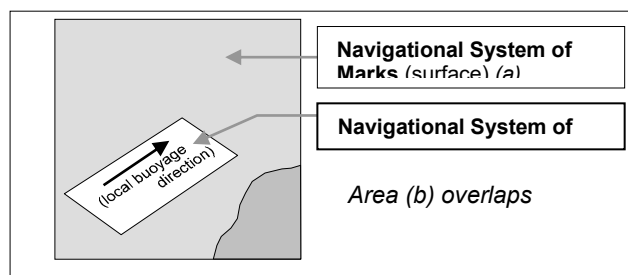
The IALA Maritime Buoyage System details, including the extent of Regions A and B, are given in other publications (e.g. UK's booklet NP 735 "IALA Maritime Buoyage System"). Although it is called a buoyage system, it applies to all fixed and floating marks except lighthouses, some sector lights, leading lights and marks, major floating lights and lights on offshore structures. Five types of marks are provided by the system: Lateral, Cardinal, Isolated danger, Safe water and Special marks, which may be used in any combination. Emergency Wreck Marking Buoys were added on a trial basis in 2006.

3.3.1 Buoyage systems and direction of buoyage

The buoyage system of the dataset and, where necessary, the direction of buoyage, must be encoded using the meta feature **Navigational System of Marks**:

All parts of the dataset containing data must be covered by **Navigational System of Marks** features, with the attribute **marks navigational – system of** indicating the buoyage system in operation. **Navigational System of Marks** features with a value encoded for **marks navigational – system of** must not overlap.

Within a dataset, there may be some areas where the direction of buoyage is defined by local rules and must, therefore, be specified. These areas should be encoded as separate **Navigational System of Marks** surface features, with the complex attribute **orientation** indicating the direction of buoyage (**marks navigational – system of** must not be encoded for these features). **Navigational System of Marks** features with a value encoded for **orientation (orientation value)** must not overlap, but in areas where local buoyage directions apply, **Navigational System of Marks** with a value encoded for **orientation (orientation value)** may overlap **Navigational System of Marks** with a value encoded for **marks navigational – system of**.



Buoyage system and direction

Individual buoys and beacons may not be part of the general buoyage system. This should be encoded using **marks navigational – system of** on these buoy and beacon features.

Remarks:

- At least one of the attributes **marks navigational – system of** or **orientation** must be populated for **Navigational System of Marks** features.

3.3.1.1 Encoding IALA marks within IALA A or B

In the following tables the symbol '/' indicates that this attribute is not relevant for that particular feature. The tables contain the most common examples of encoding; other encoding combinations are possible. For encoding of buoys, substitute **Buoy** for **Beacon** in the Feature column.

Real World Feature	Feature	category of cardinal mark	colour	colour pattern	marks navigational – system of
North cardinal beacon	Cardinal Beacon	1	2,6	1	1 and 2 (IALA A and B)
East cardinal beacon	Cardinal Beacon	2	2,6,2	1	1 and 2 (IALA A and B)
South cardinal beacon	Cardinal Beacon	3	6,2	1	1 and 2 (IALA A and B)
West cardinal beacon	Cardinal Beacon	4	6,2,6	1	1 and 2 (IALA A and B)
Real World Feature	Feature	category of lateral mark	colour	colour pattern	marks navigational – system of
Port lateral beacon	Lateral Beacon	1	3	/	1 (IALA A)
Starboard lateral beacon	Lateral Beacon	2	4	/	1 (IALA A)
Preferred channel to starboard lateral beacon	Lateral Beacon	3	3,4,3	1	1 (IALA A)
Preferred channel to port lateral beacon	Lateral Beacon	4	4,3,4	1	1 (IALA A)
Port lateral beacon	Lateral Beacon	1	4	/	2 (IALA B)
Starboard lateral beacon	Lateral Beacon	2	3	/	2 (IALA B)
Preferred channel to starboard lateral beacon	Lateral Beacon	3	4,3,4	1	2 (IALA B)

Preferred channel to port lateral beacon	Lateral Beacon	4	3,4,3	1	2 (IALA B)
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Real World Feature	Feature	colour	colour pattern	marks navigational – system of
Isolated danger beacon	Isolated Danger Beacon	2,3,2	1	1 and 2 (IALA A and B)
Safe water beacon	Safe Water Beacon	3,1 or 1,3	2	1 and 2 (IALA A and B)
Special purpose beacon	Special Purpose Beacon	6	/	1 and 2 (IALA A and B)
Emergency wreck marking buoy	Emergency Wreck Marking Buoy	5,6 or 6,5	2	1 and 2 (IALA A and B)

Real World Feature	Feature	topmark / daymark shape	colour	colour pattern	marks navigational – system of
North cardinal topmark	Topmark	13	2	/	1 and 2 (IALA A and B)
East cardinal topmark	Topmark	11	2	/	1 and 2 (IALA A and B)
South cardinal topmark	Topmark	14	2	/	1 and 2 (IALA A and B)
West cardinal topmark	Topmark	10	2	/	1 and 2 (IALA A and B)
Isolated danger topmark	Topmark	4	2	/	1 and 2 (IALA A and B)
Port lateral topmark	Topmark	5	3	/	1 (IALA A)
Starboard lateral topmark	Topmark	1	4	/	1 (IALA A)
Port lateral topmark	Topmark	5	4	/	2 (IALA B)
Starboard lateral topmark	Topmark	1	3	/	2 (IALA B)
Safe water topmark	Topmark	3	3	2	1 and 2 (IALA A and B)
Special purpose topmark	Topmark	7	6	/	1 and 2 (IALA A and B)
Emergency wreck marking topmark	Topmark	8	6	/	1 or 2 (IALA A or B)

4 AtoN Fixed structures

If it is required to encode a beacon, it must be done using the object classes Beacon cardinal, Beacon isolated danger, Beacon lateral, Beacon safe water or Beacon special purpose.

Remarks:

- If it is required to encode the altitude of the ground level above the vertical datum at the position of a beacon, it must be done using the attribute Elevation, but only for beacons built on land.
- If it is required to encode the total altitude of a beacon, including any equipment objects (e.g. topmark, light), above the vertical datum, it must be done using the attribute Height.
- The attribute value Beacon shape = 1 (stake, pole, perch, post) implies that stakes, poles, perches and posts must be embedded in the water. Encoders should note that stakes, poles, perches or posts may also exist on land, and where this is the case, they should be encoded using the relevant beacon object class.
- If it is required to encode a cairn that bears the colour(s) specified by a navigational system of marks, it must be done using a beacon object.
- If it is required to encode a beacon that has more than one colour, the attributes Colour and Colour pattern must be encoded, according to the rules

In the following table, the symbol '/' indicates that this attribute does not exist for that particular object class. A blank indicates that the encoder may choose a relevant value for the attribute. The table contains the most common examples of coding; other coding combinations are possible.

Real World Feature	Feature	Beacon shape	Category of special purpose mark	Other attributes
Minor not permanent mark	*** Beacon	1		
	Lateral Beacon	1	/	
	Lateral Beacon	2	/	
Cairn	*** Beacon	6		
Coloured or white mark	Daymark	/		Nature of construction = 9
Coloured topmark with function of beacon	Daymark	/		Nature of construction = 9
Painted board with function of leading beacon	Daymark	/	16	Nature of construction = 9, Topmark/daymark shape = 6
Beacon tower	*** Beacon	3		
Lattice beacon	*** Beacon	4		
Leading beacon	Special Purpose Beacon		16	
Beacon marking a clearing line	Special Purpose Beacon		41	
	Special Purpose			

Real World Feature	Feature	Beacon shape	Category of special purpose mark	Other attributes
Beacon marking measured distance	Beacon		<i>17</i>	
Cable landing beacon	Special Purpose Beacon		<i>6</i>	
Outfall landing beacon	Special Purpose Beacon		<i>8</i>	
Pipeline landing beacon	Special Purpose Beacon		<i>39</i>	
Refuge beacon	Special Purpose Beacon		<i>44</i>	
Firing practice area beacon	Special Purpose Beacon		<i>1</i>	
Notice board	Special Purpose Beacon		<i>18</i>	
Buoyant beacon	*** Beacon	<i>7</i>		

table 12.2

4.1 Special Purpose/General Beacon (SpecialPurposeGeneralBeacon)

Definition: **SPECIAL PURPOSE/GENERAL BEACON.** A beacon is a prominent specially constructed object forming a conspicuous mark as a fixed aid to navigation or for use in hydrographic survey. (IHO Dictionary – S-32).

Beacon in general: A beacon whose appearance or purpose is not adequately known. (S-57 Edition 3.1, Appendix A – Chapter 1, Page 1.12, November 2000).

S-201 Geo Feature: Special Purpose/General

Primitives: Point

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *

AtoN Number		TE	1, 1
Aid Availability Category	1: Category 1 2: Category 2 3: Category 3	EN	0, 1
Condition	1: Under Construction 2: Ruined 3: Under Reclamation 4: Wingless 5: Planned Construction	EN	0, 1
Contact Address		C	0, 1
Delivery Point		(S) TE	0, 1
City Name		(S) TE	0, 1
Administrative Division		(S) TE	0, 1
Country Name		(S) TE	0, 1
Postal Code		(S) TE	0, 1
Beacon Shape	1: Stake, Pole, Perch, Post 2: Withy 3: Beacon Tower 4: Lattice Beacon 5: Pile Beacon 6: Cairn 7: Buoyant Beacon	EN	1, 1
Colour	1 : white 2 : black 3 : red 4 : green 5 : blue 6 : yellow 7 : grey 8 : brown 9 : amber 10 : violet 11 : orange 12 : magenta 13 : pink	EN	1, *
Colour Pattern	1: Horizontal Stripes 2: Vertical Stripes 3: Diagonal Stripes 4: Squared 5: Stripes (Direction Unknown) 6: Border Stripe 7: Single Colour 8: Rectangle 9: Triangle	EN	0, *
Elevation		RE	0, 1
Height		RE	0, 1
Marks Navigational - System Of	1: IALA A 2: IALA B 9: No System 10: Other System 11: CEVNI 12: Russian Inland Waterway Regulations 13: Brazilian National Inland Waterway Regulations - Two Sides 15: Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	0, 1
Nature of Construction	1: Masonry	EN	0, *

	2: Concreted 3: Loose Boulders 4: Hard Surfaced 5: Unsurfaced 6: Wooden 7: Metal 8: Glass Reinforced Plastic 9: Painted 10: Framework 11: Latticed 12: Glass 13: Fiberglass 14: Plastic		
Radar Conspicuous		BO	0, 1
Status	1: Permanent 2: Occasional 3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 8: Private 9: Mandatory 11: Extinguished 12: Illuminated 13: Historic 14: Public 15: Synchronized 16: Watched 17: Unwatched 18: Existence Doubtful 19: On Request 20: Drop Away 21: Rising 22: Increasing 23: Decreasing 24: Strong 25: Good 26: Moderately 27: Poor 28: Buoyed 29: Fully Operational 30: Partially Operational 31: Drifting 32: Broken 33: Offline 34: Discontinued 35: Manual Observation 36: Unknown Status 37: Confirmed 38: Candidate 39: Under Modification 40: Experimental 41: Under Removal / Deletion 42: Removed / Deleted 43: Candidate for Modification	EN	0, *
Vertical Length		RE	0, 1
Visual Prominence	1: Visually Conspicuous 2: Not Visually Conspicuous 3: Prominent	EN	0, 1

Vertical Accuracy		RE	0, 1
Category of special purpose mark	1 : firing danger area mark 2 : target mark 3 : marker ship mark 4 : degaussing range mark 5 : barge mark 6 : cable mark 7 : spoil ground mark 8 : outfall mark 10 : recording mark 11 : seaplane anchorage mark 12 : recreation zone mark 14 : mooring mark 15 : LANBY (Large Automatic Navigational Buoy) 16 : leading mark 17 : measured distance mark 18 : notice mark 19 : TSS mark (Traffic Separation Scheme) 20 : anchoring prohibited mark 21 : berthing prohibited mark 22 : overtaking prohibited mark 23 : two-way traffic prohibited mark 24 : reduced wake mark 25 : speed limit mark 26 : stop mark 27 : general warning mark 28 : sound ship's siren mark 29 : restricted vertical clearance mark 30 : maximum vessel's draught mark 31 : restricted horizontal clearance mark 32 : strong current warning mark 33 : berthing permitted mark 34 : overhead power cable mark 35 : channel edge gradient' mark 36 : telephone mark 37 : ferry crossing mark 39 : pipeline mark 40 : anchorage mark 41 : clearing mark 42 : control mark 43 : diving mark 44 : refuge beacon 45 : foul ground mark 46 : yachting mark 47 : heliport mark 48 : GNSS mark 49 : seaplane landing mark 50 : control mark 51 : work in progress mark 52 : mark with unknown purpose 53 : wellhead mark 54 : channel separation mark 55 : marine farm mark 56 : artificial reef mark 57 : ice mark 58 : nature reserve mark 59 : Fish Aggregating Device 60 : Wreck Mark 61 : Customs Mark 62 : Causeway Mark 63 : Wave Recorder 64 : Jetski Prohibited 65 : Facility Protection Mark	EN	1,*

	66 : Oil Pipeline Protection Mark		
	67 : Marine Cable Protection Mark		

4.1.1 Special purpose/general beacons

Special beacons are used to indicate to the mariner a special area or feature, the nature of which is usually apparent from the chart or associated publication.

To conform to the IALA Maritime Buoyage System, the body of the beacon is yellow. The topmark (if fitted) is a yellow diagonal 'X' (St Andrew's cross). Lights (if fitted) are yellow and of any rhythm except those used for cardinal, isolated danger and safe water marks.

If it is required to encode a beacon having the function of a special purpose mark, or a beacon whose appearance or purpose is inadequately known, it must be done using the feature **Special Purpose/General Beacon**.

In the following table, a blank indicates that the encoder may choose a relevant value for the attribute. The table contains the most common examples of coding; other coding combinations are possible.

Feature	INT1	Feature	beacon shape	category of special purpose mark	Other attributes
Minor not permanent mark	Q90	Beacon ***	1		
Cairn	Q100	Beacon ***	6		
Beacon tower	Q110	Beacon ***	3		
Lattice beacon	Q111	Beacon ***	4		
Leading beacon	Q120	Beacon Special Purpose/General		16	
Beacon marking a clearing line	Q121	Beacon Special Purpose/General		41	
Beacon marking measured distance	Q122	Beacon Special Purpose/General		17	
Cable landing beacon	Q123	Beacon Special Purpose/General		6	
Outfall landing beacon	Q123	Beacon Special Purpose/General		8	
Pipeline landing beacon	Q123	Beacon Special Purpose/General		39	
Refuge beacon	Q124	Beacon Special Purpose/General		44	
Firing practice area beacon	Q125	Beacon Special Purpose/General		1	
Notice board	Q126	Beacon Special Purpose/General		18	
Buoyant beacon	P5	Beacon Special Purpose/General	7		

Remarks:

- The attribute **colour pattern** must be populated if more than one value is populated for the mandatory attribute **colour**.
- If it is required to encode the altitude of the ground level above the vertical datum at the position of a beacon, it must be done using the attribute **elevation**, but only for beacons built on land.
- If it is required to encode the total altitude of a beacon, including the topmark and any equipment features (e.g. light), above the vertical datum, it must be done using the attribute **height**.
- If it is required to encode the total vertical length of a beacon, including the topmark and any equipment features (e.g. light), above the seabed or ground, it must be done using the attribute **vertical length**.
- If it is required to encode a cairn that bears the colour(s) specified by a navigational system of marks, it

must be done using a beacon feature.

4.1.2 Signs and notice boards

If it is required to encode a fixed or floating sign or notice board, it must be done using a **Special Purpose/General Beacon** feature or **Special Purpose/General Buoy** feature, with attribute **category of special purpose mark = 18** (notice mark), or using the feature **Daymark**.

Remarks:

- If it is required to encode a sign or notice board that has more than one colour, the attributes **colour** and **colour pattern** must be used, according to the rules.
- If it is required to encode any text shown on a notice board or sign, it must be done using the complex attribute **information**.
- If it is required to encode the shape and colour of a notice board, it must be done by encoding the board as a **Daymark** feature.

Distinction: Beacon, cardinal; beacon, isolated danger; beacon, lateral; beacon, safe water; daymark.

4.2 Lateral beacons (LateralBeacon)

Definition: LATERAL BEACON. A beacon is a prominent specially constructed object forming a conspicuous mark as a fixed aid to navigation or for use in hydrographic survey. (IHO Dictionary – S-32).

S-201 Geo Feature: Lateral Beacon

Primitives: Point

<i>Real World</i>	<i>Paper Chart Symbol</i>	<i>ECDIS Symbol</i>		
S-201 Attribute	Allowable Encoding Value	Type	Multiplicity	
ID Code		TE	0, 1	
Information		C	0, *	
File Locator		(S) TE	0, 1	
File Reference		(S) TE	0, 1	
Headline		(S) TE	0, 1	
Language		(S) TE	1, 1	
Text		(S) TE	0, 1	
Feature Name		C	0, *	
Display Name		(S) BO	0, 1	
Language		(S) TE	0, 1	
Name		(S) TE	1, 1	
Scale Minimum		IN	0, 1	
Source Date		DA	0, 1	
Source		TE	0, 1	
Pictorial Representation		TE	0, 1	
Inspection Frequency		TE	0, 1	
Inspection Requirements		TE	0, 1	
AtoN Maintenance Record		TE	0, 1	
Installation Date		DA	0, 1	
Fixed Date Range		C	0, 1	
Date End		(S) TD	0, 1	
Date Start		(S) TD	0, 1	
Periodic Date Range		C	0, 1	
Date End		(S) TD	1, 1	
Date Start		(S) TD	1, 1	
Seasonal Action Required		TE	0, *	
AtoN Number		TE	1,1	
Aid Availability Category	1: Category 1 2: Category 2 3: Category 3	EN	0, 1	

Condition	1: Under Construction 2: Ruined 3: Under Reclamation 4: Wingless 5: Planned Construction	EN	0, 1
Contact Address		C	0, 1
Delivery Point		(S) TE	0, 1
City Name		(S) TE	0, 1
Administrative Division		(S) TE	0, 1
Country Name		(S) TE	0, 1
Postal Code		(S) TE	0, 1
Beacon Shape	1: Stake, Pole, Perch, Post 2: Withy 3: Beacon Tower 4: Lattice Beacon 5: Pile Beacon 6: Cairn 7: Buoyant Beacon	EN	1, 1
Colour	1: White 2: Black 3: Red 4: Green 5: Blue 6: Yellow 7: Grey 8: Brown 9: Amber 10: Violet 11: Orange 12: Magenta 13: Pink	EN	1, *
Colour Pattern	1: Horizontal Stripes 2: Vertical Stripes 3: Diagonal Stripes 4: Squared 5: Stripes (Direction Unknown) 6: Border Stripe 7: Single Colour 8: Rectangle 9: Triangle	EN	0, *
Elevation		RE	0, 1
Height		RE	0, 1
Marks Navigational - System Of	1: IALA A 2: IALA B 9: No System 10: Other System 11: CEVNI 12: Russian Inland Waterway Regulations 13: Brazilian National Inland Waterway Regulations - Two Sides 15: Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	0, 1
Nature of Construction	1: Masonry 2: Concreted 3: Loose Boulders 4: Hard Surfaced	EN	0, *

	5: Unsurfaced 6: Wooden 7: Metal 8: Glass Reinforced Plastic 9: Painted 10: Framework 11: Latticed 12: Glass 13: Fiberglass 14: Plastic		
Radar Conspicuous		BO	0, 1
Status	1: Permanent 2: Occasional 3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 8: Private 9: Mandatory 11: Extinguished 12: Illuminated 13: Historic 14: Public 15: Synchronized 16: Watched 17: Unwatched 18: Existence Doubtful 19: On Request 20: Drop Away 21: Rising 22: Increasing 23: Decreasing 24: Strong 25: Good 26: Moderately 27: Poor 28: Buoyed 29: Fully Operational 30: Partially Operational 31: Drifting 32: Broken 33: Offline 34: Discontinued 35: Manual Observation 36: Unknown Status 37: Confirmed 38: Candidate 39: Under Modification 40: Experimental 41: Under Removal / Deletion 42: Removed / Deleted 43: Candidate for Modification	EN	0, *
Vertical Length		RE	0, 1
Visual Prominence	1: Visually Conspicuous 2: Not Visually Conspicuous 3: Prominent	EN	0, 1
Vertical Accuracy		RE	0, 1
Category of lateral mark	1 : Port-Hand Lateral Mark 2 : Starboard-Hand Lateral Mark	EN	1,1

3 : Preferred Channel to Starboard Lateral Mark	
4 : Preferred Channel to Port Lateral Mark	

4.2.1 Lateral Beacons

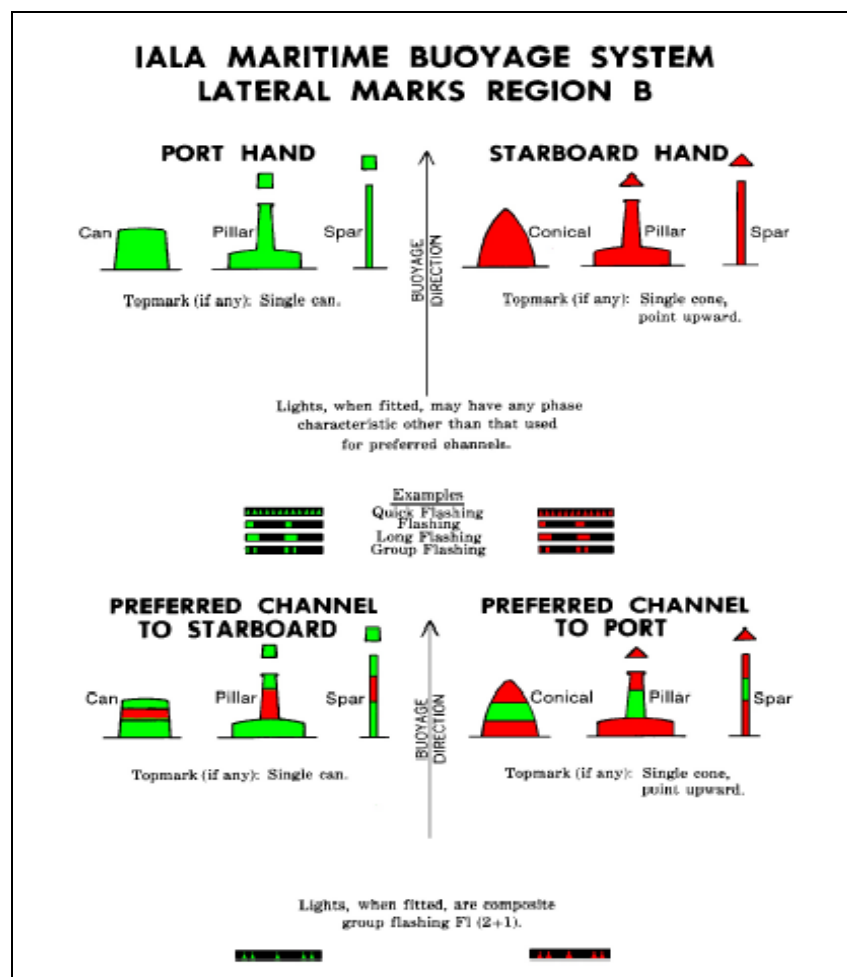
Lateral beacons are generally used for well-defined channels, in conjunction with a direction of buoyage. They indicate the port and starboard sides of the route to be followed.

To conform to the IALA Maritime Buoyage System, port hand beacons have a can shaped topmark. The colour of port hand beacons, topmarks and lights (if fitted) will be red in IALA region A and green in IALA region B.

To conform to the IALA Maritime Buoyage System, starboard hand beacons have a conical shaped topmark. The colour of starboard hand beacons, topmarks and lights (if fitted) will be green in IALA region A and red in IALA region B.

A preferred channel beacon is a modified lateral beacon, with horizontal colour bands. The predominant colour indicates which side is the preferred channel, the other colour indicates the secondary channel. If fitted, the light is Fl(2+1), the colour indicating the preferred channel.

If it is required to encode a beacon having the function of a lateral mark, it must be done using the feature **Lateral Beacon**.



Remarks:

- The attribute **colour pattern** must be populated if more than one value is populated for the mandatory attribute **colour**.
- If it is required to encode the altitude of the ground level above the vertical datum at the position of a beacon, it must be done using the attribute **elevation**, but only for beacons built on land.

- If it is required to encode the total altitude of a beacon, including the topmark and any equipment features (e.g. light), above the vertical datum, it must be done using the attribute **height**.
- If it is required to encode the total vertical length of a beacon, including the topmark and any equipment features (e.g. light), above the seabed or ground, it must be done using the attribute **vertical length**.
- If it is required to encode a cairn that bears the colour(s) specified by a navigational system of marks, it must be done using a beacon feature.

Distinction: Beacon, cardinal; beacon, isolated danger; beacon, safe water; beacon, special purpose/general; daymark.

4.3 Cardinal beacons (CardinalBeacon)

Definition: **CARDINAL BEACON**,. A beacon is a prominent specially constructed object forming a conspicuous mark as a fixed aid to navigation or for use in hydrographic survey. (IHO Dictionary – S-32).

S-201 Geo Feature: Cardinal Beacon

Primitives: Point

<i>Real World</i>	<i>Paper Chart Symbol</i>	<i>ECDIS Symbol</i>	
S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
AtoN Number		TE	1,1
Aid Availability Category	1: Category 1 2: Category 2 3: Category 3	EN	0, 1

Condition	1: Under Construction 2: Ruined 3: Under Reclamation 4: Wingless 5: Planned Construction	EN	0, 1
Contact Address		C	0, 1
Delivery Point		(S) TE	0, 1
City Name		(S) TE	0, 1
Administrative Division		(S) TE	0, 1
Country Name		(S) TE	0, 1
Postal Code		(S) TE	0, 1
Beacon Shape	1: Stake, Pole, Perch, Post 2: Withy 3: Beacon Tower 4: Lattice Beacon 5: Pile Beacon 6: Cairn 7: Buoyant Beacon	EN	1, 1
Colour	1: White 2: Black 3: Red 4: Green 5: Blue 6: Yellow 7: Grey 8: Brown 9: Amber 10: Violet 11: Orange 12: Magenta 13: Pink	EN	1, *
Colour Pattern	1: Horizontal Stripes 2: Vertical Stripes 3: Diagonal Stripes 4: Squared 5: Stripes (Direction Unknown) 6: Border Stripe 7: Single Colour 8: Rectangle 9: Triangle	EN	0, *
Elevation		RE	0, 1
Height		RE	0, 1
Marks Navigational - System Of	1: IALA A 2: IALA B 9: No System 10: Other System 11: CEVNI 12: Russian Inland Waterway Regulations 13: Brazilian National Inland Waterway Regulations - Two Sides 15: Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	0, 1
Nature of Construction	1: Masonry 2: Concreted 3: Loose Boulders 4: Hard Surfaced	EN	0, *

	5: Unsurfaced 6: Wooden 7: Metal 8: Glass Reinforced Plastic 9: Painted 10: Framework 11: Latticed 12: Glass 13: Fiberglass 14: Plastic		
Radar Conspicuous		BO	0, 1
Status	1: Permanent 2: Occasional 3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 8: Private 9: Mandatory 11: Extinguished 12: Illuminated 13: Historic 14: Public 15: Synchronized 16: Watched 17: Unwatched 18: Existence Doubtful 19: On Request 20: Drop Away 21: Rising 22: Increasing 23: Decreasing 24: Strong 25: Good 26: Moderately 27: Poor 28: Buoyed 29: Fully Operational 30: Partially Operational 31: Drifting 32: Broken 33: Offline 34: Discontinued 35: Manual Observation 36: Unknown Status 37: Confirmed 38: Candidate 39: Under Modification 40: Experimental 41: Under Removal / Deletion 42: Removed / Deleted 43: Candidate for Modification	EN	0, *
Vertical Length		RE	0, 1
Visual Prominence	1: Visually Conspicuous 2: Not Visually Conspicuous 3: Prominent	EN	0, 1
Vertical Accuracy		RE	0, 1
Category of Cardinal Mark	1 : North Cardinal Mark 2 : East Cardinal Mark	EN	1,1

3 : South Cardinal Mark

4 : West Cardinal Mark

4.3.1 Cardinal beacons

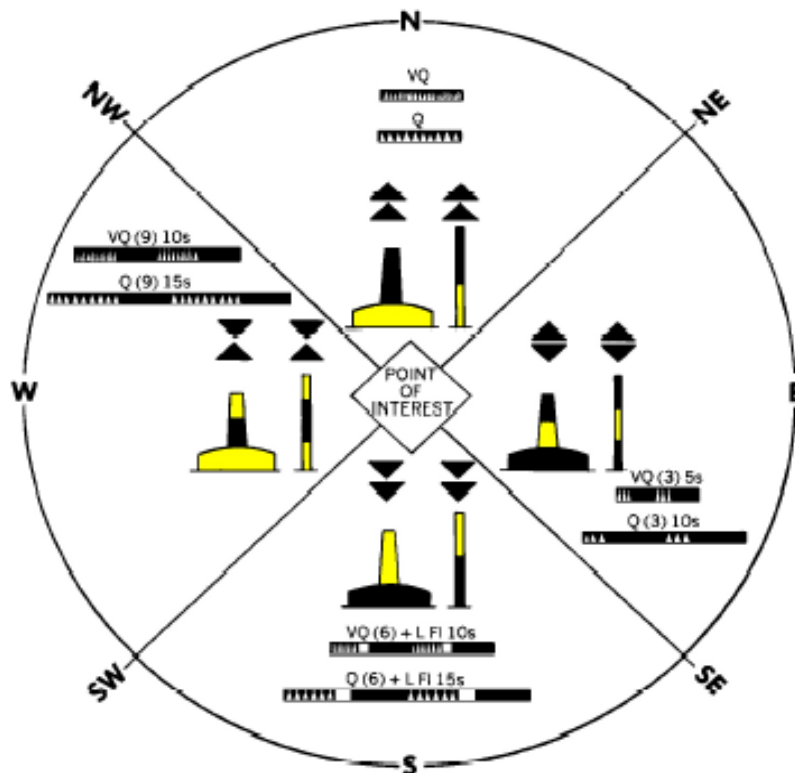
Cardinal marks are used in conjunction with the compass to indicate where a mariner may find best navigable water, taking their name from the quadrant in which they are placed in relation to the point marked. The mariner should pass N of a North mark, E of an East mark, etc.

To conform to the IALA Maritime Buoyage System, the body of the beacon has black and yellow bands, configured with black reflecting the points of the topmark cones (e.g. black above yellow for north). Black double-cone topmarks are an important feature of cardinal marks and are carried wherever practicable. The points are up for a north mark, down for a south mark, apart for an east mark and together for a west mark. Lights (if fitted) are white Q or VQ, uninterrupted for the north, 3 flashes for east, 6 flashes + LFI for south and 9 flashes for west (resembling an analogue clock).

If it is required to encode a beacon having the function of a cardinal mark, it must be done using the feature **Cardinal Beacon**.

IALA MARITIME BUOYAGE SYSTEM CARDINAL MARKS REGIONS A AND B

Topmarks are always fitted (when practicable),
Buoy shapes are pillar or spar.



Lights, when fitted, are **white**. Very Quick Flashing or Quick Flashing; a South mark also has a Long Flash immediately following the quick flashes.

Remarks:

- The attribute **colour pattern** must be populated if more than one value is populated for the mandatory attribute **colour**.
- If it is required to encode the altitude of the ground level above the vertical datum at the position of a beacon, it must be done using the attribute **elevation**, but only for beacons built on land.
- If it is required to encode the total altitude of a beacon, including the topmark and any equipment features (e.g. light), above the vertical datum, it must be done using the attribute **height**.
- If it is required to encode the total vertical length of a beacon, including the topmark and any equipment features (e.g. light), above the seabed or ground, it must be done using the attribute **vertical length**.
- If it is required to encode a cairn that bears the colour(s) specified by a navigational system of marks, it must be done using a beacon feature.

Distinction: Beacon, isolated danger; beacon, lateral; beacon, safe water; beacon, special purpose/general; daymark.

4.4 Isolated danger beacons (IsolatedDangerBeacon)

Definition: **ISOLATED DANGER BEACON.** A beacon is a prominent, specially constructed object forming a conspicuous mark as a fixed aid to navigation or for use in hydrographic survey. (IHO Dictionary – S-32).

S-201 Geo Feature: Isolated danger Beacon

Primitives: Point

Real World	Paper Chart Symbol	ECDIS Symbol	
S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
AtoN Number		TE	1,1
Aid Availability Category	1: Category 1 2: Category 2 3: Category 3	EN	0, 1
Condition	1: Under Construction	EN	0, 1

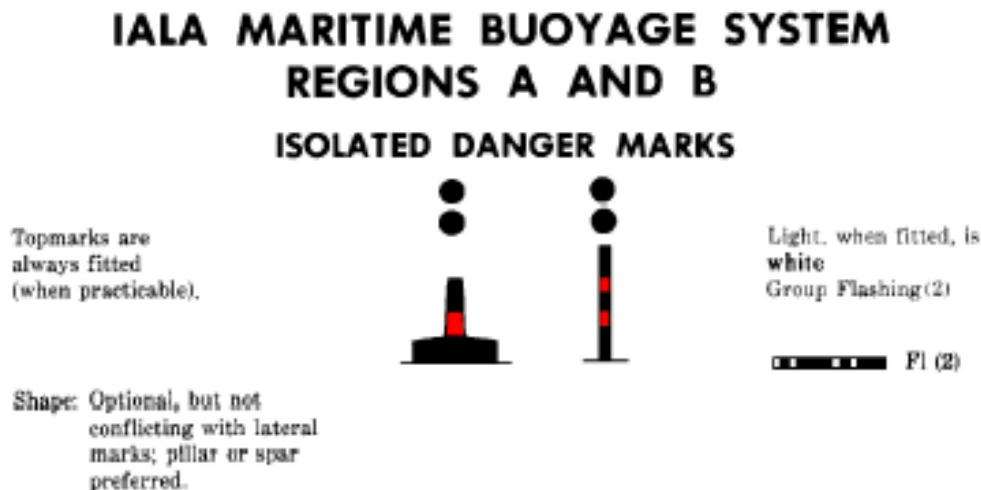
	2: Ruined 3: Under Reclamation 4: Wingless 5: Planned Construction		
Contact Address		C	0, 1
Delivery Point		(S) TE	0, 1
City Name		(S) TE	0, 1
Administrative Division		(S) TE	0, 1
Country Name		(S) TE	0, 1
Postal Code		(S) TE	0, 1
Beacon Shape	1: Stake, Pole, Perch, Post 2: Withy 3: Beacon Tower 4: Lattice Beacon 5: Pile Beacon 6: Cairn 7: Buoyant Beacon	EN	1, 1
Colour	1: White 2: Black 3: Red 4: Green 5: Blue 6: Yellow 7: Grey 8: Brown 9: Amber 10: Violet 11: Orange 12: Magenta 13: Pink	EN	1, *
Colour Pattern	1: Horizontal Stripes 2: Vertical Stripes 3: Diagonal Stripes 4: Squared 5: Stripes (Direction Unknown) 6: Border Stripe 7: Single Colour 8: Rectangle 9: Triangle	EN	0, *
Elevation		RE	0, 1
Height		RE	0, 1
Marks Navigational - System Of	1: IALA A 2: IALA B 9: No System 10: Other System 11: CEVNI 12: Russian Inland Waterway Regulations 13: Brazilian National Inland Waterway Regulations - Two Sides 15: Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	0, 1
Nature of Construction	1: Masonry 2: Concreted 3: Loose Boulders 4: Hard Surfaced 5: Unsurfaced	EN	0, *

	6: Wooden 7: Metal 8: Glass Reinforced Plastic 9: Painted 10: Framework 11: Latticed 12: Glass 13: Fiberglass 14: Plastic		
Radar Conspicuous		BO	0, 1
Status	1: Permanent 2: Occasional 3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 8: Private 9: Mandatory 11: Extinguished 12: Illuminated 13: Historic 14: Public 15: Synchronized 16: Watched 17: Unwatched 18: Existence Doubtful 19: On Request 20: Drop Away 21: Rising 22: Increasing 23: Decreasing 24: Strong 25: Good 26: Moderately 27: Poor 28: Buoyed 29: Fully Operational 30: Partially Operational 31: Drifting 32: Broken 33: Offline 34: Discontinued 35: Manual Observation 36: Unknown Status 37: Confirmed 38: Candidate 39: Under Modification 40: Experimental 41: Under Removal / Deletion 42: Removed / Deleted 43: Candidate for Modification	EN	0, *
Vertical Length		RE	0, 1
Visual Prominence	1: Visually Conspicuous 2: Not Visually Conspicuous 3: Prominent	EN	0, 1
Vertical Accuracy		RE	0, 1
4.4.1 Isolated danger beacons			

Isolated danger beacons are placed on isolated dangers of limited extent with navigable water all around them.

To conform to the IALA Maritime Buoyage System, the body of an isolated danger beacon is black, with one or more red bands. Black double-sphere topmarks are an important feature of isolated danger beacons and carried wherever practicable. The light (if fitted) is white Fl(2).

If it is required to encode a beacon having the function of an isolated danger mark, it must be done using the feature **Isolated Danger Beacon**.



Remarks:

- The attribute **colour pattern** must be populated if more than one value is populated for the mandatory attribute **colour**.
- If it is required to encode the altitude of the ground level above the vertical datum at the position of a beacon, it must be done using the attribute **elevation**, but only for beacons built on land.
- If it is required to encode the total altitude of a beacon, including the topmark and any equipment features (e.g. light), above the vertical datum, it must be done using the attribute **height**.
- If it is required to encode the total vertical length of a beacon, including the topmark and any equipment features (e.g. light), above the seabed or ground, it must be done using the attribute **vertical length**.
- If it is required to encode a cairn that bears the colour(s) specified by a navigational system of marks, it must be done using a beacon feature.

Distinction: Beacon, cardinal; beacon, lateral; beacon, safe water; beacon, special purpose/general; daymark.

4.5 Safe water beacons (SafeWaterBeacon)

Definition: **SAFE WATER BEACON.** A safe water beacon is a prominent specially constructed object forming a conspicuous mark as a fixed aid to navigation or for use in hydrographic survey. (IHO Dictionary – S-32, Edition 5).

S-201 Geo Feature: Safe water Beacon

Primitives: Point

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
AtoN Number		TE	1, 1
Aid Availability Category	1: Category 1 2: Category 2	EN	0, 1

	3: Category 3		
Condition	1: Under Construction 2: Ruined 3: Under Reclamation 4: Wingless 5: Planned Construction	EN	0, 1
Contact Address		C	0, 1
Delivery Point		(S) TE	0, 1
City Name		(S) TE	0, 1
Administrative Division		(S) TE	0, 1
Country Name		(S) TE	0, 1
Postal Code		(S) TE	0, 1
Beacon Shape	1: Stake, Pole, Perch, Post 2: Withy 3: Beacon Tower 4: Lattice Beacon 5: Pile Beacon 6: Cairn 7: Buoyant Beacon	EN	1, 1
Colour	1: White 2: Black 3: Red 4: Green 5: Blue 6: Yellow 7: Grey 8: Brown 9: Amber 10: Violet 11: Orange 12: Magenta 13: Pink	EN	1, *
Colour Pattern	1: Horizontal Stripes 2: Vertical Stripes 3: Diagonal Stripes 4: Squared 5: Stripes (Direction Unknown) 6: Border Stripe 7: Single Colour 8: Rectangle 9: Triangle	EN	0, *
Elevation		RE	0, 1
Height		RE	0, 1
Marks Navigational - System Of	1: IALA A 2: IALA B 9: No System 10: Other System 11: CEVNI 12: Russian Inland Waterway Regulations 13: Brazilian National Inland Waterway Regulations - Two Sides 15: Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	0, 1
Nature of Construction	1: Masonry 2: Concreted 3: Loose Boulders	EN	0, *

	4: Hard Surfaced 5: Unsurfaced 6: Wooden 7: Metal 8: Glass Reinforced Plastic 9: Painted 10: Framework 11: Latticed 12: Glass 13: Fiberglass 14: Plastic		
Radar Conspicuous		BO	0, 1
Status	1: Permanent 2: Occasional 3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 8: Private 9: Mandatory 11: Extinguished 12: Illuminated 13: Historic 14: Public 15: Synchronized 16: Watched 17: Unwatched 18: Existence Doubtful 19: On Request 20: Drop Away 21: Rising 22: Increasing 23: Decreasing 24: Strong 25: Good 26: Moderately 27: Poor 28: Buoyed 29: Fully Operational 30: Partially Operational 31: Drifting 32: Broken 33: Offline 34: Discontinued 35: Manual Observation 36: Unknown Status 37: Confirmed 38: Candidate 39: Under Modification 40: Experimental 41: Under Removal / Deletion 42: Removed / Deleted 43: Candidate for Modification	EN	0, *
Vertical Length		RE	0, 1
Visual Prominence	1: Visually Conspicuous 2: Not Visually Conspicuous 3: Prominent	EN	0, 1
Vertical Accuracy		RE	0, 1

4.5.1 Safe water beacons

Safe water marks are used to indicate there is safe water all around the mark. It may be used as a centre-line, mid-channel or landfall beacon, or to mark the best point of passage under a bridge.

To conform to the IALA Maritime Buoyage System, the body of the mark has red and white vertical stripes. A red spherical topmark is an important feature if the beacon is not spherical and carried wherever practicable. The light (if fitted) is white Oc, Iso, LFI or Mo(A) with a period of 10s.

If it is required to encode a beacon having the function of a safe water mark, it must be done using the feature **Safe Water Beacon**.



Remarks:

- The attribute **colour pattern** must be populated if more than one value is populated for the mandatory attribute **colour**.
- If it is required to encode the altitude of the ground level above the vertical datum at the position of a beacon, it must be done using the attribute **elevation**, but only for beacons built on land.
- If it is required to encode the total altitude of a beacon, including the topmark and any equipment features (e.g. light), above the vertical datum, it must be done using the attribute **height**.
- If it is required to encode the total vertical length of a beacon, including the topmark and any equipment features (e.g. light), above the seabed or ground, it must be done using the attribute **vertical length**.
- If it is required to encode a cairn that bears the colour(s) specified by a navigational system of marks, it must be done using a beacon feature.

Distinction: Beacon, cardinal; beacon, isolated danger; beacon, lateral; beacon, special purpose/general; daymark.

5 AtoN floating structures

If it is required to encode a buoy, it must be done using the object classes Buoy cardinal, Buoy installation, Buoy isolated danger, Buoy lateral, Buoy safe water or Buoy special purpose.

Remarks:

- If it is required to encode a buoy that has more than one colour, the attributes Colour and Colour pattern must be encoded, according to the rules laid out in clause 2.4.
- The optional attribute IALA_typeOfBuoy on buoy feature types is used to encode the type of buoy equipment used.

In the following table, the symbol '/' indicates that this attribute does not exist for that particular object class. A blank indicates that the encoder may choose a relevant value for the attribute. The table contains the most common examples of coding; other coding combinations are possible.

Real Feature Type	Feature	Buoy Shape	Category of special purpose mark	Other attributes
Oil or gas installation buoy	Installation Buoy	7	/	Product = 1 or 2, Category of installation buoy = 1 or 2
Firing danger area buoy	Special purpose/general Buoy		1	
Target	Special purpose/general Buoy		2	
Marker ship	Special purpose/general Buoy		3	
Barge	Special purpose/general Buoy		5	
Degaussing range buoy	Special purpose/general Buoy		4	
Buoy marking cable	Special purpose/general Buoy		6	
Spoil ground buoy	Special purpose/general Buoy		7	
Buoy marking outfall	Special purpose/general Buoy		8	
Buoy marking pipeline	Special purpose/general Buoy		39	
Superbuoy	Buoy ***	7		
Large automatic navigational buoy	Special purpose/general Buoy	7	15	
Data-collecting buoy of superbuoy size	Special purpose/general Buoy	7	9	
Buoy marking wave recorder (or current meter)	Special purpose/general Buoy		10	INFORM = wave recorder (e.g.)
Seaplane anchorage buoy	Special purpose/general Buoy		11	
Buoy marking traffic separation scheme	Special purpose/general Buoy		19	
Buoy marking recreation zone	Special purpose/general Buoy		12	

table 12.3

5.1 Installation buoy (InstallationBuoy)

Definition: **INSTALLATION BUOY**. A buoy is a floating object moored to the bottom in a particular place, as an aid to navigation or for other specific purposes. (IHO Dictionary – S-32).

An installation buoy is a buoy used for loading tankers with gas or oil. (S-57 Edition 3.1, Appendix A – Chapter 1, Page 1.20, November 2000).

S-201 Geo Feature: Installation Buoy

Primitives: Point

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
AtoN Number		TE	1, 1

Aid Availability Category	1: Category 1 2: Category 2 3: Category 3	EN	0, 1
Condition	1: Under Construction 2: Ruined 3: Under Reclamation 4: Wingless 5: Planned Construction	EN	0, 1
Contact Address		C	0, 1
Delivery Point		(S) TE	0, 1
City Name		(S) TE	0, 1
Administrative Division		(S) TE	0, 1
Country Name		(S) TE	0, 1
Postal Code		(S) TE	0, 1
Buoy Shape	1: Conical 2: Can 3: Spherical 4: Pillar 5: Spar 6: Barrel 7: Superbuoy 8: Ice Buoy	EN	1, 1
Colour	1: White 2: Black 3: Red 4: Green 5: Blue 6: Yellow 7: Grey 8: Brown 9: Amber 10: Violet 11: Orange 12: Magenta 13: Pink	EN	1, *
Colour Pattern	1: Horizontal Stripes 2: Vertical Stripes 3: Diagonal Stripes 4: Squared 5: Stripes (Direction Unknown) 6: Border Stripe 7: Single Colour 8: Rectangle 9: Triangle	EN	0, *
Marks Navigational - System Of	1: IALA A 2: IALA B 9: No System 10: Other System 11: CEVNI 12: Russian Inland Waterway Regulations 13: Brazilian National Inland Waterway Regulations - Two Sides 15: Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	0, 1

Nature of Construction	1: Masonry 2: Concreted 3: Loose Boulders 4: Hard Surfaced 5: Unsurfaced 6: Wooden 7: Metal 8: Glass Reinforced Plastic 9: Painted 10: Framework 11: Latticed 12: Glass 13: Fiberglass 14: Plastic	EN	0, *
Radar Conspicuous		BO	0, 1
Category of Installation Buoy	1: Catenary Anchor Leg Mooring 2: Single Buoy Mooring	EN	0,1

INT 1 Reference: L 16

5.1.1 Installation buoys

Although the oil and gas from some fields are sent ashore by submarine pipeline, a variety of mooring systems have been developed for use in deep water and in the vicinity of certain ports, to allow the loading of large vessels and the permanent mooring of floating storage vessels or units. These offshore systems include large mooring buoys, designed for mooring vessels up to 500,000 tonnes, and platforms on structures fixed at their lower ends to the sea floor. They allow a vessel to moor forward or aft to them, and to swing to the wind or stream, and are termed installation buoys.

If it is required to encode an installation buoy, it must be done using the feature **Installation Buoy**.

Remarks:

- The attribute **colour pattern** must be populated if more than one value is populated for the mandatory attribute **colour**.

Distinction: Buoy, special purpose/general; mooring/warping facility; offshore platform.

5.2 Lateral buoys (LateralBuoy)

Definition: **LATERAL MARKS BUOY.** A buoy is a floating object moored to the bottom in a particular place, as an aid to navigation or for other specific purposes. (IHO Dictionary – S-32).

S-201 Geo Feature: Lateral Buoy

Primitives: Point

Real World	Paper Chart Symbol	ECDIS Symbol		
S-201 Attribute	Allowable Encoding Value	Type	Multiplicity	
ID Code		TE	0, 1	
Information		C	0, *	
File Locator		(S) TE	0, 1	
File Reference		(S) TE	0, 1	
Headline		(S) TE	0, 1	
Language		(S) TE	1, 1	
Text		(S) TE	0, 1	
Feature Name		C	0, *	
Display Name		(S) BO	0, 1	
Language		(S) TE	0, 1	
Name		(S) TE	1, 1	
Scale Minimum		IN	0, 1	
Source Date		DA	0, 1	
Source		TE	0, 1	
Pictorial Representation		TE	0, 1	
Inspection Frequency		TE	0, 1	
Inspection Requirements		TE	0, 1	
AtoN Maintenance Record		TE	0, 1	
Installation Date		DA	0, 1	
Fixed Date Range		C	0, 1	
Date End		(S) TD	0, 1	
Date Start		(S) TD	0, 1	
Periodic Date Range		C	0, 1	
Date End		(S) TD	1, 1	
Date Start		(S) TD	1, 1	
Seasonal Action Required		TE	0, *	
AtoN Number		TE	1,1	
Aid Availability Category	1: Category 1 2: Category 2 3: Category 3	EN	0, 1	

Condition	1: Under Construction 2: Ruined 3: Under Reclamation 4: Wingless 5: Planned Construction	EN	0, 1
Contact Address		C	0, 1
Delivery Point		(S) TE	0, 1
City Name		(S) TE	0, 1
Administrative Division		(S) TE	0, 1
Country Name		(S) TE	0, 1
Postal Code		(S) TE	0, 1
Buoy Shape	1: Conical 2: Can 3: Spherical 4: Pillar 5: Spar 6: Barrel 7: Superbuoy 8: Ice Buoy	EN	1, 1
Colour	1: White 2: Black 3: Red 4: Green 5: Blue 6: Yellow 7: Grey 8: Brown 9: Amber 10: Violet 11: Orange 12: Magenta 13: Pink	EN	1, *
Colour Pattern	1: Horizontal Stripes 2: Vertical Stripes 3: Diagonal Stripes 4: Squared 5: Stripes (Direction Unknown) 6: Border Stripe 7: Single Colour 8: Rectangle 9: Triangle	EN	0, *
Marks Navigational - System Of	1: IALA A 2: IALA B 9: No System 10: Other System 11: CEVNI 12: Russian Inland Waterway Regulations 13: Brazilian National Inland Waterway Regulations - Two Sides 15: Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	0, 1
Nature of Construction	1: Masonry 2: Concreted 3: Loose Boulders 4: Hard Surfaced 5: Unsurfaced 6: Wooden 7: Metal 8: Glass Reinforced Plastic	EN	0, *

	9: Painted 10: Framework 11: Latticed 12: Glass 13: Fiberglass 14: Plastic		
Radar Conspicuous		BO	0, 1
Category of lateral mark	1 : Port-Hand Lateral Mark 2 : Starboard-Hand Lateral Mark 3 : Preferred Channel to Starboard Lateral Mark 4 : Preferred Channel to Port Lateral Mark	EN	1,1

5.2.1 Lateral buoys

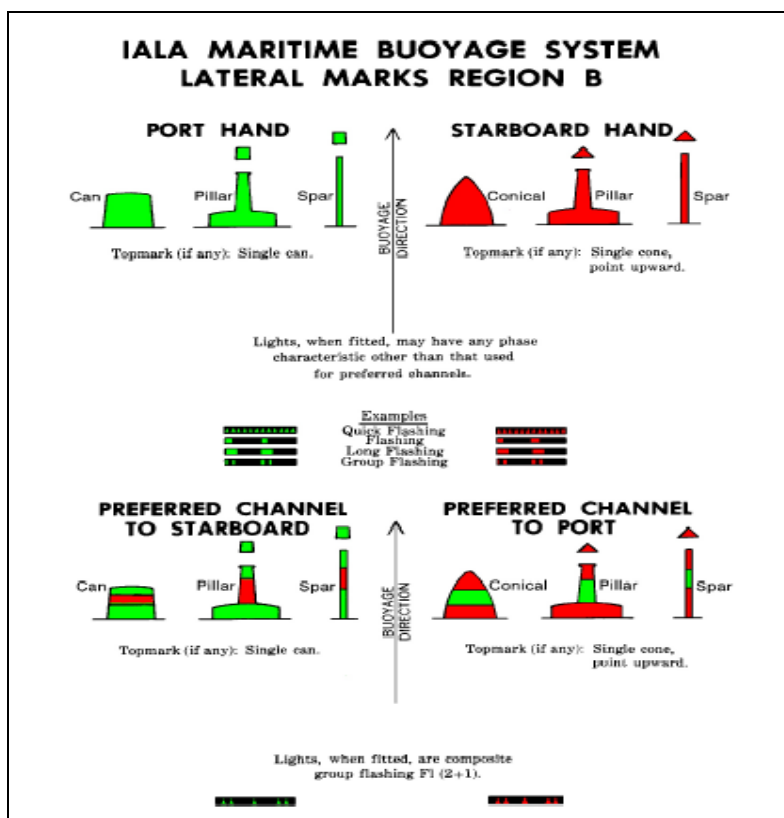
Lateral buoys are generally used for well defined channels, in conjunction with a direction of buoyage. They indicate the port and starboard sides of the route to be followed.

To conform to the IALA Maritime Buoyage System, port hand buoys are usually can shaped, but may be another shape (except conical or spherical). Other shaped buoys have a can topmark. The colour of port hand buoys, topmarks and lights (if fitted) will be red in IALA region A and green in IALA region B.

To conform to the IALA Maritime Buoyage System, starboard hand buoys are usually conical shaped, but may be another shape (except can or spherical). Other shaped buoys have a conical topmark. The colour of starboard hand buoys, topmarks and lights (if fitted) will be green in IALA region A and red in IALA region B.

A preferred channel mark is a modified lateral mark, with horizontal colour bands. The shape and predominant colour indicates which side is the preferred channel, the other colour indicates the secondary channel. If fitted, the light is Fl(2+1), the colour indicating the preferred channel.

If it is required to encode a buoy having the function of a lateral mark, it must be done using the feature **Lateral Buoy**.



Remarks:

- The attribute **colour pattern** must be populated if more than one value is populated for the mandatory attribute **colour**.

Distinction: Buoy, cardinal; buoy, emergency wreck marking; buoy, installation; buoy, isolated danger; buoy, safe water; buoy, special purpose/general; mooring/warping facility.

5.3 Cardinal buoy (CardinalBuoy)

Definition: **CARDINAL BUOY**. A buoy is a floating object moored to the bottom in a particular place, as an aid to navigation or for other specific purposes. (IHO Dictionary – S-32).

S-201 Geo Feature: Cardinal Buoy

Primitives: Point

Real World	Paper Chart Symbol	ECDIS Symbol		
S-201 Attribute		Allowable Encoding Value	Type	Multiplicity
ID Code			TE	0, 1
Information			C	0, *
File Locator			(S) TE	0, 1
File Reference			(S) TE	0, 1
Headline			(S) TE	0, 1
Language			(S) TE	1, 1
Text			(S) TE	0, 1
Feature Name			C	0, *
Display Name			(S) BO	0, 1
Language			(S) TE	0, 1
Name			(S) TE	1, 1
Scale Minimum			IN	0, 1
Source Date			DA	0, 1
Source			TE	0, 1
Pictorial Representation			TE	0, 1
Inspection Frequency			TE	0, 1
Inspection Requirements			TE	0, 1
AtoN Maintenance Record			TE	0, 1
Installation Date			DA	0, 1
Fixed Date Range			C	0, 1
Date End			(S) TD	0, 1
Date Start			(S) TD	0, 1
Periodic Date Range			C	0, 1
Date End			(S) TD	1, 1
Date Start			(S) TD	1, 1
Seasonal Action Required			TE	0, *
AtoN Number			TE	1,1
Aid Availability Category		1: Category 1 2: Category 2 3: Category 3	EN	0, 1

Condition	1: Under Construction 2: Ruined 3: Under Reclamation 4: Wingless 5: Planned Construction	EN	0, 1
Contact Address		C	0, 1
Delivery Point		(S) TE	0, 1
City Name		(S) TE	0, 1
Administrative Division		(S) TE	0, 1
Country Name		(S) TE	0, 1
Postal Code		(S) TE	0, 1
Buoy Shape	1: Conical 2: Can 3: Spherical 4: Pillar 5: Spar 6: Barrel 7: Superbuoy 8: Ice Buoy	EN	1, 1
Colour	1: White 2: Black 3: Red 4: Green 5: Blue 6: Yellow 7: Grey 8: Brown 9: Amber 10: Violet 11: Orange 12: Magenta 13: Pink	EN	1, *
Colour Pattern	1: Horizontal Stripes 2: Vertical Stripes 3: Diagonal Stripes 4: Squared 5: Stripes (Direction Unknown) 6: Border Stripe 7: Single Colour 8: Rectangle 9: Triangle	EN	0, *
Marks Navigational - System Of	1: IALA A 2: IALA B 9: No System 10: Other System 11: CEVNI 12: Russian Inland Waterway Regulations 13: Brazilian National Inland Waterway Regulations - Two Sides 15: Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	0, 1
Nature of Construction	1: Masonry 2: Concreted 3: Loose Boulders 4: Hard Surfaced 5: Unsurfaced 6: Wooden 7: Metal 8: Glass Reinforced Plastic	EN	0, *

	9: Painted 10: Framework 11: Latticed 12: Glass 13: Fiberglass 14: Plastic		
Radar Conspicuous		BO	0, 1
Category of Cardinal Mark	1 : North Cardinal Mark 2 : East Cardinal Mark 3 : South Cardinal Mark 4 : West Cardinal Mark	EN	1,1

5.3.1 Cardinal buoys

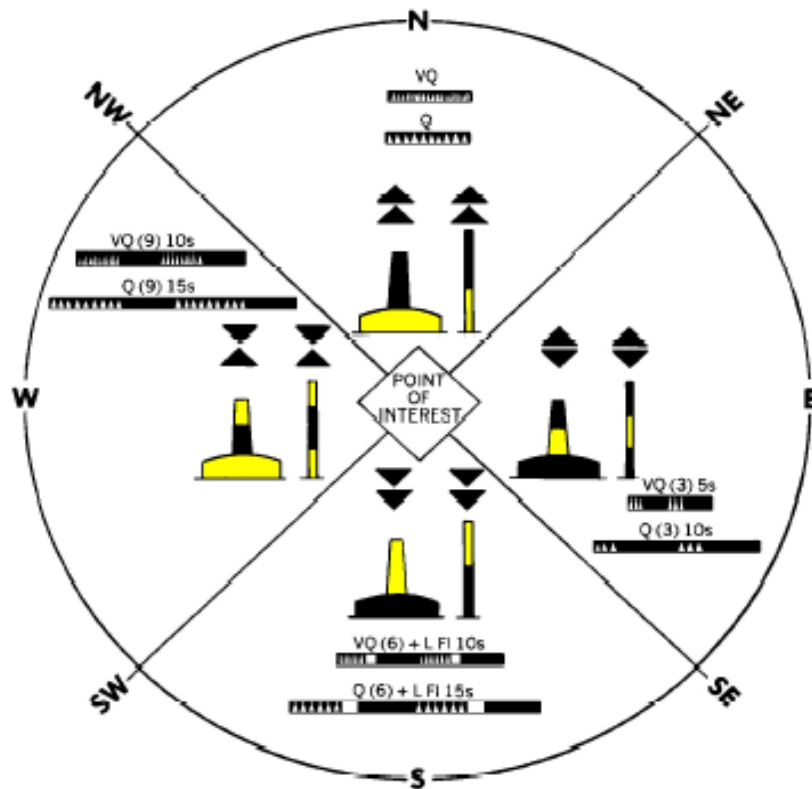
Cardinal marks are used in conjunction with the compass to indicate where a mariner may find best navigable water, taking their name from the quadrant in which they are placed in relation to the point marked. The mariner should pass N of a North mark, E of an East mark, etc. The shape of cardinal buoys is not significant (although they are usually pillar or spar).

To conform to the IALA Maritime Buoyage System, the body has black and yellow bands, configured with black reflecting the points of the topmark cones (e.g. black above yellow for north). Black double-cone topmarks are an important feature of cardinal marks and are carried wherever practicable. The points are up for a north mark, down for a south mark, apart for an east mark and together for a west mark. Lights (if fitted) are white Q or VQ, uninterrupted for the north, 3 flashes for east, 6 flashes + LFI for south and 9 flashes for west (resembling an analogue clock).

If it is required to encode a buoy having the function of a cardinal mark, it must be done using the feature **Cardinal Buoy**.

IALA MARITIME BUOYAGE SYSTEM CARDINAL MARKS REGIONS A AND B

Topmarks are always fitted (when practicable),
Buoy shapes are pillar or spar.



Lights, when fitted, are **white**. Very Quick Flashing or Quick Flashing; a South mark also has a Long Flash immediately following the quick flashes.

Remarks:

- The attribute **colour pattern** must be populated if more than one value is populated for the mandatory attribute **colour**.

Distinction: Buoy, emergency wreck marking; buoy, installation; buoy, isolated danger; buoy, lateral; buoy, safe water; buoy, special purpose/general; mooring/warping facility.

5.4 Safe water buoys (SafeWaterBuoy)

Definition: **SAFE WATER BUOY.** A buoy is a floating object moored to the bottom in a particular place, as an aid to navigation or for other specific purposes. (IHO Dictionary – S-32).

S-201 Geo Feature: Safe water Buoy

Primitives: Point

Real World	Paper Chart Symbol	ECDIS Symbol		
S-201 Attribute		Allowable Encoding Value	Type	Multiplicity
ID Code			TE	0, 1
Information			C	0, *
File Locator			(S) TE	0, 1
File Reference			(S) TE	0, 1
Headline			(S) TE	0, 1
Language			(S) TE	1, 1
Text			(S) TE	0, 1
Feature Name			C	0, *
Display Name			(S) BO	0, 1
Language			(S) TE	0, 1
Name			(S) TE	1, 1
Scale Minimum			IN	0, 1
Source Date			DA	0, 1
Source			TE	0, 1
Pictorial Representation			TE	0, 1
Inspection Frequency			TE	0, 1
Inspection Requirements			TE	0, 1
AtoN Maintenance Record			TE	0, 1
Installation Date			DA	0, 1
Fixed Date Range			C	0, 1
Date End			(S) TD	0, 1
Date Start			(S) TD	0, 1
Periodic Date Range			C	0, 1
Date End			(S) TD	1, 1
Date Start			(S) TD	1, 1
Seasonal Action Required			TE	0, *
AtoN Number			TE	1,1
Aid Availability Category		1: Category 1 2: Category 2 3: Category 3	EN	0, 1

Condition	1: Under Construction 2: Ruined 3: Under Reclamation 4: Wingless 5: Planned Construction	EN	0, 1
Contact Address		C	0, 1
Delivery Point		(S) TE	0, 1
City Name		(S) TE	0, 1
Administrative Division		(S) TE	0, 1
Country Name		(S) TE	0, 1
Postal Code		(S) TE	0, 1
Buoy Shape	1: Conical 2: Can 3: Spherical 4: Pillar 5: Spar 6: Barrel 7: Superbuoy 8: Ice Buoy	EN	1, 1
Colour	1: White 2: Black 3: Red 4: Green 5: Blue 6: Yellow 7: Grey 8: Brown 9: Amber 10: Violet 11: Orange 12: Magenta 13: Pink	EN	1, *
Colour Pattern	1: Horizontal Stripes 2: Vertical Stripes 3: Diagonal Stripes 4: Squared 5: Stripes (Direction Unknown) 6: Border Stripe 7: Single Colour 8: Rectangle 9: Triangle	EN	0, *
Marks Navigational - System Of	1: IALA A 2: IALA B 9: No System 10: Other System 11: CEVNI 12: Russian Inland Waterway Regulations 13: Brazilian National Inland Waterway Regulations - Two Sides 15: Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	0, 1
Nature of Construction	1: Masonry 2: Concreted 3: Loose Boulders 4: Hard Surfaced 5: Unsurfaced 6: Wooden 7: Metal 8: Glass Reinforced Plastic	EN	0, *

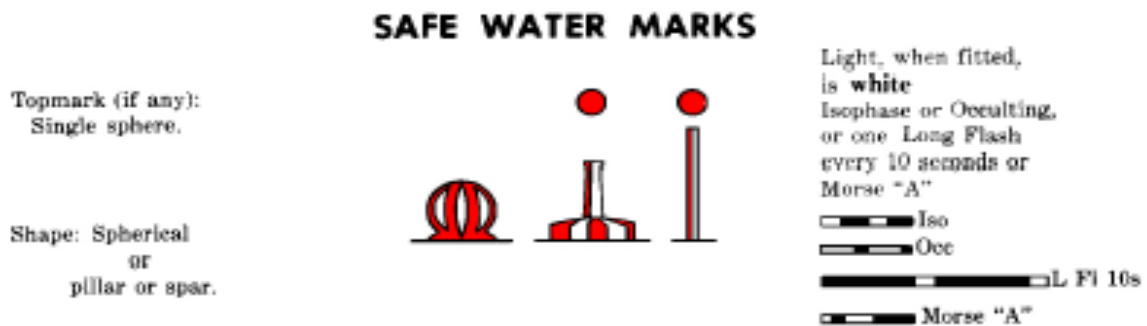
	9: Painted 10: Framework 11: Latticed 12: Glass 13: Fiberglass 14: Plastic		
Radar Conspicuous		BO	0, 1

5.4.1 Safe water buoys

Safe water marks are used to indicate there is safe water all around the mark. It may be used as a centre-line, mid-channel or landfall buoy, or to mark the best point of passage under a bridge.

To conform to the IALA Maritime Buoyage System, the shape of a safe water buoy is spherical, pillar or spar. The body of the mark has red and white vertical stripes. If the shape of the buoy is not spherical a red spherical topmark is carried wherever practicable. The light (if fitted) is white Oc, Iso, LFI or Mo(A) with a period of 10s.

If it is required to encode a buoy having the function of a safe water mark, it must be done using the feature **Safe Water Buoy**.



Remarks:

- The attribute **colour pattern** must be populated if more than one value is populated for the mandatory attribute **colour**.

Distinction: Buoy, cardinal; buoy, emergency wreck marking; buoy, installation; buoy, isolated danger; buoy, lateral; buoy, special purpose/general; mooring/warping facility.

5.5 Isolated danger buoys (IsolatedDangerBuoy)

Definition: **ISOLATED DANGER BUOY**. A buoy is a floating object moored to the bottom in a particular place, as an aid to navigation or for other specific purposes. (IHO Dictionary – S-32).

S-201 Geo Feature: Isolated danger Buoy

Primitives: Point

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
AtoN Number		TE	1, 1
Aid Availability Category	1: Category 1 2: Category 2 3: Category 3	EN	0, 1

Condition	1: Under Construction 2: Ruined 3: Under Reclamation 4: Wingless 5: Planned Construction	EN	0, 1
Contact Address		C	0, 1
Delivery Point		(S) TE	0, 1
City Name		(S) TE	0, 1
Administrative Division		(S) TE	0, 1
Country Name		(S) TE	0, 1
Postal Code		(S) TE	0, 1
Buoy Shape	1: Conical 2: Can 3: Spherical 4: Pillar 5: Spar 6: Barrel 7: Superbuoy 8: Ice Buoy	EN	1, 1
Colour	1: White 2: Black 3: Red 4: Green 5: Blue 6: Yellow 7: Grey 8: Brown 9: Amber 10: Violet 11: Orange 12: Magenta 13: Pink	EN	1, *
Colour Pattern	1: Horizontal Stripes 2: Vertical Stripes 3: Diagonal Stripes 4: Squared 5: Stripes (Direction Unknown) 6: Border Stripe 7: Single Colour 8: Rectangle 9: Triangle	EN	0, *
Marks Navigational - System Of	1: IALA A 2: IALA B 9: No System 10: Other System 11: CEVNI 12: Russian Inland Waterway Regulations 13: Brazilian National Inland Waterway Regulations - Two Sides 15: Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	0, 1
Nature of Construction	1: Masonry 2: Concreted 3: Loose Boulders 4: Hard Surfaced 5: Unsurfaced 6: Wooden 7: Metal 8: Glass Reinforced Plastic	EN	0, *

	9: Painted 10: Framework 11: Latticed 12: Glass 13: Fiberglass 14: Plastic		
Radar Conspicuous		BO	0, 1

5.5.1 Isolated danger buoys

Isolated danger buoys are moored above isolated dangers of limited extent with navigable water all around them.

The shape of isolated danger buoys is not significant (although they are usually pillar or spar shaped). To conform to the IALA Maritime Buoyage System, the body is black, with one or more red bands. Black double-sphere topmarks are an important feature of isolated danger buoys and carried wherever practicable. The light (if fitted) is white Fl(2).

If it is required to encode a buoy having the function of an isolated danger mark, it must be done using the feature **Isolated Danger Buoy**.

IALA MARITIME BUOYAGE SYSTEM REGIONS A AND B ISOLATED DANGER MARKS

Topmarks are
always fitted
(when practicable).

Shape: Optional, but not
conflicting with lateral
marks; pillar or spar
preferred.



Light, when fitted, is
white
Group Flashing(2)

 Fl (2)

Remarks:

- The attribute **colour pattern** must be populated if more than one value is populated for the mandatory attribute **colour**.

Distinction: Buoy, cardinal; buoy, emergency wreck marking; buoy, installation; buoy, lateral; buoy, safe water; buoy, special purpose/general; mooring/warping facility.

5.6 Special purpose/general buoys (SpecialPurposeGeneralBuoy)

Definition: **SPECIAL PURPOSE/GENERAL BUOY.** A buoy is a floating object moored to the bottom in a particular place, as an aid to navigation or for other specific purposes. (IHO Dictionary – S-32).

Buoy in general: A buoy whose appearance or purpose is not adequately known. (S-57 Edition 3.1, Appendix A – Chapter 1, Page 1.24, November 2000).

S-201 Geo Feature: Special purpose/general Buoy

Primitives: Point

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
AtoN Number		TE	1, 1

Aid Availability Category	1: Category 1 2: Category 2 3: Category 3	EN	0, 1
Condition	1: Under Construction 2: Ruined 3: Under Reclamation 4: Wingless 5: Planned Construction	EN	0, 1
Contact Address		C	0, 1
Delivery Point		(S) TE	0, 1
City Name		(S) TE	0, 1
Administrative Division		(S) TE	0, 1
Country Name		(S) TE	0, 1
Postal Code		(S) TE	0, 1
Buoy Shape	1: Conical 2: Can 3: Spherical 4: Pillar 5: Spar 6: Barrel 7: Superbuoy 8: Ice Buoy	EN	1, 1
Colour	1: White 2: Black 3: Red 4: Green 5: Blue 6: Yellow 7: Grey 8: Brown 9: Amber 10: Violet 11: Orange 12: Magenta 13: Pink	EN	1, *
Colour Pattern	1: Horizontal Stripes 2: Vertical Stripes 3: Diagonal Stripes 4: Squared 5: Stripes (Direction Unknown) 6: Border Stripe 7: Single Colour 8: Rectangle 9: Triangle	EN	0, *
Marks Navigational - System Of	1: IALA A 2: IALA B 9: No System 10: Other System 11: CEVNI 12: Russian Inland Waterway Regulations 13: Brazilian National Inland Waterway Regulations - Two Sides 15: Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	0, 1
Nature of Construction	1: Masonry 2: Concreted 3: Loose Boulders 4: Hard Surfaced 5: Unsurfaced	EN	0, *

	6: Wooden 7: Metal 8: Glass Reinforced Plastic 9: Painted 10: Framework 11: Latticed 12: Glass 13: Fiberglass 14: Plastic		
Radar Conspicuous		BO	0, 1
Category of special purpose mark	1 : firing danger area mark 2 : target mark 3 : marker ship mark 4 : degaussing range mark 5 : barge mark 6 : cable mark 7 : spoil ground mark 8 : outfall mark 10 : recording mark 11 : seaplane anchorage mark 12 : recreation zone mark 14 : mooring mark 15 : LANBY (Large Automatic Navigational Buoy) 16 : leading mark 17 : measured distance mark 18 : notice mark 19 : TSS mark (Traffic Separation Scheme) 20 : anchoring prohibited mark 21 : berthing prohibited mark 22 : overtaking prohibited mark 23 : two-way traffic prohibited mark 24 : reduced wake mark 25 : speed limit mark 26 : stop mark 27 : general warning mark 28 : sound ship's siren mark 29 : restricted vertical clearance mark 30 : maximum vessel's draught mark 31 : restricted horizontal clearance mark 32 : strong current warning mark 33 : berthing permitted mark 34 : overhead power cable mark 35 : channel edge gradient' mark 36 : telephone mark 37 : ferry crossing mark 39 : pipeline mark 40 : anchorage mark 41 : clearing mark 42 : control mark 43 : diving mark 44 : refuge beacon 45 : foul ground mark 46 : yachting mark 47 : heliport mark 48 : GNSS mark 49 : seaplane landing mark 50 : control mark 51 : work in progress mark 52 : mark with unknown purpose 53 : wellhead mark 54 : channel separation mark 55 : marine farm mark	EN	1,*

	56 : artificial reef mark 57 : ice mark 58 : nature reserve mark 59 : Fish Aggregating Device 60 : Wreck Mark 61 : Customs Mark 62 : Causeway Mark 63 : Wave Recorder 64 : Jetski Prohibited 65 : Facility Protection Mark 66 : Oil Pipeline Protection Mark 67 : Marine Cable Protection Mark		
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5.6.1 Special purpose/general buoys

Special marks are used to indicate to the mariner a special area or feature, the nature of which is usually apparent from the ENC, paper chart or associated publication. Special marks may also be used to mark a channel within a channel (e.g. a Deep Water route), using yellow buoys of the appropriate lateral shape, or yellow spherical buoys to mark the centreline. A special buoy may be any shape but must not conflict with lateral or safe water marks (e.g. an outfall buoy on the port-side of a channel could be can but should not be conical).

To conform to the IALA Maritime Buoyage System, the body of the buoy is yellow. The topmark (if fitted) is a yellow diagonal 'X' (St Andrew's cross). Lights (if fitted) are yellow and of any rhythm except those used for cardinal, isolated danger and safe water marks.

If it is required to encode a buoy having the function of a special purpose mark, or a buoy whose appearance or purpose is inadequately known, it must be done using the feature **Special Purpose/General Buoy**.

In the following table, a blank indicates that the encoder may choose a relevant value for the attribute. The table contains the most common examples of coding; other coding combinations are possible for **Special Purpose/General Buoy** features.

Feature	INT1	Feature	buoy shape	category of special purpose mark	Other attributes
Firing danger area buoy	Q50	Special Purpose/General Buoy		1	
Target	Q51	Special Purpose/General Buoy		2	
Marker ship	Q52	Special Purpose/General Buoy		3	
Barge	Q53	Special Purpose/General Buoy		5	
Degaussing range buoy	Q54	Special Purpose/General Buoy		4	
Buoy marking cable	Q55	Special Purpose/General Buoy		6	
Spoil ground buoy	Q56	Special Purpose/General Buoy		7	
Buoy marking outfall	Q57	Special Purpose/General Buoy		8	
Buoy marking pipeline		Special Purpose/General Buoy		39	
Superbuoy	Q26	Buoy ***	7		

Large automatic navigational buoy	P6	Special Purpose/ General Buoy	7	15	
Ocean data acquisition system (ODAS) buoy	Q58	Special Purpose/ General Buoy	7	9	Subsurface ODAS encoded as Obstruction
Buoy marking wave recorder (or current meter)	Q59	Special Purpose/ General Buoy		10	information = wave recorder (e.g.)
Seaplane anchorage buoy	Q60	Special Purpose/ General Buoy		11	
Buoy marking traffic separation scheme	Q61	Special Purpose/ General Buoy		19	
Buoy marking recreation zone	Q62	Special Purpose/ General Buoy		12	
Floating waste bin		Special Purpose/ General Buoy		<i>Empty (null) value</i>	information = waste bin (e.g.)
Floating Fish Aggregating Device (FAD)		Special Purpose/ General Buoy	/	59	Subsurface FAD encoded as Obstruction

Remarks:

- The attribute **colour pattern** must be populated if more than one value is populated for the mandatory attribute **colour**.

Distinction: Buoy, cardinal; buoy, emergency wreck marking; buoy, installation; buoy, isolated danger; buoy, lateral; buoy, safe water; mooring/warping facility.

5.7 Buoy emergency wreck marking (EmergencyWreckMarkingBuoy)

Definition: **EMERGENCY WRECK MARKING BUOY**, An emergency wreck marking buoy is a buoy moored on or above a new wreck, designed to provide a prominent (both visual and radio) and easily identifiable temporary (24-72 hours) first response. (IHO Registry)

S-201 Geo Feature: Emergency wreck marking Buoy

Primitives: Point

<i>Real World</i>	<i>Paper Chart Symbol</i>	<i>ECDIS Symbol</i>
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S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
AtoN Number		TE	1,1
Aid Availability Category	1: Category 1 2: Category 2	EN	0, 1

	3: Category 3		
Condition	1: Under Construction 2: Ruined 3: Under Reclamation 4: Wingless 5: Planned Construction	EN	0, 1
Contact Address		C	0, 1
Delivery Point		(S) TE	0, 1
City Name		(S) TE	0, 1
Administrative Division		(S) TE	0, 1
Country Name		(S) TE	0, 1
Postal Code		(S) TE	0, 1
Buoy Shape	1: Conical 2: Can 3: Spherical 4: Pillar 5: Spar 6: Barrel 7: Superbuoy 8: Ice Buoy	EN	1, 1
Colour	1: White 2: Black 3: Red 4: Green 5: Blue 6: Yellow 7: Grey 8: Brown 9: Amber 10: Violet 11: Orange 12: Magenta 13: Pink	EN	1, *
Colour Pattern	1: Horizontal Stripes 2: Vertical Stripes 3: Diagonal Stripes 4: Squared 5: Stripes (Direction Unknown) 6: Border Stripe 7: Single Colour 8: Rectangle 9: Triangle	EN	0, *
Marks Navigational - System Of	1: IALA A 2: IALA B 9: No System 10: Other System 11: CEVNI 12: Russian Inland Waterway Regulations 13: Brazilian National Inland Waterway Regulations - Two Sides 15: Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	0, 1
Nature of Construction	1: Masonry 2: Concreted 3: Loose Boulders 4: Hard Surfaced 5: Unsurfaced 6: Wooden 7: Metal	EN	0, *

	8: Glass Reinforced Plastic 9: Painted 10: Framework 11: Latticed 12: Glass 13: Fiberglass 14: Plastic		
Radar Conspicuous		BO	0, 1

5.7.1 Emergency wreck marking buoys

Emergency wreck marking buoys are used to mark new wrecks until a permanent form of marking has been established and the wreck itself has been promulgated by Notice to Mariners, or removed.

To conform to the IALA Maritime Buoyage System, the shape of an emergency wreck marking buoy is pillar or spar. The body of the mark has blue and yellow vertical stripes. The topmark (if fitted) is a standing/upright yellow '+' (St. George's cross). Lights (if fitted) are Al.Oc.BuY.3s.

If it is required to encode a buoy having the function of an emergency wreck mark, it must be done using the feature Emergency Wreck Marking Buoy.

Remarks:

- If it is required to encode a buoy or topmark that has more than one colour, the attributes colour and colour pattern must be encoded.
- If it is required to encode the total vertical length, including the topmark and any equipment features (for example light), of the buoy above the water level, it must be done using the attribute vertical length.
- An IALA compliant emergency wreck marking buoy topmark should be populated using the complex attribute topmark, with sub-attributes topmark shape = 8 (upright cross) and colour = 6 (yellow).
- An IALA compliant emergency wreck marking buoy should also have the following associated equipment features:
 - A Light All Around feature, with attributes colour = 5,6 (blue, yellow), light characteristic = 17 (occulting alternating), signal group = (1) and signal period = 3. The complex attribute signal sequence, sub-attributes signal duration and signal status, should be populated equivalent to (00.50)+01.00+(00.50)+01.00 (bracketed values corresponding to periods of eclipse and non-bracketed values corresponding to periods of light); and the attribute value of nominal range should be populated as 4.
 - A Radar Transponder Beacon feature, with attributes category of radar transponder beacon = 2 (racon, radar transponder beacon) and signal group = (D).

Distinction: Cardinal Buoy; Installation Buoy; Lateral Buoy; Mooring Buoy; Safe Water Buoy; Special Purpose/General Buoy.

6 AtoN relevant structures

6.1 Offshore platform (OffshorePlatform)

Definition: **OFFSHORE PLATFORM.** A permanent offshore structure, either fixed or floating. (Adapted from IHO Dictionary – S-32).

S-201 Geo Feature: Offshore platform

Primitives: Point

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
AtoN Number		TE	1, 1

Aid Availability Category	1:Category 1 2:Category 2 3:Category 3	EN	0, 1
Condition	1:Under Construction 2:Ruined 3:Under Reclamation 4:Wingless 5:Planned Construction	EN	0, 1
Contact Address		C	0, 1
Delivery Point		(S) TE	0, 1
City Name		(S) TE	0, 1
Administrative Division		(S) TE	0, 1
Country Name		(S) TE	0, 1
Postal Code		(S) TE	0, 1
Category of Offshore Platform	1: Oil Rig 2: Production Platform 3: Observation/Research Platform 4: Articulated Loading Platform 5: Single Anchor Leg Mooring 6: Mooring Tower 7: Artificial Island 8: Floating Production, Storage and Off-Loading Vessel 9: Accommodation Platform 10: Navigation, Communication and Control Buoy 11: Floating Oil Tank	EN	0, *
Colour	1: White 2: Black 3: Red 4: Green 5: Blue 6: Yellow 7: Grey 8: Brown 9: Amber 10: Violet 11: Orange 12: Magenta 13: Pink	EN	0, *
Colour Pattern	1: Horizontal Stripes 2: Vertical Stripes 3: Diagonal Stripes 4: Squared 5: Stripes (Direction Unknown) 6: Border Stripe 7: Single Colour 8: Rectangle 9: Triangle	EN	0, *

Height		RE	0, 1
Manned Structure		BO	0, 1
Nature of Construction	1: Masonry 2: Concreted 3: Loose Boulders 4: Hard Surfaced 5: Unsurfaced 6: Wooden 7: Metal 8: Glass Reinforced Plastic 9: Painted 10: Framework 11: Latticed 12: Glass 13: Fiberglass 14: Plastic	EN	0, *

6.1.1 Offshore platforms

Several different types of platforms are in use. They are normally piled steel or concrete structures, the latter held in position on the sea floor by gravity. Tension Leg Platforms (TLP) consist of semi-submersible platforms secured to flooded caissons on the sea floor vertically below them by wires kept in tension by the buoyancy of the platform.

Platforms may serve a number of purposes. They may carry any of the following equipment: drilling and production equipment; oil and gas separation and treatment plants; pump-line stations; and electricity generators. They may be fitted with cranes, a helicopter landing deck, and accommodation for up to 350 people. Platforms may stand singly or in groups connected by pipelines. Some stand close together in a complex, with bridges and underwater cables connecting them. Unwanted gas or oil is sometimes burnt from a flaring boom extending from the platform or from a nearby flare stack.

If it is required to encode a permanent offshore platform, it must be done using the feature **Offshore Platform**.

Remarks:

- The attribute **height** is only relevant for fixed platforms, and is referred to the vertical datum.
- Platforms may carry lights, fog signals and helicopter platforms. Where fitted, lights should be encoded as described, with the **Offshore Platform** being used as the structure feature for the **Light** equipment feature(s).

6.2 Light vessels (LightVessel)

Definition: **LIGHT VESSEL.** A distinctively marked vessel anchored or moored at a charted point, to serve as an aid to navigation. By night, it displays a characteristic light(s) and is usually equipped with other devices, such as fog signal, submarine sound signal, and radio-beacon, to assist navigation. Also called light ship. (IHO Dictionary – S-32).

S-201 Geo Feature: Light vessel

Primitives: Point

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
AtoN Number		TE	1,1

Aid Availability Category	1:Category 1 2:Category 2 3:Category 3	EN	0, 1
Condition	1:Under Construction 2:Ruined 3:Under Reclamation 4:Wingless 5:Planned Construction	EN	0, 1
Contact Address		C	0, 1
Delivery Point		(S) TE	0, 1
City Name		(S) TE	0, 1
Administrative Division		(S) TE	0, 1
Country Name		(S) TE	0, 1
Postal Code		(S) TE	0, 1
Colour	1: White 2: Black 3: Red 4: Green 5: Blue 6: Yellow 7: Grey 8: Brown 9: Amber 10: Violet 11: Orange 12: Magenta 13: Pink	EN	1, *
Colour Pattern	1: Horizontal Stripes 2: Vertical Stripes 3: Diagonal Stripes 4: Squared 5: Stripes (Direction Unknown) 6: Border Stripe 7: Single Colour 8: Rectangle 9: Triangle	EN	0, *
Horizontal Length		RE	0, 1
Horizontal Width		RE	0, 1
Manned Structure		BO	0, 1
Nature of Construction	1: Masonry 2: Concreted 3: Loose Boulders 4: Hard Surfaced 5: Unsurfaced 6: Wooden 7: Metal 8: Glass Reinforced Plastic	EN	0, *

	9: Painted 10: Framework 11: Latticed 12: Glass 13: Fiberglass 14: Plastic		
Radar Conspicuous		BO	0, 1
6.2.1 Lights vessels Major floating lights are generally classed as those with a nominal range in excess of 10 nautical miles. Special circumstances, e.g. an isolated location, may mean that a floating light of lower range is given this status. The structure on which the light is fixed will be a light vessel, a major light float or a LANBY (Large Automatic Navigational Buoy, which is a type of superbuoy). If it is required to encode a light vessel, it must be done using the feature Light Vessel. <u>Remarks:</u> ● The attribute colour pattern must be populated if more than one value is populated for the mandatory attribute colour. <u>Distinction:</u> Beacon, cardinal; beacon, isolated danger; beacon, lateral; beacon, safe water; beacon special purpose/general; buoy, cardinal; buoy, emergency wreck marking; buoy, installation; buoy, isolated danger; buoy, lateral; buoy, safe water; buoy, special purpose/general; light float.			

6.3 Piles (Pile)

Definition: **PILE.** A long heavy timber or section of steel, wood, concrete, etc., forced into the earth or seabed which may serve as a support, as for a pier, or a free standing pole within a marine environment. (IHO Dictionary – S-32).

S-201 Geo Feature: Pile

Primitives: Point

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
AtoN Number		TE	1, 1
Aid Availability Category	1:Category 1	EN	0, 1

	2:Category 2 3:Category 3		
Condition	1:Under Construction 2:Ruined 3:Under Reclamation 4:Wingless 5:Planned Construction	EN	0, 1
Contact Address		C	0, 1
Delivery Point		(S) TE	0, 1
City Name		(S) TE	0, 1
Administrative Division		(S) TE	0, 1
Country Name		(S) TE	0, 1
Postal Code		(S) TE	0, 1
Category of Pile	1: Stake 3: Post 4: Tripodal 5: Piling 6: Area of Piles 7: Pipe	EN	0, 1
Colour	1: White 2: Black 3: Red 4: Green 5: Blue 6: Yellow 7: Grey 8: Brown 9: Amber 10: Violet 11: Orange 12: Magenta 13: Pink	EN	0, *
Colour Pattern	1: Horizontal Stripes 2: Vertical Stripes 3: Diagonal Stripes 4: Squared 5: Stripes (Direction Unknown) 6: Border Stripe 7: Single Colour 8: Rectangle 9: Triangle	EN	0, *
Height		RE	0, 1

6.3.1 Piles

If it is required to encode a pile or post that is not used as a mooring/warping facility or an aid to navigation, it must be done using the feature **Pile**.

Remarks:

- Stumps of piles or posts that are dangerous to navigation must be encoded, where required, using

Obstruction features, with attribute **category of obstruction** = 1 (snag/stump), and must not be encoded using **Pile**.

- The value **category of pile** = 5 (piling), sometimes termed “row of piles” or “sheet piling”, should only be used for **Pile** of type curve. Point primitive may be used for smaller maximum display scale ENC data.
- Stakes and posts that are identified on the source to serve the purpose of aids to navigation must be encoded, where required, using the appropriate beacon feature (e.g. **Special Purpose/General Beacon**), with attribute **beacon shape** = 1 (stake, pole, perch, post).

Distinction: Beacon, cardinal; beacon, isolated danger; beacon, lateral; beacon, safe water; beacon special purpose/general; mooring/warping facility.

6.4 Silo tank (SiloTank)

Definition: **SILO/TANK.** A silo is a large storage structure used for storing loose materials. (Defence Geospatial Information Working Group; Feature Data Dictionary Register, 2012).

A tank is a container used for the storage of liquids and/or gases. (Adapted from Defence Geospatial Information Working Group; Feature Data Dictionary Register, 2012).

S-201 Geo Feature: Silo/tank

Primitives: Point

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
AtoN Number		TE	1, 1

Aid Availability Category	1:Category 1 2:Category 2 3:Category 3	EN	0, 1
Condition	1:Under Construction 2:Ruined 3:Under Reclamation 4:Wingless 5:Planned Construction	EN	0, 1
Contact Address		C	0, 1
Delivery Point		(S) TE	0, 1
City Name		(S) TE	0, 1
Administrative Division		(S) TE	0, 1
Country Name		(S) TE	0, 1
Postal Code		(S) TE	0, 1
Building Shape	5: High-Rise Building 6: Pyramid 7: Cylindrical 8: Spherical 9: Cubic	EN	0, 1
Category of Silo/Tank	1: Silo in General 2: Tank in General 3: Grain Elevator 4: Water Tower	EN	0, 1
Colour	1: White 2: Black 3: Red 4: Green 5: Blue 6: Yellow 7: Grey 8: Brown 9: Amber 10: Violet 11: Orange 12: Magenta 13: Pink	EN	0, *
Colour Pattern	1: Horizontal Stripes 2: Vertical Stripes 3: Diagonal Stripes 4: Squared 5: Stripes (Direction Unknown) 6: Border Stripe 7: Single Colour 8: Rectangle 9: Triangle	EN	0, *
Elevation		RE	0, 1
Height		RE	0, 1
Nature of Construction	1: Masonry	EN	0, *

	2: Concreted 3: Loose Boulders 4: Hard Surfaced 5: Unsurfaced 6: Wooden 7: Metal 8: Glass Reinforced Plastic 9: Painted 10: Framework 11: Latticed 12: Glass 13: Fiberglass 14: Plastic		
Radar Conspicuous		BO	0, 1
Status	1: Permanent 2: Occasional 3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 8: Private 9: Mandatory 11: Extinguished 12: Illuminated 13: Historic 14: Public 15: Synchronized 16: Watched 17: Unwatched 18: Existence Doubtful 19: On Request 20: Drop Away 21: Rising 22: Increasing 23: Decreasing 24: Strong 25: Good 26: Moderately 27: Poor 28: Buoyed 29: Fully Operational 30: Partially Operational 31: Drifting 32: Broken 33: Offline 34: Discontinued 35: Manual Observation 36: Unknown Status 37: Confirmed 38: Candidate 39: Under Modification 40: Experimental 41: Under Removal / Deletion 42: Removed / Deleted 43: Candidate for Modification	EN	0, *
Vertical Datum	1: Mean Low Water Springs 2: Mean Lower Low Water Springs 3: Mean Sea Level 4: Lowest Low Water 5: Mean Low Water	EN	0, 1

	6: Lowest Low Water Springs 7: Approximate Mean Low Water Springs 8: Indian Spring Low Water 9: Low Water Springs 10: Approximate Lowest Astronomical Tide 11: Nearly Lowest Low Water 12: Mean Lower Low Water 13: Low Water 14: Approximate Mean Low Water 15: Approximate Mean Lower Low Water 16: Mean High Water 17: Mean High Water Springs 18: High Water 19: Approximate Mean Sea Level 20: High Water Springs 21: Mean Higher High Water 22: Equinoctial Spring Low Water 23: Lowest Astronomical Tide 24: Local Datum 25: International Great Lakes Datum 1985 26: Mean Water Level 27: Lower Low Water Large Tide 28: Higher High Water Large Tide 29: Nearly Highest High Water 30: Highest Astronomical Tide 31: Local Low Water Reference Level 32: Local High Water Reference Level 33: Local Mean Water Reference Level 34: Equivalent Height of Water (German GIW) 35: Highest Shipping Height of Water (German HSW) 36: Reference Low Water Level According to Danube Commission 37: Highest Shipping Height of Water According to Danube Commission 38: Dutch River Low Water Reference Level (OLR) 39: Russian Project Water Level 40: Russian Normal Backwater Level 41: Ohio River Datum 43: Dutch High Water Reference Level 44: Baltic Sea Chart Datum 2000 45: Dutch Estuary Low Water Reference Level (OLW) 46: International Great Lakes Datum 2020 47: Sea Floor 48: Sea Surface 49: Hydrographic Zero		
Vertical Length		RE	0, 1
Visual Prominence	1: Visually Conspicuous 2: Not Visually Conspicuous 3: Prominent	EN	0, 1

Vertical Accuracy		RE	0, 1
6.4.1 Tanks, silos Isolated tanks or gasholders may be good landmarks and should be represented true to scale (i.e. as surface) where possible, to enable them to be used as fixing marks. Groups of tanks, as at a refinery, may be useful for general identification of position but cannot usually be used for precise position-fixing because of uncertainty of the location of individual tanks. If it is required to encode a tank or silo, it must be done using the feature Silo/Tank. <u>Remarks:</u> Groups of tanks in close proximity (tank farm) must be encoded, where required, using the feature Production/Storage Area. Individual, visually conspicuous tanks within a tank farm may be encoded as Silo/Tank within the Production/Storage Area. <u>Distinction:</u> Building, single; control point; landmark; production/storage area.			

6.5 Landmarks (Landmark)

Definition: **LANDMARK.** Any prominent object on land which can be used in determining a location or a direction. (IHO Dictionary – S-32).

S-201 Geo Feature: Landmark

Primitives: Point

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
AtoN Number		TE	1, 1
Aid Availability Category	1:Category 1 2:Category 2	EN	0, 1

	3:Category 3		
Condition	1:Under Construction 2:Ruined 3:Under Reclamation 4:Wingless 5:Planned Construction	EN	0, 1
Contact Address		C	0, 1
Delivery Point		(S) TE	0, 1
City Name		(S) TE	0, 1
Administrative Division		(S) TE	0, 1
Country Name		(S) TE	0, 1
Postal Code		(S) TE	0, 1
Category of Landmark	1: Cairn 2: Cemetery 3: Chimney 4: Dish Aerial 5: Flagstaff 6: Flare Stack 7: Mast 8: Windsock 9: Monument 10: Column/Pillar 11: Memorial Plaque 12: Obelisk 13: Statue 14: Cross 15: Dome 16: Radar Scanner 17: Tower 18: Windmill 19: Windmotor 20: Spire/Minaret 21: Large Rock or Boulder on Land 22: Triangulation Mark 23: Boundary Mark 24: Observation Wheel 25: Torii 26: Bridge 27: Dam	EN	1, *
Colour	1: White 2: Black 3: Red 4: Green 5: Blue 6: Yellow 7: Grey 8: Brown 9: Amber 10: Violet 11: Orange 12: Magenta 13: Pink	EN	0, *

Colour Pattern	1: Horizontal Stripes 2: Vertical Stripes 3: Diagonal Stripes 4: Squared 5: Stripes (Direction Unknown) 6: Border Stripe 7: Single Colour 8: Rectangle 9: Triangle	EN	0, *
Function	2: Harbour-Masters Office 3: Customs Office 4: Health Office 5: Hospital 6: Post Office 7: Hotel 8: Railway Station 9: Police Station 10: Water-Police Station 11: Pilot Office 12: Pilot Lookout 13: Bank Office 14: Headquarters for District Control 15: Transit Shed/Warehouse 16: Factory 17: Power Station 18: Administrative 19: Educational Facility 20: Church 21: Chapel 22: Temple 23: Pagoda 24: Shinto Shrine 25: Buddhist Temple 26: Mosque 27: Marabout 28: Lookout 29: Communication 30: Television 31: Radio 32: Radar 33: Light Support 34: Microwave 35: Cooling 36: Observation 37: Timeball 38: Clock 39: Control 40: Airship Mooring 41: Stadium 42: Bus Station 43: Passenger Terminal Building 44: Sea Rescue Control 45: Observatory	EN	0, *

	46: Ore Crusher 47: Boathouse 48: Pumping Station		
Nature of Construction	1: Masonry 2: Concreted 3: Loose Boulders 4: Hard Surfaced 5: Unsurfaced 6: Wooden 7: Metal 8: Glass Reinforced Plastic 9: Painted 10: Framework 11: Latticed 12: Glass 13: Fiberglass 14: Plastic	EN	0, *
Radar Conspicuous		BO	0, 1
Status	1: Permanent 2: Occasional 3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 8: Private 9: Mandatory 11: Extinguished 12: Illuminated 13: Historic 14: Public 15: Synchronized 16: Watched 17: Unwatched 18: Existence Doubtful 19: On Request 20: Drop Away 21: Rising 22: Increasing 23: Decreasing 24: Strong 25: Good 26: Moderately 27: Poor 28: Buoyed 29: Fully Operational 30: Partially Operational 31: Drifting 32: Broken 33: Offline 34: Discontinued 35: Manual Observation 36: Unknown Status 37: Confirmed 38: Candidate 39: Under Modification 40: Experimental 41: Under Removal / Deletion 42: Removed / Deleted 43: Candidate for Modification	EN	0, *

Vertical Datum	1: Mean Low Water Springs 2: Mean Lower Low Water Springs 3: Mean Sea Level 4: Lowest Low Water 5: Mean Low Water 6: Lowest Low Water Springs 7: Approximate Mean Low Water Springs 8: Indian Spring Low Water 9: Low Water Springs 10: Approximate Lowest Astronomical Tide 11: Nearly Lowest Low Water 12: Mean Lower Low Water 13: Low Water 14: Approximate Mean Low Water 15: Approximate Mean Lower Low Water 16: Mean High Water 17: Mean High Water Springs 18: High Water 19: Approximate Mean Sea Level 20: High Water Springs 21: Mean Higher High Water 22: Equinoctial Spring Low Water 23: Lowest Astronomical Tide 24: Local Datum 25: International Great Lakes Datum 1985 26: Mean Water Level 27: Lower Low Water Large Tide 28: Higher High Water Large Tide 29: Nearly Highest High Water 30: Highest Astronomical Tide 31: Local Low Water Reference Level 32: Local High Water Reference Level 33: Local Mean Water Reference Level 34: Equivalent Height of Water (German GIW) 35: Highest Shipping Height of Water (German HSW) 36: Reference Low Water Level According to Danube Commission 37: Highest Shipping Height of Water According to Danube Commission 38: Dutch River Low Water Reference Level (OLR) 39: Russian Project Water Level 40: Russian Normal Backwater Level 41: Ohio River Datum 43: Dutch High Water Reference Level 44: Baltic Sea Chart Datum 2000 45: Dutch Estuary Low Water Reference Level (OLW) 46: International Great Lakes Datum 2020 47: Sea Floor 48: Sea Surface 49: Hydrographic Zero	EN	0, 1
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Visual Prominence	1: Visually Conspicuous 2: Not Visually Conspicuous 3: Prominent	EN	1, 1
Elevation		RE	0, 1
Height		RE	0, 1
Manned Structure		BO	0, 1
Vertical Length		RE	0, 1
Vertical Accuracy		RE	0, 1

6.5.1 Buildings, landmarks, tanks, silos

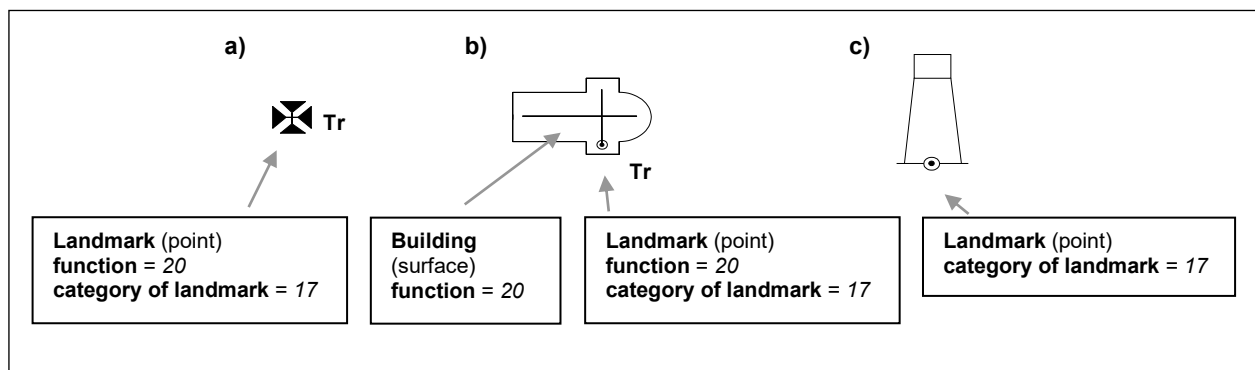
Depending on height and the topographic relief, structures considered to be landmarks should be encoded up to several miles/kilometres inland.

Waterfront, landmark and some public buildings should be encoded precisely and individually on the larger maximum display scale ENC data. When representing buildings generally, forming urban and suburban areas, villages, and other built-up areas, the aim of the compiler must be to create the correct impression of the extent of the built-up area and the density of the buildings.

If it is required to encode a landmark (other than a tank or silo), it must be done using the feature **Landmark**.

Remarks:

- A ruined landmark should be encoded in the same way as the feature in good condition, but with attribute **condition** = 2 (ruined).
- Radio and television masts and towers are likely to be visible over long distances and should be encoded as landmarks, even when well inland. They will usually carry air obstruction lights.
- To aid identification of landmarks by the mariner it may be useful to add the height of the top of the structure above ground level (**vertical length**) or above the general height datum (**height**).
- Buildings constructed as places of worship often form significant landmarks; their size and structure incorporating towers, spires, cupolas, etc often render them conspicuous. These buildings when known to be prominent or conspicuous should be encoded up to several miles inland.
- The attribute **category of special purpose mark** should only be used if the **Landmark** is used as the front or rear lead for a transit, clearing line or measured distance, or for a leading line. Values for **category of special purpose mark** such as 16 (leading mark), 17 (measured distance mark) or 41 (clearing mark) in particular should be used for these purposes.
- When a building is shown as a surface, indicating its true shape, and it is required to encode a prominent feature such as a tower or spire that is part of the structure, two features must be created:
 - a **Building** feature of type surface for the main building,
 - a **Landmark** feature of type point for the prominent feature.



- Not all landmarks are visually conspicuous. If a feature is visually conspicuous (i.e. it is distinctly and noticeably visible from seaward), the attribute **visually conspicuous** must be encoded.

Distinction: Beacon, special purpose/general; building single; control point; daymark; pylon/bridge support; silo/tank ; topmark.

6.6 Light floats (LightFloat)

Definition: **LIGHT FLOAT.** A boat-like structure used instead of a light buoy in waters where strong streams or currents are experienced, or when a greater elevation than that of a light buoy is necessary. (IHO Dictionary – S-32).

S-201 Geo Feature: Light float

Primitives: Point

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
AtoN Number		TE	1, 1
Aid Availability Category	1:Category 1	EN	0, 1

	2:Category 2 3:Category 3		
Condition	1:Under Construction 2:Ruined 3:Under Reclamation 4:Wingless 5:Planned Construction	EN	0, 1
Contact Address		C	0, 1
Delivery Point		(S) TE	0, 1
City Name		(S) TE	0, 1
Administrative Division		(S) TE	0, 1
Country Name		(S) TE	0, 1
Postal Code		(S) TE	0, 1
Colour	1: White 2: Black 3: Red 4: Green 5: Blue 6: Yellow 7: Grey 8: Brown 9: Amber 10: Violet 11: Orange 12: Magenta 13: Pink	EN	1, *
Colour Pattern	1: Horizontal Stripes 2: Vertical Stripes 3: Diagonal Stripes 4: Squared 5: Stripes (Direction Unknown) 6: Border Stripe 7: Single Colour 8: Rectangle	EN	0, *
Horizontal Length		RE	0, 1
Horizontal Width		RE	0, 1
Manned Structure		BO	0, 1
Nature of Construction	1: Masonry 2: Concreted 3: Loose Boulders 4: Hard Surfaced 5: Unsurfaced 6: Wooden 7: Metal 8: Glass Reinforced Plastic 9: Painted 10: Framework 11: Latticed	EN	0, *

	12: Glass 13: Fiberglass 14: Plastic		
Radar Conspicuous		BO	0, 1
Status	1: Permanent 2: Occasional 3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 8: Private 9: Mandatory 11: Extinguished 12: Illuminated 13: Historic 14: Public 15: Synchronized 16: Watched 17: Unwatched 18: Existence Doubtful 19: On Request 20: Drop Away 21: Rising 22: Increasing 23: Decreasing 24: Strong 25: Good 26: Moderately 27: Poor 28: Buoyed 29: Fully Operational 30: Partially Operational 31: Drifting 32: Broken 33: Offline 34: Discontinued 35: Manual Observation 36: Unknown Status 37: Confirmed 38: Candidate 39: Under Modification 40: Experimental 41: Under Removal / Deletion 42: Removed / Deleted 43: Candidate for Modification	EN	0, *
Vertical Length		RE	0, 1
Visual Prominence	1: Visually Conspicuous 2: Not Visually Conspicuous 3: Prominent	EN	0, 1
Vertical Accuracy		RE	0, 1

Horizontal Accuracy		RE	0, 1
6.6.1 Lights floats If it is required to encode a light float, it must be done using the feature Light Float. <u>Remarks:</u> • The light on a light float is a separate feature, handled as with buoys, beacons, etc. <u>Distinction:</u> Buoy, cardinal; buoy, emergency wreck marking; buoy, installation; buoy, isolated danger; buoy, lateral; buoy, safe water; buoy, special purpose/general; light vessel.			

6.7 Building (Building)

Definition: **Building.** : A free-standing self-supporting construction that is roofed, usually walled, and is intended for human occupancy (for example: a place of work or recreation) and/or habitation.

S-201 Geo Feature: Building

Primitives: Point

<i>Real World</i>	<i>Paper Chart Symbol</i>	<i>ECDIS Symbol</i>
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S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
AtoN Number		TE	1, 1
Aid Availability Category	1:Category 1 2:Category 2 3:Category 3	EN	0, 1

Condition	1:Under Construction 2:Ruined 3:Under Reclamation 4:Wingless 5:Planned Construction	EN	0, 1
Contact Address		C	0, 1
Delivery Point		(S) TE	0, 1
City Name		(S) TE	0, 1
Administrative Division		(S) TE	0, 1
Country Name		(S) TE	0, 1
Postal Code		(S) TE	0, 1

6.7.1 Building

Waterfront, landmark and some public buildings should be encoded precisely and individually on the larger optimum display scale data. When representing buildings generally, forming urban and suburban areas, villages, and other built-up areas, the aim of the compiler must be to create the correct impression of the extent of the built-up area and the density of the buildings.

Within built-up areas, only waterfront, landmark, and certain public buildings of interest should be encoded individually.

Scattered buildings of no individual importance must be omitted when more than about 1 mile (or 2 kilometres) inland. Nearer the shore they may be generalised by encoding a few representative buildings, sufficient to give the correct impression of building density.

Public buildings, with the possible exception of Post Offices and Hospitals, are charted mainly as visual features or points of reference ashore, not for their interest for particular functions.

If it is required to encode a building (other than a landmark, tank, silo or roofed structure erected or extending over navigable water), it must be done using the feature Building.

6.8 Bridge (Bridge)

Definition: Bridge. : (1) An elevated structure extending across or over the weather deck of a vessel, or part of such a structure. The term is sometimes modified to indicate the intended use, such as navigating bridge or signal bridge. (2) A structure erected over a depression or an obstacle such as a body of water, railroad, etc., to provide a roadway for vehicles or pedestrians.

S-201 Geo Feature: Bridge

Primitives: Point, Curve, Surface

<i>Real World</i>	<i>Paper Chart Symbol</i>	<i>ECDIS Symbol</i>
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S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
AtoN Number		TE	1, 1
Aid Availability Category	1:Category 1	EN	0, 1

	2:Category 2 3:Category 3		
Condition	1:Under Construction 2:Ruined 3:Under Reclamation 4:Wingless 5:Planned Construction	EN	0, 1
Contact Address		C	0, 1
Delivery Point		(S) TE	0, 1
City Name		(S) TE	0, 1
Administrative Division		(S) TE	0, 1
Country Name		(S) TE	0, 1
Postal Code		(S) TE	0, 1
6.8.1 Bridge If it is required to encode a bridge, it should be done using the feature Bridge.			

6.9 Lighthouse (Lighthouse)

Definition: **Lighthouse.** A distinctive structure on or off a coast exhibiting a major light designed to serve as an aid to navigation.

S-201 Geo Feature: Lighthouse

Primitives: Point

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
AtoN Number		TE	1,1
Aid Availability Category	1:Category 1 2:Category 2 3:Category 3	EN	0, 1

Condition	1:Under Construction 2:Ruined 3:Under Reclamation 4:Wingless 5:Planned Construction	EN	0, 1
Contact Address		C	0, 1
Delivery Point		(S) TE	0, 1
City Name		(S) TE	0, 1
Administrative Division		(S) TE	0, 1
Country Name		(S) TE	0, 1
Postal Code		(S) TE	0, 1
6.9.1 Lighthouse If it is required to encode a lighthouse, it should be done using the feature Lighthouse.			

7 Electronic Aton

7.1 Virtual aid aid to navigation (VirtualAISaidToNavigation)

Definition: **Virtual AIS Aid To Navigation.** An Automatic Identification System (AIS) message 21 transmitted from an AIS station to simulate on navigation systems an Aid to Navigation which does not physically exist.

S-201 Geo Feature: Virtual AIS Aid To Navigation

Primitives: Point

Real World	Paper Chart Symbol	ECDIS Symbol	
S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
AtoN Number		TE	0,1

Aid Availability Category	1:Category 1 2:Category 2 3:Category 3	EN	0, 1
Condition	1:Under Construction 2:Ruined 3:Under Reclamation 4:Wingless 5:Planned Construction	EN	0, 1
Contact Address		C	0, 1
Delivery Point		(S) TE	0, 1
City Name		(S) TE	0, 1
Administrative Division		(S) TE	0, 1
Country Name		(S) TE	0, 1
Postal Code		(S) TE	0, 1
MMSI Code		TE	1, 1
Status	1: Permanent 2: Occasional 3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 8: Private 9: Mandatory 11: Extinguished 12: Illuminated 13: Historic 14: Public 15: Synchronized 16: Watched 17: Unwatched 18: Existence Doubtful 19: On Request 20: Drop Away 21: Rising 22: Increasing 23: Decreasing 24: Strong 25: Good 26: Moderately 27: Poor 28: Buoyed 29: Fully Operational 30: Partially Operational 31: Drifting 32: Broken 33: Offline 34: Discontinued 35: Manual Observation	EN	0, *

	36: Unknown Status 37: Confirmed 38: Candidate 39: Under Modification 40: Experimental 41: Under Removal / Deletion 42: Removed / Deleted 43: Candidate for Modification		
Virtual AIS Aid to Navigation Type	1: North Cardinal 2: East Cardinal 3: South Cardinal 4: West Cardinal 5: Port Lateral 6: Starboard Lateral 7: Preferred Channel to Port 8: Preferred Channel to Starboard 9: Isolated Danger 10: Safe Water 11: Special Purpose 12: New Danger Marking	EN	1, 1
7.1.1 Virtual AIS Aid To Navigation If it is required to encode a virtual AIS Aton, it should be done using the feature VirtualAIS Aid To Navigation.			

7.2 Physical aid to navigation (PhysicalAISaidToNavigation)

Definition: Physical AIS Aid To Navigation. An Automatic Identification System (AIS) message 21 transmitted from a physical Aid to Navigation, or transmitted from an AIS station for an Aid to Navigation which physically exists.

S-201 Geo Feature: Physial AIS Aid To Navigation

Primitives: Point

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
AtoN Number		TE	0, 1
Aid Availability Category	1:Category 1 2:Category 2	EN	0, 1

	3:Category 3		
Condition	1:Under Construction 2:Ruined 3:Under Reclamation 4:Wingless 5:Planned Construction	EN	0, 1
Contact Address		C	0, 1
Delivery Point		(S) TE	0, 1
City Name		(S) TE	0, 1
Administrative Division		(S) TE	0, 1
Country Name		(S) TE	0, 1
Postal Code		(S) TE	0, 1
MMSI Code		TE	1, 1
Status	1: Permanent 2: Occasional 3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 8: Private 9: Mandatory 11: Extinguished 12: Illuminated 13: Historic 14: Public 15: Synchronized 16: Watched 17: Unwatched 18: Existence Doubtful 19: On Request 20: Drop Away 21: Rising 22: Increasing 23: Decreasing 24: Strong 25: Good 26: Moderately 27: Poor 28: Buoyed 29: Fully Operational 30: Partially Operational 31: Drifting 32: Broken 33: Offline 34: Discontinued 35: Manual Observation 36: Unknown Status 37: Confirmed	EN	0, *

	38: Candidate 39: Under Modification 40: Experimental 41: Under Removal / Deletion 42: Removed / Deleted 43: Candidate for Modification		
Category Of Physical AIS Aid To Navigation	1: Physical AIS Type 1 2: Physical AIS Type 2 3: Physical AIS Type 3	EN	1, 1
7.2.1 Physical AIS Aid To Navigation If it is required to encode a physical AIS Aton, it should be done using the feature PhysicalAISaidToNavigation.			

7.3 Synthetic ais aid to navigation (SyntheticAISaidToNavigation)

Definition: **Synthetic AIS Aid To Navigation.** A virtual or synthetic Aid to Navigation (AtoN) transmitted via the Automatic Identification System (AIS). Unlike a physical AtoN, which exists in the real world (like buoys or lighthouses), a synthetic AIS AtoN is created in the AIS network

S-201 Geo Feature: Synthetic AIS Aid To Navigation

Primitives: Point

Real World		Paper Chart Symbol	ECDIS Symbol	
S-201 Attribute		Allowable Encoding Value	Type	Multiplicity
ID Code			TE	0, 1
Information			C	0, *
File Locator			(S) TE	0, 1
File Reference			(S) TE	0, 1
Headline			(S) TE	0, 1
Language			(S) TE	1, 1
Text			(S) TE	0, 1
Feature Name			C	0, *
Display Name			(S) BO	0, 1
Language			(S) TE	0, 1
Name			(S) TE	1, 1
Scale Minimum			IN	0, 1
Source Date			DA	0, 1
Source			TE	0, 1
Pictorial Representation			TE	0, 1
Inspection Frequency			TE	0, 1
Inspection Requirements			TE	0, 1
AtoN Maintenance Record			TE	0, 1
Installation Date			DA	0, 1
Fixed Date Range			C	0, 1
Date End			(S) TD	0, 1
Date Start			(S) TD	0, 1
Periodic Date Range			C	0, 1
Date End			(S) TD	1, 1
Date Start			(S) TD	1, 1
Seasonal Action Required			TE	0, *
AtoN Number			TE	0,1
Aid Availability Category		1: Category 1 2: Category 2	EN	0, 1

	3: Category 3		
Condition	1: Under Construction 2: Ruined 3: Under Reclamation 4: Wingless 5: Planned Construction	EN	0, 1
Contact Address		C	0, 1
Delivery Point		(S) TE	0, 1
City Name		(S) TE	0, 1
Administrative Division		(S) TE	0, 1
Country Name		(S) TE	0, 1
Postal Code		(S) TE	0, 1
MMSI Code		TE	1, 1
Status	1: Permanent 2: Occasional 3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 8: Private 9: Mandatory 11: Extinguished 12: Illuminated 13: Historic 14: Public 15: Synchronized 16: Watched 17: Unwatched 18: Existence Doubtful 19: On Request 20: Drop Away 21: Rising 22: Increasing 23: Decreasing 24: Strong 25: Good 26: Moderately 27: Poor 28: Buoyed 29: Fully Operational 30: Partially Operational 31: Drifting 32: Broken 33: Offline 34: Discontinued 35: Manual Observation 36: Unknown Status 37: Confirmed	EN	0, *

	38: Candidate 39: Under Modification 40: Experimental 41: Under Removal / Deletion 42: Removed / Deleted 43: Candidate for Modification		
Virtual AIS Aid to Navigation Type	1: North Cardinal 2: East Cardinal 3: South Cardinal 4: West Cardinal 5: Port Lateral 6: Starboard Lateral 7: Preferred Channel to Port 8: Preferred Channel to Starboard 9: Isolated Danger 10: Safe Water 11: Special Purpose 12: New Danger Marking	EN	1, 1
Category Of Synthetic AIS Aid To Navigation	1: predicted 2: monitored	EN	1, 1
7.3.1 Synthetic AIS Aid To Navigation If it is required to encode a physical AIS Aton, it should be done using the feature SyntheticAISaidToNavigation.			

8 AtoN Equipment

8.1 Topmarks (Topmark)

Definition: **TOPMARK.** A characteristic shape secured at the top of a BUOY or BEACON to aid in its identification. (IHO Dictionary – S-32).

S-201 Geo Feature: Tomark

Primitives: Point

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
Remote Monitoring System		TE	0, *

Colour	1: White 2: Black 3: Red 4: Green 5: Blue 6: Yellow 7: Grey 8: Brown 9: Amber 10: Violet 11: Orange 12: Magenta 13: Pink	EN	0, *
Colour Pattern	1: Horizontal Stripes 2: Vertical Stripes 3: Diagonal Stripes 4: Squared 5: Stripes (Direction Unknown) 6: Border Stripe 7: Single Colour 8: Rectangle	EN	0, *
Status	1: Permanent 2: Occasional 3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 8: Private 9: Mandatory 11: Extinguished 12: Illuminated 13: Historic 14: Public 15: Synchronized 16: Watched 17: Unwatched 18: Existence Doubtful 19: On Request 20: Drop Away 21: Rising 22: Increasing 23: Decreasing 24: Strong 25: Good 26: Moderately 27: Poor 28: Buoyed 29: Fully Operational 30: Partially Operational 31: Drifting 32: Broken	EN	0, *

	33: Offline 34: Discontinued 35: Manual Observation 36: Unknown Status 37: Confirmed 38: Candidate 39: Under Modification 40: Experimental 41: Under Removal / Deletion 42: Removed / Deleted 43: Candidate for Modification		
Topmark/daymark shape	1 : cone, point up 2 : cone, point down 3 : sphere 4 : 2 spheres 5 : cylinder (can) 6 : board 7 : x-shape (St. Andrew's cross) 8 : upright cross (St George's cross) 9 : cube, point up 10 : 2 cones, point to point 11 : 2 cones, base to base 12 : rhombus (diamond) 13 : 2 cones (points upward) 14 : 2 cones (points downward) 15 : besom, point up (broom or perch) 16 : besom, point down (broom or perch) 17 : flag 18 : sphere over rhombus 19 : square 20 : rectangle, horizontal 21 : rectangle, vertical 22 : trapezium, up 23 : trapezium, down 24 : triangle, point up 25 : triangle, point down 26 : circle 27 : two upright crosses (one over the other) 28 : T-shape 29 : triangle pointing up over a circle 30 : upright cross over a circle 31 : rhombus over a circle 32 : circle over a triangle pointing up 33 : other shape	EN	1,1
Vertical Length		RE	0, 1
8.1.1 Topmark If it is required to encode a topmark, it should be done using the feature Topmark.			

8.2 Light sectored (Light Sectored)

Definition: **LIGHT SECTORED:** A light presenting different appearances (in particular, different colours) over various parts of the horizon of interest to maritime navigation.

S-201 Geo Feature: Light sectored

Primitives: Point

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
Remote Monitoring System		TE	0, *
Colour	1 : white 3 : red	EN	1, *

	4 : green 5 : blue 6 : yellow 9 : amber 10 : violet 11 : orange 14: Green A 15: Green B 16: White Temporary 17: Red Temporary 18: Yellow Temporary 19: Green Preferred 20: Green Temporary		
Height		RE	0, 1
Status	1: Permanent 2: Occasional 3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 8: Private 9: Mandatory 11: Extinguished 12: Illuminated 13: Historic 14: Public 15: Synchronized 16: Watched 17: Unwatched 18: Existence Doubtful 19: On Request 20: Drop Away 21: Rising 22: Increasing 23: Decreasing 24: Strong 25: Good 26: Moderately 27: Poor 28: Buoyed 29: Fully Operational 30: Partially Operational 31: Drifting 32: Broken 33: Offline 34: Discontinued 35: Manual Observation 36: Unknown Status 37: Confirmed 38: Candidate 39: Under Modification 40: Experimental 41: Under Removal / Deletion 42: Removed / Deleted 43: Candidate for Modification	EN	0, *
Vertical Datum	1: Mean Low Water Springs 2: Mean Lower Low Water Springs 3: Mean Sea Level 4: Lowest Low Water 5: Mean Low Water	EN	0, 1

	6: Lowest Low Water Springs 7: Approximate Mean Low Water Springs 8: Indian Spring Low Water 9: Low Water Springs 10: Approximate Lowest Astronomical Tide 11: Nearly Lowest Low Water 12: Mean Lower Low Water 13: Low Water 14: Approximate Mean Low Water 15: Approximate Mean Lower Low Water 16: Mean High Water 17: Mean High Water Springs 18: High Water 19: Approximate Mean Sea Level 20: High Water Springs 21: Mean Higher High Water 22: Equinoctial Spring Low Water 23: Lowest Astronomical Tide 24: Local Datum 25: International Great Lakes Datum 1985 26: Mean Water Level 27: Lower Low Water Large Tide 28: Higher High Water Large Tide 29: Nearly Highest High Water 30: Highest Astronomical Tide 31: Local Low Water Reference Level 32: Local High Water Reference Level 33: Local Mean Water Reference Level 34: Equivalent Height of Water (German GIW) 35: Highest Shipping Height of Water (German HSW) 36: Reference Low Water Level According to Danube Commission 37: Highest Shipping Height of Water According to Danube Commission 38: Dutch River Low Water Reference Level (OLR) 39: Russian Project Water Level 40: Russian Normal Backwater Level 41: Ohio River Datum 43: Dutch High Water Reference Level 44: Baltic Sea Chart Datum 2000 45: Dutch Estuary Low Water Reference Level (OLW) 46: International Great Lakes Datum 2020 47: Sea Floor 48: Sea Surface 49: Hydrographic Zero		
Vertical Length		RE	0, 1
Peak Intensity		RE	0, 1
Effective Intensity		RE	0, 1
Category of Light	1: Directional Function	EN	0, *

	4: Leading Light 5: Aero Light 6: Air Obstruction Light 8: Flood Light 9: Strip Light 10: Subsidiary Light 11: Spotlight 12: Front 13: Rear 14: Lower 15: Upper 17: Emergency 18: Bearing Light 19: Horizontally Disposed 20: Vertically Disposed		
Exhibition Condition of Light	1: Light Shown Without Change of Character 2: Daytime Light 3: Fog Light 4: Night Light	EN	0, *
Marks Navigational - System Of	1: IALA A 2: IALA B 9: No System 10: Other System 11: CEVNI 12: Russian Inland Waterway Regulations 13: Brazilian National Inland Waterway Regulations - Two Sides 15: Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	0, 1
Signal Generation	1: Automatically 2: By Wave Action 3: By Hand 4: By Wind 5: Radio Activated 6: Call Activated	EN	0, 1
Obscured Sector		C	0, *
Sector Limit		(S) C	1, 1
Sector Limit One		(S) C	1, 1
Sector Bearing		(S) RE	1, 1
Sector Line Length		(S) IN	0, 1
Sector Limit Two		(S) C	1, 1
Sector Bearing		(S) RE	1, 1
Sector Line Length		(S) IN	0, 1
Sector Information		(S) C	0, 1
Language		(S) TE	0, 1
Text		(S) TE	1, 1
Sector Characteristics		C	1, *
Light Characteristic	1: Fixed 2: Flashing	(S) EN	1, 1

	3: Long-Flashing 4: Quick-Flashing 5: Very Quick-Flashing 6: Continuous Ultra Quick-Flashing 7: Isophased 8: Occulting 12: Morse 13: Fixed and Flash 14: Flash and Long-Flash 15: Occulting and Flash 16: Fixed and Long-Flash 17: Occulting Alternating 18: Long-Flash Alternating 19: Flash Alternating 20: Group Alternating 25: Quick-Flash Plus Long-Flash 26: Very Quick-Flash Plus Long-Flash 27: Ultra Quick-Flash Plus Long-Flash 28: Alternating 29: Fixed and Alternating Flashing 30: group-occulting light 31: composite group-occulting light 32: group flashing light 33: composite group-flashing light 34: group quick light 35: group very quick light		
Light Sector		(S) C	1, 10
Colour	1 : white 3 : red 4 : green 5 : blue 6 : yellow 9 : amber 10 : violet 11 : orange 14: Green A 15: Green B 16: White Temporary 17: Red Temporary 18: Yellow Temporary 19: Green Preferred 20: Green Temporary	(S) EN	1, *
Directional Character		(S) C	0, 1
Moire Effect		(S) BO	0, 1
Orientation		(S) C	1, 1
Orientation Uncertainty		(S) RE	0, 1
Orientation Value		(S) RE	1, 1
Light Visibility	1: High Intensity 2: Low Intensity 3: Faint	(S) EN	0, *

	4: Intensified 5: Unintensified 6: Visibility Deliberately Restricted 7: Obscured 8: Partially Obscured 9: Visible in Line of Range		
Sector Limit		(S) C	0, 1
Sector Limit One		(S) C	1, 1
Sector Bearing		(S) RE	1, 1
Sector Line Length		(S) IN	0, 1
Sector Limit Two		(S) C	1, 1
Sector Bearing		(S) RE	1, 1
Sector Line Length		(S) IN	0, 1
Value of Nominal Range		(S) RE	0, 1
Sector Information		(S) C	0, *
Language		(S) TE	0, 1
Text		(S) TE	1, 1
Sector Arc Extension		(S) BO	0, 1
Signal Group		(S) TE	0, *
Signal Period		(S) RE	0, 1
Signal Sequence		(S) C	0, *
Signal Duration		(S) RE	1, 1
Signal Status		(S) EN	1, 1

8.2.1 Light sectored

If it is required to encode a light that consists of one or more sectors, it must be done using the feature Light Sectored. This feature must be an equipment feature of a structure feature, which may be another light feature at the same position.

The IALA Maritime Buoyage System rules do not apply for most landfall lights and will apply to minor lights, but not to leading lights, some sector lights or major floating lights. In general, sector lights follow IALA convention when used for marking a channel.

Remarks:

- The complex attribute sector characteristics, sub-complex attribute light sector is used to populate each sector for the light, except for sectors in which there is no light exhibited. Where there is a different rhythm of light between sectors (for example, for complex lights), separate instances of sector characteristics must be populated.
- Population of the sub-complex attribute sector limit having sub-attributes sector limit one = 0 and sector limit two = 360 (that is, encoding an all around light as a sectored light) is prohibited.
- If a sector of a sectored light is intended to have a directional function, this must be encoded using the light sector complex sub-attribute directional character. If the light is intensified in this sector, light sector sub-attribute light visibility = 4 (intensified) must be populated. The sub-complex attribute sector limit is optional for directional light sectors.
- The fairway defined by the succession of navigable areas in the white sectors of a series of Light Sectored

features may be encoded using the feature Fairway.

- If there is additional information required to be encoded that is relevant to all sectors of the light, this must be done using the information. If the additional information is relevant to individual sectors of the light only (for example, for complex (oscillating) light sectors, this must be encoded using the complex sub-attribute sector information for the sub-complex attribute light sector.
- If it is required to encode details of the lighting technology (for example neon), it must be done using the complex attribute information.
- Names of major lights are very important. If a light has a name which is unrelated to any other encoded feature, the name must be populated using the complex attribute feature name on at least the largest scale optimum display scale ENC data. If the name of a light is obviously that of the named feature on which the light stands, for example Saint Catherine's Point, the name of the light need not be repeated for the light.

8.3 Light all around (LightAllAround)

Definition: **LIGHT ALL AROUND:** An all around light is a light that is visible over the whole horizon of interest to marine navigation and having no change in the characteristics of the light.

S-201 Geo Feature: Light all around

Primitives: Point

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
Remote Monitoring System		TE	0, *
Colour	1 : white 3 : red 4 : green	EN	1, *

	5 : blue 6 : yellow 9 : amber 10 : violet 11 : orange 14: Green A 15: Green B 16: White Temporary 17: Red Temporary 18: Yellow Temporary 19: Green Preferred 20: Green Temporary		
Height		RE	0, 1
Status	1: Permanent 2: Occasional 3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 8: Private 9: Mandatory 11: Extinguished 12: Illuminated 13: Historic 14: Public 15: Synchronized 16: Watched 17: Unwatched 18: Existence Doubtful 19: On Request 20: Drop Away 21: Rising 22: Increasing 23: Decreasing 24: Strong 25: Good 26: Moderately 27: Poor 28: Buoyed 29: Fully Operational 30: Partially Operational 31: Drifting 32: Broken 33: Offline 34: Discontinued 35: Manual Observation 36: Unknown Status 37: Confirmed 38: Candidate 39: Under Modification 40: Experimental 41: Under Removal / Deletion 42: Removed / Deleted 43: Candidate for Modification	EN	0, *
Vertical Datum	1: Mean Low Water Springs 2: Mean Lower Low Water Springs 3: Mean Sea Level 4: Lowest Low Water 5: Mean Low Water 6: Lowest Low Water Springs	EN	0, 1

	7: Approximate Mean Low Water Springs 8: Indian Spring Low Water 9: Low Water Springs 10: Approximate Lowest Astronomical Tide 11: Nearly Lowest Low Water 12: Mean Lower Low Water 13: Low Water 14: Approximate Mean Low Water 15: Approximate Mean Lower Low Water 16: Mean High Water 17: Mean High Water Springs 18: High Water 19: Approximate Mean Sea Level 20: High Water Springs 21: Mean Higher High Water 22: Equinoctial Spring Low Water 23: Lowest Astronomical Tide 24: Local Datum 25: International Great Lakes Datum 1985 26: Mean Water Level 27: Lower Low Water Large Tide 28: Higher High Water Large Tide 29: Nearly Highest High Water 30: Highest Astronomical Tide 31: Local Low Water Reference Level 32: Local High Water Reference Level 33: Local Mean Water Reference Level 34: Equivalent Height of Water (German GIW) 35: Highest Shipping Height of Water (German HSW) 36: Reference Low Water Level According to Danube Commission 37: Highest Shipping Height of Water According to Danube Commission 38: Dutch River Low Water Reference Level (OLR) 39: Russian Project Water Level 40: Russian Normal Backwater Level 41: Ohio River Datum 43: Dutch High Water Reference Level 44: Baltic Sea Chart Datum 2000 45: Dutch Estuary Low Water Reference Level (OLW) 46: International Great Lakes Datum 2020 47: Sea Floor 48: Sea Surface 49: Hydrographic Zero		
Vertical Length		RE	0, 1
Rhythm of Light		C	1, 1
Light Characteristic	1: Fixed 2: Flashing 3: Long-Flashing	(S) EN	1, 1

	4: Quick-Flashing 5: Very Quick-Flashing 6: Continuous Ultra Quick-Flashing 7: Isophased 8: Occulting 12: Morse 13: Fixed and Flash 14: Flash and Long-Flash 15: Occulting and Flash 16: Fixed and Long-Flash 17: Occulting Alternating 18: Long-Flash Alternating 19: Flash Alternating 20: Group Alternating 25: Quick-Flash Plus Long-Flash 26: Very Quick-Flash Plus Long-Flash 27: Ultra Quick-Flash Plus Long-Flash 28: Alternating 29: Fixed and Alternating Flashing 30: group-occulting light 31: composite group-occulting light 32: group flashing light 33: composite group-flashing light 34: group quick light 35: group very quick light		
Signal Group		(S) TE	0, 10
Signal Period		(S) RE	0, 1
Signal Sequence		(S) C	0, 10
Signal Duration		(S) RE	1, 1
Signal Status	1: Lit/Sound 2: Eclipsed/Silent	(S) EN	1, 1
Peak Intensity		RE	0, 1
Effective Intensity		RE	0, 1
Category of Light	1: Directional Function 4: Leading Light 5: Aero Light 6: Air Obstruction Light 8: Flood Light 9: Strip Light 10: Subsidiary Light 11: Spotlight 12: Front 13: Rear 14: Lower 15: Upper 17: Emergency 18: Bearing Light 19: Horizontally Disposed 20: Vertically Disposed	EN	0, *
Exhibition Condition of Light	1: Light Shown Without Change of Character	EN	0, *

	2: Daytime Light 3: Fog Light 4: Night Light		
Light Visibility	1: High Intensity 2: Low Intensity 3: Faint 4: Intensified 5: Unintensified 6: Visibility Deliberately Restricted 7: Obscured 8: Partially Obscured 9: Visible in Line of Range	EN	0, 1
Major Light		BO	0, 1
Marks Navigational - System Of	1: IALA A 2: IALA B 9: No System 10: Other System 11: CEVNI 12: Russian Inland Waterway Regulations 13: Brazilian National Inland Waterway Regulations - Two Sides 15: Paraguay-Parana Waterway - Brazilian Complementary Aids 4 : other system	EN	0, 1
Signal Generation	1: Automatically 2: By Wave Action 3: By Hand 4: By Wind 5: Radio Activated 6: Call Activated	EN	0, 1
Value of Nominal Range		RE	0, 1
Multiplicity of Features		C	0, 1
Multiplicity Known		(S) BO	1, 1
Number of Features		(S) IN	0, 1

8.3.1 Light all around

If it is required to encode an all around light (excluding fog detector and air obstruction lights), it must be done using the feature Light All Around. This feature must be an equipment feature of a structure feature, which may be another light feature at the same position (if it exists and there is no structure feature available).

The IALA Maritime Buoyage System rules do not apply for most landfall lights and will apply to minor lights, but not to leading lights, some sector lights or major floating lights. In general, sector lights follow IALA convention when used for marking a channel.

Remarks:

- All sector lights, whether single sector, multi-sectored or having sectors that are deliberately obscured or completely or partially obscured by obstructions, must be encoded, where required, using the feature Light Sector.
- Fog detector and air obstruction lights must be encoded, where required, using the features Light Fog Detector and Light Air Obstruction.
- If it is required to encode details of the lighting technology (for example neon), it must be done using the

complex attribute information.

- If it is required to encode the purpose of a marine spotlight, it must be done using the complex attribute information.
- Lights on land encoded as major lights (Boolean attribute major light = True) must have a structure feature encoded in order for the position of the light to be clearly indicated in the ECDIS.
- The indication that a light is a “major” light through the population of the Boolean attribute major light with a True value determines the way the light is displayed in ECDIS, and is not based on any legal or formal classification of the importance of lights. Generally, a major light may be considered to be a light intended for use at sea, usually with a range of 15 miles or more, and in outer approaches to harbours. However the determination of what is a major light may be based on a number of additional factors, including the number and characteristics of navigational (and non-navigational) lights in the geographical area, and specific navigational requirements for the area. Indication in a dataset that a light is a major light should be based only on the requirements for ECDIS display, at the discretion of the Producing Authority.
- Names of major lights are very important. If a light has a name which is unrelated to any other encoded feature, the name must be populated using the complex attribute feature name on at least the largest optimum display scale ENC data. If the name of a light is obviously that of the named feature on which the light stands, for example Saint Catherine’s Point, the name of the light need not be repeated for the light.

8.4 Light air obstruction (LightAirObstruction)

Definition: **LIGHT AIR OBSTRUCTION:** An air obstruction light is a light marking an obstacle which constitutes a danger to air navigation.

S-201 Geo Feature: Light air obstruction

Primitives: Point

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
Remote Monitoring System		TE	0, *
Colour	1 : white 3 : red 4 : green	EN	1, *

	5 : blue 6 : yellow 9 : amber 10 : violet 11 : orange 14: Green A 15: Green B 16: White Temporary 17: Red Temporary 18: Yellow Temporary 19: Green Preferred 20: Green Temporary		
Height		RE	0, 1
Status	1: Permanent 2: Occasional 3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 8: Private 9: Mandatory 11: Extinguished 12: Illuminated 13: Historic 14: Public 15: Synchronized 16: Watched 17: Unwatched 18: Existence Doubtful 19: On Request 20: Drop Away 21: Rising 22: Increasing 23: Decreasing 24: Strong 25: Good 26: Moderately 27: Poor 28: Buoyed 29: Fully Operational 30: Partially Operational 31: Drifting 32: Broken 33: Offline 34: Discontinued 35: Manual Observation 36: Unknown Status 37: Confirmed 38: Candidate 39: Under Modification 40: Experimental 41: Under Removal / Deletion 42: Removed / Deleted 43: Candidate for Modification	EN	0, *
Vertical Datum	1: Mean Low Water Springs 2: Mean Lower Low Water Springs 3: Mean Sea Level 4: Lowest Low Water 5: Mean Low Water 6: Lowest Low Water Springs	EN	0, 1

	7: Approximate Mean Low Water Springs 8: Indian Spring Low Water 9: Low Water Springs 10: Approximate Lowest Astronomical Tide 11: Nearly Lowest Low Water 12: Mean Lower Low Water 13: Low Water 14: Approximate Mean Low Water 15: Approximate Mean Lower Low Water 16: Mean High Water 17: Mean High Water Springs 18: High Water 19: Approximate Mean Sea Level 20: High Water Springs 21: Mean Higher High Water 22: Equinoctial Spring Low Water 23: Lowest Astronomical Tide 24: Local Datum 25: International Great Lakes Datum 1985 26: Mean Water Level 27: Lower Low Water Large Tide 28: Higher High Water Large Tide 29: Nearly Highest High Water 30: Highest Astronomical Tide 31: Local Low Water Reference Level 32: Local High Water Reference Level 33: Local Mean Water Reference Level 34: Equivalent Height of Water (German GIW) 35: Highest Shipping Height of Water (German HSW) 36: Reference Low Water Level According to Danube Commission 37: Highest Shipping Height of Water According to Danube Commission 38: Dutch River Low Water Reference Level (OLR) 39: Russian Project Water Level 40: Russian Normal Backwater Level 41: Ohio River Datum 43: Dutch High Water Reference Level 44: Baltic Sea Chart Datum 2000 45: Dutch Estuary Low Water Reference Level (OLW) 46: International Great Lakes Datum 2020 47: Sea Floor 48: Sea Surface 49: Hydrographic Zero		
Vertical Length		RE	0, 1
Rhythm of Light		C	1, 1
Light Characteristic	1: Fixed 2: Flashing 3: Long-Flashing	(S) EN	1, 1

	4: Quick-Flashing 5: Very Quick-Flashing 6: Continuous Ultra Quick-Flashing 7: Isophased 8: Occulting 12: Morse 13: Fixed and Flash 14: Flash and Long-Flash 15: Occulting and Flash 16: Fixed and Long-Flash 17: Occulting Alternating 18: Long-Flash Alternating 19: Flash Alternating 20: Group Alternating 25: Quick-Flash Plus Long-Flash 26: Very Quick-Flash Plus Long-Flash 27: Ultra Quick-Flash Plus Long-Flash 28: Alternating 29: Fixed and Alternating Flashing 30: group-occulting light 31: composite group-occulting light 32: group flashing light 33: composite group-flashing light 34: group quick light 35: group very quick light		
Signal Group		(S) TE	0, 10
Signal Period		(S) RE	0, 1
Signal Sequence		(S) C	0, 10
Signal Duration		(S) RE	1, 1
Signal Status	1: Lit/Sound 2: Eclipsed/Silent	(S) EN	1, 1
Peak Intensity		RE	0, 1
Effective Intensity		RE	0, 1
Exhibition Condition of Light	1: Light Shown Without Change of Character 2: Daytime Light 3: Fog Light 4: Night Light	EN	0, *
Light Visibility	1: High Intensity 2: Low Intensity 3: Faint 4: Intensified 5: Unintensified 6: Visibility Deliberately Restricted 7: Obscured 8: Partially Obscured 9: Visible in Line of Range	EN	0, *
Value of Geographic Range		RE	0, 1
Multiplicity of Features		C	0, 1
Multiplicity Known		(S) BO	1, 1

Number of Features		(S) IN	0, 1
8.4.1 Light air obstruction If it is required to encode a light marking an obstacle which constitutes a danger to air navigation, which may also be used as a marine navigational aid, it must be done using the feature Light Air Obstruction. This feature must be an equipment feature of a structure feature using a feature association. Remarks: • If it is required to encode details of the lighting technology (for example neon), it must be done using the complex attribute information.			

8.5 Light fog detector (Light Fog Detector)

Definition: **LIGHT FOG DETECTOR:** A fog detector light is a light used to automatically determine conditions of visibility which warrant the turning on or off of a sound signal

S-201 Geo Feature: Light fog detector

Primitives: Point

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
Remote Monitoring System		TE	0, *
Height		RE	0, 1
Status	1: Permanent 2: Occasional 3: Recommended	EN	0, *

	4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 8: Private 9: Mandatory 11: Extinguished 12: Illuminated 13: Historic 14: Public 15: Synchronized 16: Watched 17: Unwatched 18: Existence Doubtful 19: On Request 20: Drop Away 21: Rising 22: Increasing 23: Decreasing 24: Strong 25: Good 26: Moderately 27: Poor 28: Buoyed 29: Fully Operational 30: Partially Operational 31: Drifting 32: Broken 33: Offline 34: Discontinued 35: Manual Observation 36: Unknown Status 37: Confirmed 38: Candidate 39: Under Modification 40: Experimental 41: Under Removal / Deletion 42: Removed / Deleted 43: Candidate for Modification		
Vertical Datum	1: Mean Low Water Springs 2: Mean Lower Low Water Springs 3: Mean Sea Level 4: Lowest Low Water 5: Mean Low Water 6: Lowest Low Water Springs 7: Approximate Mean Low Water Springs 8: Indian Spring Low Water 9: Low Water Springs 10: Approximate Lowest Astronomical Tide 11: Nearly Lowest Low Water 12: Mean Lower Low Water 13: Low Water 14: Approximate Mean Low Water 15: Approximate Mean Lower Low Water 16: Mean High Water 17: Mean High Water Springs 18: High Water 19: Approximate Mean Sea Level 20: High Water Springs 21: Mean Higher High Water	EN	0, 1

	22: Equinoctial Spring Low Water 23: Lowest Astronomical Tide 24: Local Datum 25: International Great Lakes Datum 1985 26: Mean Water Level 27: Lower Low Water Large Tide 28: Higher High Water Large Tide 29: Nearly Highest High Water 30: Highest Astronomical Tide 31: Local Low Water Reference Level 32: Local High Water Reference Level 33: Local Mean Water Reference Level 34: Equivalent Height of Water (German GIW) 35: Highest Shipping Height of Water (German HSW) 36: Reference Low Water Level According to Danube Commission 37: Highest Shipping Height of Water According to Danube Commission 38: Dutch River Low Water Reference Level (OLR) 39: Russian Project Water Level 40: Russian Normal Backwater Level 41: Ohio River Datum 43: Dutch High Water Reference Level 44: Baltic Sea Chart Datum 2000 45: Dutch Estuary Low Water Reference Level (OLW) 46: International Great Lakes Datum 2020 47: Sea Floor 48: Sea Surface 49: Hydrographic Zero		
Vertical Length		RE	0, 1
Rhythm of Light		C	1, 1
Light Characteristic	1: Fixed 2: Flashing 3: Long-Flashing 4: Quick-Flashing 5: Very Quick-Flashing 6: Continuous Ultra Quick-Flashing 7: Isophased 8: Occulting 12: Morse 13: Fixed and Flash 14: Flash and Long-Flash 15: Occulting and Flash 16: Fixed and Long-Flash 17: Occulting Alternating 18: Long-Flash Alternating 19: Flash Alternating 20: Group Alternating 25: Quick-Flash Plus Long-Flash	(S) EN	1, 1

	26: Very Quick-Flash Plus Long-Flash 27: Ultra Quick-Flash Plus Long-Flash 28: Alternating 29: Fixed and Alternating Flashing 30: group-occulting light 31: composite group-occulting light 32: group flashing light 33: composite group-flashing light 34: group quick light 35: group very quick light		
Signal Group		(S) TE	0, 10
Signal Period		(S) RE	0, 1
Signal Sequence		(S) C	0, 10
Signal Duration		(S) RE	1, 1
Signal Status	1: Lit/Sound 2: Eclipsed/Silent	(S) EN	1, 1
Peak Intensity		RE	0, 1
Effective Intensity		RE	0, 1
Signal Generation	1: Automatically 2: By Wave Action 3: By Hand 4: By Wind 5: Radio Activated 6: Call Activated	EN	0, 1

8.5.1 Light fog detector

If it is required to encode a light used to automatically determine conditions of visibility which warrant the turning on or off of a sound signal, it must be done using the feature Light Fog Detector. This feature must be an equipment feature of a structure feature, if it exists, using a feature association.

Remarks:

- If it is required to encode details of the lighting technology (for instance neon), it must be done using the complex attribute information.
- The attribute vertical length only applies to lights attached to floating structures.

8.6 Fog signals (FogSignal)

Definition: FOG SIGNALS. A warning signal transmitted by a vessel, or aid to navigation, during periods of low visibility. Also, the device producing such a signal. (IHO Dictionary – S-32).			
S-201 Geo Feature: Fog signal			
Primitives: Point			
<i>Real World</i>	<i>Paper Chart Symbol</i>	<i>ECDIS Symbol</i>	
S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
Remote Monitoring System		TE	0, *
Category of fog signal	1 : explosive 2 : diaphone 3 : siren	EN	1,1

	4 : nautophone 5 : reed 6 : tyfon 7 : bell 8 : whistle 9 : gong 10 : horn		
Signal frequency		IN	0,1
Signal generation	1 : automatically 2 : by wave action 3 : by hand 4 : by wind	EN	0,1
Signal group		TE	0,1
Signal period		RE	0,1
Signal sequence		TE	0,1
Status	1: Permanent 2: Occasional 3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 8: Private 9: Mandatory 11: Extinguished 12: Illuminated 13: Historic 14: Public 15: Synchronized 16: Watched 17: Unwatched 18: Existence Doubtful 19: On Request 20: Drop Away 21: Rising 22: Increasing 23: Decreasing 24: Strong 25: Good 26: Moderately 27: Poor 28: Buoyed 29: Fully Operational 30: Partially Operational 31: Drifting 32: Broken 33: Offline 34: Discontinued 35: Manual Observation 36: Unknown Status 37: Confirmed 38: Candidate	EN	0,*

	39: Under Modification 40: Experimental 41: Under Removal / Deletion 42: Removed / Deleted 43: Candidate for Modification		
Value of maximum range		RE	0,1
Signal Sequence		C	0, 1
Signal Duration		(S) RE	1, 1
Signal Status	1: Lit/Sound 2: Eclipsed/Silent	(S) EN	1, 1

8.6.1 Fog signals

The term “fog signal” refers to the sound emitted, not the apparatus. Fog signals are short range aids to navigation, principally used as hazard warnings. For various reasons they are unreliable as indicators of position. Their importance relative to other aids to navigation has declined but they are still considered useful for the safe navigation of vessels with very limited (or non-functioning) electronic equipment. A fog signal should be shown on ENC's at an maximum display scale on which vessels may navigate within range.

The position from which a fog signal is emitted is usually on a buoy, or close enough to a light to be treated as sounded from the same position as the light.

If it is required to encode a fog signal, it must be done using the feature **Fog Signal**.

Remarks:

- The characteristic rhythm of fog signals (other than those actuated by waves, which are irregular) may be more important than their type when mariners are attempting to identify them. The number of sound emissions (e.g. blasts, strokes) and the period must therefore be encoded, where known, using the attributes **signal group**, **signal period** and **signal sequence**.
- Where required, the attribute **signal frequency** must be quoted in Hertz, e.g. a signal frequency of 950 MHz must be encoded as *950000000*.

Distinction: Signal station, warning.

8.7 Retroreflectors (RetroReflector)

Definition: **RETROREFLECTOR.** A means of distinguishing unlighted marks at night. Retroreflective material is secured to the mark in a particular pattern to reflect back light. (UKHO NP 735, 5th Edition).

S-201 Geo Feature: Retroreflector

Primitives: Point

Real World	Paper Chart Symbol	ECDIS Symbol	
S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
Remote Monitoring System		TE	0, *
Colour	1 : white	EN	0,*

	3 : red 4 : green 5 : blue 6 : yellow 9 : amber 10 : violet 11 : orange 14: Green A 15: Green B 16: White Temporary 17: Red Temporary 18: Yellow Temporary 19: Green Preferred 20: Green Temporary		
Colour pattern	1 : horizontal stripes 2 : vertical stripes 3 : diagonal stripes 4 : squared 5 : stripes (direction unknown) 6 : border stripe	EN	0,1
Height		RE	0,1
Marks Navigational - System Of	1: IALA A 2: IALA B 9: No System 10: Other System 11: CEVNI 12: Russian Inland Waterway Regulations 13: Brazilian National Inland Waterway Regulations - Two Sides 15: Paraguay-Parana Waterway - Brazilian Complementary Aids 4 : other system	EN	0,1
Status	1: Permanent 2: Occasional 3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 8: Private 9: Mandatory 11: Extinguished 12: Illuminated 13: Historic 14: Public 15: Synchronized 16: Watched 17: Unwatched 18: Existence Doubtful 19: On Request 20: Drop Away 21: Rising 22: Increasing 23: Decreasing	EN	0,*

	24: Strong 25: Good 26: Moderately 27: Poor 28: Buoyed 29: Fully Operational 30: Partially Operational 31: Drifting 32: Broken 33: Offline 34: Discontinued 35: Manual Observation 36: Unknown Status 37: Confirmed 38: Candidate 39: Under Modification 40: Experimental 41: Under Removal / Deletion 42: Removed / Deleted 43: Candidate for Modification		
Vertical accuracy		RE	0,1
Vertical datum	1: Mean Low Water Springs 2: Mean Lower Low Water Springs 3: Mean Sea Level 4: Lowest Low Water 5: Mean Low Water 6: Lowest Low Water Springs 7: Approximate Mean Low Water Springs 8: Indian Spring Low Water 9: Low Water Springs 10: Approximate Lowest Astronomical Tide 11: Nearly Lowest Low Water 12: Mean Lower Low Water 13: Low Water 14: Approximate Mean Low Water 15: Approximate Mean Lower Low Water 16: Mean High Water 17: Mean High Water Springs 18: High Water 19: Approximate Mean Sea Level 20: High Water Springs 21: Mean Higher High Water 22: Equinoctial Spring Low Water 23: Lowest Astronomical Tide 24: Local Datum 25: International Great Lakes Datum 1985 26: Mean Water Level 27: Lower Low Water Large Tide 28: Higher High Water Large Tide 29: Nearly Highest High Water 30: Highest Astronomical Tide 31: Local Low Water Reference Level	EN	0,1

	32: Local High Water Reference Level 33: Local Mean Water Reference Level 34: Equivalent Height of Water (German GIW) 35: Highest Shipping Height of Water (German HSW) 36: Reference Low Water Level According to Danube Commission 37: Highest Shipping Height of Water According to Danube Commission 38: Dutch River Low Water Reference Level (OLR) 39: Russian Project Water Level 40: Russian Normal Backwater Level 41: Ohio River Datum 43: Dutch High Water Reference Level 44: Baltic Sea Chart Datum 2000 45: Dutch Estuary Low Water Reference Level (OLW) 46: International Great Lakes Datum 2020 47: Sea Floor 48: Sea Surface 49: Hydrographic Zero		
8.7.1 Retroreflectors Retroreflective material may be secured to unlit marks to aid their identification at night. The material is coloured according to one of two recognized IALA codes ("Standard" and "Comprehensive"). In any specified area only one of the codes will be used and this may be given in nautical publications. If it is required to encode a retroreflector, it must be done using the feature Retroreflector. <u>Remarks:</u> • The complex attribute information is used to describe letters, patterns or numerals shown on the retroreflector. • The body carrying the retroreflector is a separate feature. <u>Distinction:</u> Beacon, cardinal; beacon, isolated danger; beacon, lateral; beacon, safe water; beacon, special purpose/general; buoy, cardinal; buoy, emergency wreck marking; buoy, installation; buoy, isolated danger; buoy, lateral; buoy, safe water; buoy, special purpose/general; radar reflector.			

8.8 Radar reflectors (RadarReflector)

Definition: **RADAR REFLECTOR.** A device capable of, or intended for, reflecting radar signals. (IHO Dictionary – S-32).

A radar reflector is usually a “tetrahedron or pentagonal corner reflector (...) to facilitate reflection towards the sender”. (International Maritime Dictionary, 2nd Ed.).

S-201 Geo Feature: Radar reflector

Primitives: Point

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
Remote Monitoring System		TE	0, *

Height		RE	0,1
Status	1: Permanent 2: Occasional 3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 8: Private 9: Mandatory 11: Extinguished 12: Illuminated 13: Historic 14: Public 15: Synchronized 16: Watched 17: Unwatched 18: Existence Doubtful 19: On Request 20: Drop Away 21: Rising 22: Increasing 23: Decreasing 24: Strong 25: Good 26: Moderately 27: Poor 28: Buoyed 29: Fully Operational 30: Partially Operational 31: Drifting 32: Broken 33: Offline 34: Discontinued 35: Manual Observation 36: Unknown Status 37: Confirmed 38: Candidate 39: Under Modification 40: Experimental 41: Under Removal / Deletion 42: Removed / Deleted 43: Candidate for Modification	EN	0,*
Vertical accuracy		RE	0,1
Vertical datum	1: Mean Low Water Springs 2: Mean Lower Low Water Springs 3: Mean Sea Level 4: Lowest Low Water 5: Mean Low Water 6: Lowest Low Water Springs 7: Approximate Mean Low Water Springs	EN	0,1

	8: Indian Spring Low Water 9: Low Water Springs 10: Approximate Lowest Astronomical Tide 11: Nearly Lowest Low Water 12: Mean Lower Low Water 13: Low Water 14: Approximate Mean Low Water 15: Approximate Mean Lower Low Water 16: Mean High Water 17: Mean High Water Springs 18: High Water 19: Approximate Mean Sea Level 20: High Water Springs 21: Mean Higher High Water 22: Equinoctial Spring Low Water 23: Lowest Astronomical Tide 24: Local Datum 25: International Great Lakes Datum 1985 26: Mean Water Level 27: Lower Low Water Large Tide 28: Higher High Water Large Tide 29: Nearly Highest High Water 30: Highest Astronomical Tide 31: Local Low Water Reference Level 32: Local High Water Reference Level 33: Local Mean Water Reference Level 34: Equivalent Height of Water (German GIW) 35: Highest Shipping Height of Water (German HSW) 36: Reference Low Water Level According to Danube Commission 37: Highest Shipping Height of Water According to Danube Commission 38: Dutch River Low Water Reference Level (OLR) 39: Russian Project Water Level 40: Russian Normal Backwater Level 41: Ohio River Datum 43: Dutch High Water Reference Level 44: Baltic Sea Chart Datum 2000 45: Dutch Estuary Low Water Reference Level (OLW) 46: International Great Lakes Datum 2020 47: Sea Floor 48: Sea Surface 49: Hydrographic Zero		
8.8.1 Radar reflectors If it is required to encode radar reflectors on curve features (e.g. overhead cables), this must be done using the feature Radar Reflector. <u>Remarks:</u> • If it is required to encode a feature which has no radar reflector, but is radar conspicuous, it must be			

indicated using attribute **radar conspicuous** on the feature.

- If it is required to encode an area or point feature which is radar conspicuous because it is fitted with a radar reflector, it must be indicated using **radar conspicuous** on the feature. A **Radar Reflector** feature must not be encoded in this case.

Distinction: Retro-reflector.

8.9 Environment observation equipment (EnvironmentObservationEquipment)

Definition: **Environment Observation Equipment**. A sensor used to observe the environment.

S-201 Geo Feature: Environment Observation Equipment

Primitives: Point

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
Remote Monitoring System		TE	0, *
Height		RE	0,1
Status	1: Permanent 2: Occasional	EN	0,*

	3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 8: Private 9: Mandatory 11: Extinguished 12: Illuminated 13: Historic 14: Public 15: Synchronized 16: Watched 17: Unwatched 18: Existence Doubtful 19: On Request 20: Drop Away 21: Rising 22: Increasing 23: Decreasing 24: Strong 25: Good 26: Moderately 27: Poor 28: Buoyed 29: Fully Operational 30: Partially Operational 31: Drifting 32: Broken 33: Offline 34: Discontinued 35: Manual Observation 36: Unknown Status 37: Confirmed 38: Candidate 39: Under Modification 40: Experimental 41: Under Removal / Deletion 42: Removed / Deleted 43: Candidate for Modification		
Type of environment observation equipment		TE	1,*

8.9.1 Environment Observation Equipment

If it is required to encode environment observation equipment features (e.g. marine weather observation sensors), this must be done using the feature **Environment Observation Equipment**.

8.10 Daymark (Daymark)

Definition: **DAYMARK.** The identifying characteristics of an aid to navigation which serve to facilitate its recognition against a daylight viewing background. On those structures that do not by themselves present an adequate viewing area to be seen at the required distance, the aid is made more visible by affixing a daymark to the structure. A daymark so affixed has a distinctive colour and shape depending on the purpose of the aid. (IHO Dictionary – S-32, Edition 5).

S-201 Geo Feature: Daymark

Primitives: Point

<i>Real World</i>	<i>Paper Chart Symbol</i>	<i>ECDIS Symbol</i>
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S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
AtoN Number		UI	1, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *

Remote Monitoring System		TE	0, *
Category of special purpose mark	1 : firing danger area mark 2 : target mark 3 : marker ship mark 4 : degaussing range mark 5 : barge mark 6 : cable mark 7 : spoil ground mark 8 : outfall mark 10 : recording mark 11 : seaplane anchorage mark 12 : recreation zone mark 14 : mooring mark 15 : LANBY (Large Automatic Navigational Buoy) 16 : leading mark 17 : measured distance mark 18 : notice mark 19 : TSS mark (Traffic Separation Scheme) 20 : anchoring prohibited mark 21 : berthing prohibited mark 22 : overtaking prohibited mark 23 : two-way traffic prohibited mark 24 : reduced wake mark 25 : speed limit mark 26 : stop mark 27 : general warning mark 28 : sound ship's siren mark 29 : restricted vertical clearance mark 30 : maximum vessel's draught mark 31 : restricted horizontal clearance mark 32 : strong current warning mark 33 : berthing permitted mark 34 : overhead power cable mark 35 : channel edge gradient' mark 36 : telephone mark 37 : ferry crossing mark 39 : pipeline mark 40 : anchorage mark 41 : clearing mark 42 : control mark 43 : diving mark 44 : refuge beacon 45 : foul ground mark 46 : yachting mark 47 : heliport mark 48 : GNSS mark 49 : seaplane landing mark 50 : control mark 51 : work in progress mark 52 : mark with unknown purpose 53 : wellhead mark 54 : channel separation mark 55 : marine farm mark 56 : artificial reef mark 57 : ice mark 58 : nature reserve mark 59 : Fish Aggregating Device 60 : Wreck Mark 61 : Customs Mark 62 : Causeway Mark 63 : Wave Recorder 64 : Jetski Prohibited 65 : Facility Protection Mark	EN	0,*

	66 : Oil Pipeline Protection Mark 67 : Marine Cable Protection Mark		
Colour	1 : white 2 : black 3 : red 4 : green 5 : blue 6 : yellow 7 : grey 8 : brown 9 : amber 10 : violet 11 : orange 12 : magenta 13 : pink	EN	1,*
Colour pattern	1 : horizontal stripes 2 : vertical stripes 3 : diagonal stripes 4 : squared 5 : stripes (direction unknown) 6 : border stripe	EN	0,1
Elevation		RE	0,1
Height		RE	0,1
Nature of construction	1: Masonry 2: Concreted 3: Loose Boulders 4: Hard Surfaced 5: Unsurfaced 6: Wooden 7: Metal 8: Glass Reinforced Plastic 9: Painted 10: Framework 11: Latticed 12: Glass 13: Fiberglass 14: Plastic	EN	0,*
Status	1: Permanent 2: Occasional 3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 8: Private 9: Mandatory 11: Extinguished 12: Illuminated 13: Historic 14: Public 15: Synchronized 16: Watched 17: Unwatched 18: Existence Doubtful	EN	0,*

	19: On Request 20: Drop Away 21: Rising 22: Increasing 23: Decreasing 24: Strong 25: Good 26: Moderately 27: Poor 28: Buoyed 29: Fully Operational 30: Partially Operational 31: Drifting 32: Broken 33: Offline 34: Discontinued 35: Manual Observation 36: Unknown Status 37: Confirmed 38: Candidate 39: Under Modification 40: Experimental 41: Under Removal / Deletion 42: Removed / Deleted 43: Candidate for Modification		
Topmark/daymark shape	1 : cone, point up 2 : cone, point down 3 : sphere 4 : 2 spheres 5 : cylinder (can) 6 : board 7 : x-shape (St. Andrew's cross) 8 : upright cross (St George's cross) 9 : cube, point up 10 : 2 cones, point to point 11 : 2 cones, base to base 12 : rhombus (diamond) 13 : 2 cones (points upward) 14 : 2 cones (points downward) 15 : besom, point up (broom or perch) 16 : besom, point down (broom or perch) 17 : flag 18 : sphere over rhombus 19 : square 20 : rectangle, horizontal 21 : rectangle, vertical 22 : trapezium, up 23 : trapezium, down 24 : triangle, point up 25 : triangle, point down 26 : circle 27 : two upright crosses (one over the other) 28 : T-shape 29 : triangle pointing up over a circle 30 : upright cross over a circle 31 : rhombus over a circle 32 : circle over a triangle pointing up 33 : other shape	EN	1,1
Vertical accuracy		RE	0,1

Vertical datum	1: Mean Low Water Springs 2: Mean Lower Low Water Springs 3: Mean Sea Level 4: Lowest Low Water 5: Mean Low Water 6: Lowest Low Water Springs 7: Approximate Mean Low Water Springs 8: Indian Spring Low Water 9: Low Water Springs 10: Approximate Lowest Astronomical Tide 11: Nearly Lowest Low Water 12: Mean Lower Low Water 13: Low Water 14: Approximate Mean Low Water 15: Approximate Mean Lower Low Water 16: Mean High Water 17: Mean High Water Springs 18: High Water 19: Approximate Mean Sea Level 20: High Water Springs 21: Mean Higher High Water 22: Equinoctial Spring Low Water 23: Lowest Astronomical Tide 24: Local Datum 25: International Great Lakes Datum 1985 26: Mean Water Level 27: Lower Low Water Large Tide 28: Higher High Water Large Tide 29: Nearly Highest High Water 30: Highest Astronomical Tide 31: Local Low Water Reference Level 32: Local High Water Reference Level 33: Local Mean Water Reference Level 34: Equivalent Height of Water (German GIW) 35: Highest Shipping Height of Water (German HSW) 36: Reference Low Water Level According to Danube Commission 37: Highest Shipping Height of Water According to Danube Commission 38: Dutch River Low Water Reference Level (OLR) 39: Russian Project Water Level 40: Russian Normal Backwater Level 41: Ohio River Datum 43: Dutch High Water Reference Level 44: Baltic Sea Chart Datum 2000 45: Dutch Estuary Low Water Reference Level (OLW) 46: International Great Lakes Datum 2020 47: Sea Floor 48: Sea Surface 49: Hydrographic Zero	EN	0,1
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Vertical length		RE	0,1
Shape Information		C	0, 1
Language		(S) TE	0, 1
Text		(S) TE	1, 1
IsSlatted		BO	1, 1

8.10.1 Daymarks

If it is required to encode a daymark, it must be done using the feature **Daymark**.

The term “daymark” may also simply refer to any unlighted aid to navigation, and encoders may choose to encode **Daymark** instead of **Beacon Special Purpose/General**, particularly for leading marks. In North America, the term “daybeacon” is used for an unlit beacon.

In the following table, a blank indicates that the encoder may choose a relevant value for the attribute. The table contains the most common examples of coding; other coding combinations are possible.

Feature	INT1	Feature	category of special purpose mark	Other attributes
Coloured or white mark	Q101	Daymark		nature of construction = 9
Coloured topmark with function of beacon	Q102.1	Daymark		nature of construction = 9
Painted board with function of leading beacon	Q102.2	Daymark	16	nature of construction = 9, topmark shape = 6

Remarks:

- The attribute **colour pattern** must be populated if more than one value is populated for the mandatory attribute **colour**.
- If it is required to encode the altitude of the ground level above the vertical datum at the position of a daymark, it must be done using the attribute **elevation**, but only for beacons built on land.
- If it is required to encode the total altitude of a daymark, including any equipment features (e.g. light), above the vertical datum, it must be done using the attribute **height**.
- If it is required to encode the total vertical length of a daymark, including any equipment features (e.g. light), above the seabed or ground, it must be done using the attribute **vertical length**.
- If it is required to encode a cairn that bears the colour(s) specified by a navigational system of marks, it must be done using a beacon feature.
- If it is required to encode a daymark that has more than one colour, the attributes **colour** and **colour pattern** must be encoded, according to the rules.

Distinction: Beacon, cardinal; beacon, isolated danger; beacon, lateral; beacon, safe water; beacon, special purpose/general; topmark.

8.11 Radar transponder beacon (RadarTransponderBeacon)

Definition: **RADAR TRANSPONDER BEACON.** A transponder beacon transmitting a coded signal on radar frequency, permitting an interrogating craft to determine the bearing and range of the transponder. Also called racon. (IHO Dictionary – S-32).

S-201 Geo Feature: Radar transponder beacon

Primitives: Point

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
Remote Monitoring System		TE	0, *
Category of radar transponder beacon	1 : ramark, radar beacon transmitting	EN	1,1

	continuously 2 : racon, radar transponder beacon 3 : leading racon/radar transponder beacon		
Radar Wave Length		C	0, 1
Radar Band		(S) TE	1, 1
Wave Length Value		(S) RE	1, 1
Signal Group		TE	0, 1
Status	1: Permanent 2: Occasional 3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 8: Private 9: Mandatory 11: Extinguished 12: Illuminated 13: Historic 14: Public 15: Synchronized 16: Watched 17: Unwatched 18: Existence Doubtful 19: On Request 20: Drop Away 21: Rising 22: Increasing 23: Decreasing 24: Strong 25: Good 26: Moderately 27: Poor 28: Buoyed 29: Fully Operational 30: Partially Operational 31: Drifting 32: Broken 33: Offline 34: Discontinued 35: Manual Observation 36: Unknown Status 37: Confirmed 38: Candidate 39: Under Modification 40: Experimental 41: Under Removal / Deletion 42: Removed / Deleted 43: Candidate for Modification	EN	0, *
Value of Nominal Range		RE	0, 1
Manufacturer		TE	0, 1

Sector Limit One		C	0, 1
Sector Bearing		(S) RE	1, 1
Sector Line Length		(S) IN	0, 1
Sector Limit Two		C	0, 1
Sector Bearing		(S) RE	1, 1
Sector Line Length		(S) IN	0, 1
Signal Sequence		C	0, 1
Signal Duration		(S) RE	1, 1
Signal Status	1: Lit/Sound 2: Eclipsed/Silent	(S) EN	1, 1

8.11.1 Radar beacons

Radar beacons are transmitters operating in the marine radar frequency band. The signals produce a characteristic line on a vessel's radar display enabling the mariner to determine their position with greater certainty than would be possible by means of a normal radar display alone.

If it is required to encode a radar beacon, it must be done using the feature **Radar Transponder Beacon**.

Remarks:

- The **Radar Transponder Beacon** must only be used to encode the technical equipment itself, independent of the building or structure in which it is installed. If it is required to encode the building or structure (e.g. mast, tower, radar dome), it must be done using an appropriate feature (e.g. **Building, Landmark**).
- The attribute **signal group** is used to encode Morse identification letter(s) for the radar beacon, where known.
- Leading racons are established such that, when their bearing lines are coincident on a vessel's radar display, the bearing serves to indicate the track to be followed. If it is required to encode the bearing line and the recommended track for leading racons, it must be done. Where the bearing line coincides with a leading line defined by lights or other visual features making up a range system, navigation lines and recommended tracks must not be duplicated. The features making up the range system must be aggregated using the collection feature **Aggregation**.
- If, for some reason, the radar transponder beacon signal is obscured between certain bearings, this information should be encoded using the complex attribute **sector limit** to encode the "visible" sector, as for lights.
- The sweep period may be encoded using the complex attribute **information**.

Distinction: Radar line; radar range; radar station.

8.12 Power source (PowerSource)

Definition: **POWER SOURCE** The system or mechanism that provides the necessary energy to operate an Aid to Navigation or any other device. Common power sources for AtoNs could include solar panels, batteries, generators, or electrical grid connections.

S-201 Geo Feature: Power source

Primitives: Point

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
Remote Monitoring System		TE	0, *

Category of power source	1: battery 2: generator 3: solar panel 4: electrical service	EN	1,1
Status	1: Permanent 2: Occasional 3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 8: Private 9: Mandatory 11: Extinguished 12: Illuminated 13: Historic 14: Public 15: Synchronized 16: Watched 17: Unwatched 18: Existence Doubtful 19: On Request 20: Drop Away 21: Rising 22: Increasing 23: Decreasing 24: Strong 25: Good 26: Moderately 27: Poor 28: Buoyed 29: Fully Operational 30: Partially Operational 31: Drifting 32: Broken 33: Offline 34: Discontinued 35: Manual Observation 36: Unknown Status 37: Confirmed 38: Candidate 39: Under Modification 40: Experimental 41: Under Removal / Deletion 42: Removed / Deleted 43: Candidate for Modification	EN	0,*

8.12.1 Power source

If it is required to encode a power source, it should be done using the feature **Power Source**.

8.13 Radio station (RadioStation)

Definition: **Radio station.** : A place equipped to transmit radio waves. Such a station may be either stationary or mobile, and may also be provided with a radio receiver.

S-201 Geo Feature: Radio station

Primitives: Point

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
Remote Monitoring System		TE	0, *
Category of Radio Station	1: Circular (Non-Directional) Marine or Aero-Marine Radiobeacon 2: Directional Radiobeacon	EN	1, 1

	3: Rotating Pattern Radiobeacon 4: Consol Beacon 5: Radio Direction-Finding Station 6: Coast Radio Station Providing QTG Service 7: Aeronautical Radiobeacon 8: Decca 9: Loran C 10: Differential GNSS 11: Toran 12: Omega 13: Syledis 14: Chaika 19: Radio Telephone Station 20: AIS Base Station		
Estimated Range of Transmission		RE	0, 1
Status	1: Permanent 2: Occasional 3: Recommended 4: Not in Use 5: Periodic/Intermittent 6: Reserved 7: Temporary 8: Private 9: Mandatory 11: Extinguished 12: Illuminated 13: Historic 14: Public 15: Synchronized 16: Watched 17: Unwatched 18: Existence Doubtful 19: On Request 20: Drop Away 21: Rising 22: Increasing 23: Decreasing 24: Strong 25: Good 26: Moderately 27: Poor 28: Buoyed 29: Fully Operational 30: Partially Operational 31: Drifting 32: Broken 33: Offline 34: Discontinued 35: Manual Observation 36: Unknown Status 37: Confirmed 38: Candidate 39: Under Modification 40: Experimental 41: Under Removal / Deletion 42: Removed / Deleted 43: Candidate for Modification	EN	0, 1
8.13.1 Radio Station			

Transmissions from radio stations may provide Mariners with a line of position. Most radio position fixing systems require Radio Direction Finding (RDF) equipment to determine the bearing of the transmitting device; such equipment is generally no longer fitted on vessels. The feature “radio station” is used to encode the point of transmission of the signal.

If it is required to encode a radio station, it must be done using the feature Radio Station.

Remarks:

- The Radio Station must only be used to encode the technical equipment itself, independent of the building or structure in which it is installed. If it is required to encode the building or structure (for example mast, tower, radar dome), it must be done using an appropriate feature (for example Building, Landmark).
- If it is required to encode a DGPS station, it must be done using Radio Station, with attribute category of radio station = 10 (Differential GNSS).
- Where required, the complex attribute frequency pair, sub-attributes frequency shore station receives and frequency shore station transmits, must be quoted in Hertz, for example a signal frequency of 950 MHz must be encoded as 950000000.

9 Moorings

9.1 Bridle (Bridle)

Definition: **BRIDLE.** A Two lengths of chain connected by a central ring and used for lifting wide loads. (IALA Dictionary,8-3-195)

S-201 Geo Feature: Bridle

Primitives: NoGeometry

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *

Remote Monitoring System		TE	0, *
BridleLink Type		TE	0, 1
Legs Details		TE	0, 1
9.1.1 Bridle If it is required to encode a bridle, it should be done using the feature Bridle .			

9.2 Cable submarine (CableSubmarine)

Definition: **CABLE SUBMARINE.** An assembly of wires or fibres, or a wire rope or chain, which has been laid underwater or buried beneath the sea floor.

S-201 Geo Feature: Cable Submarine

Primitives: NoGeometry

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
Remote Monitoring System		TE	0, *
Cable Dimensions		C	0, 1

Cable Length		(S) RE	1, 1
Height Length Units	1: Metres 2: Feet 3: Kilometres 4: Hectometres 5: Statute Miles 6: Nautical Miles	(S) EN	1, 1
Diameter		(S) RE	1, 1
Category of Cable	1: Power Line 3: Transmission Line 4: Telephone 5: Telegraph 6: Mooring Cable 7: Ferry 8: Fibre Optic Cable	EN	1, 1
9.2.1 Cable submarine If it is required to encode a cable submarine, it should be done using the feature Cable Submarine.			

9.3 Mooring Shackle (MooringShackle)

Definition: **MOORING SHACKLE.** A shackle at the lower end of a mooring chain, for attachment to an anchor or sinker. (IALA Dictionary, 8-5-150)

S-201 Geo Feature: Mooring shackle

Primitives: NoGeometry

<i>Real World</i>	<i>Paper Chart Symbol</i>	<i>ECDIS Symbol</i>		
S-201 Attribute	Allowable Encoding Value	Type	Multiplicity	
ID Code		TE	0, 1	
Information		C	0, *	
File Locator		(S) TE	0, 1	
File Reference		(S) TE	0, 1	
Headline		(S) TE	0, 1	
Language		(S) TE	1, 1	
Text		(S) TE	0, 1	
Feature Name		C	0, *	
Display Name		(S) BO	0, 1	
Language		(S) TE	0, 1	
Name		(S) TE	1, 1	
Scale Minimum		IN	0, 1	
Source Date		DA	0, 1	
Source		TE	0, 1	
Pictorial Representation		TE	0, 1	
Inspection Frequency		TE	0, 1	
Inspection Requirements		TE	0, 1	
AtoN Maintenance Record		TE	0, 1	
Installation Date		DA	0, 1	
Fixed Date Range		C	0, 1	
Date End		(S) TD	0, 1	
Date Start		(S) TD	0, 1	
Periodic Date Range		C	0, 1	
Date End		(S) TD	1, 1	
Date Start		(S) TD	1, 1	
Seasonal Action Required		TE	0, *	
Remote Monitoring System		TE	0, *	
Nature of Construction	1: Masonry 2: Concreted	EN	0, 1	

	3: Loose Boulders 4: Hard Surfaced 5: Unsurfaced 6: Wooden 7: Metal 8: Glass Reinforced Plastic 9: Painted 10: Framework 11: Latticed 12: Glass 13: Fiberglass 14: Plastic		
Shackle Type	1: forelock shackles 2: clenching shackles 3: bolt shackles 4: screw pin shackles 5: kenter shackle 6: quick release link	EN	0, 1
Weight		RE	0, 1
9.3.1 Mooring Shackle If it is required to encode a mooring shackle, it should be done using the feature Mooring Shackle.			

9.4 Sinker Anchor (SinkerAnchor)

Definition: **SINKER ANCHOR** A heavy weight (of concrete, cast-iron, etc..) that rests on the sea bed and to which a mooring line can be attached. (IALA Dictionary, 8-5-025)

S-201 Geo Feature: Sinker Anchor

Primitives: NoGeometry

<i>Real World</i>	<i>Paper Chart Symbol</i>	<i>ECDIS Symbol</i>		
S-201 Attribute	Allowable Encoding Value	Type	Multiplicity	
ID Code		TE	0, 1	
Information		C	0, *	
File Locator		(S) TE	0, 1	
File Reference		(S) TE	0, 1	
Headline		(S) TE	0, 1	
Language		(S) TE	1, 1	
Text		(S) TE	0, 1	
Feature Name		C	0, *	
Display Name		(S) BO	0, 1	
Language		(S) TE	0, 1	
Name		(S) TE	1, 1	
Scale Minimum		IN	0, 1	
Source Date		DA	0, 1	
Source		TE	0, 1	
Pictorial Representation		TE	0, 1	
Inspection Frequency		TE	0, 1	
Inspection Requirements		TE	0, 1	
AtoN Maintenance Record		TE	0, 1	
Installation Date		DA	0, 1	
Fixed Date Range		C	0, 1	
Date End		(S) TD	0, 1	
Date Start		(S) TD	0, 1	
Periodic Date Range		C	0, 1	
Date End		(S) TD	1, 1	
Date Start		(S) TD	1, 1	
Seasonal Action Required		TE	0, *	
Remote Monitoring System		TE	0, *	
Nature of Construction	1: Masonry 2: Concreted	EN	0, 1	

	3: Loose Boulders 4: Hard Surfaced 5: Unsurfaced 6: Wooden 7: Metal 8: Glass Reinforced Plastic 9: Painted 10: Framework 11: Latticed 12: Glass 13: Fiberglass 14: Plastic		
Sinker Dimensions		C	0, 1
Height Length Units	1: forelock shackles 2: clenching shackles 3: bolt shackles 4: screw pin shackles 5: kenter shackle 6: quick release link	(S) EN	1, 1
Horizontal Length		(S) RE	0, 1
Horizontal Width		(S) RE	0, 1
Vertical Length		(S) RE	0, 1
Weight		RE	1, 1
Sinker Type		TE	0, 1
9.4.1 Sinker Anchor If it is required to encode a sinker anchor, it should be done using the feature Sinker Anchor .			

9.5 Swivel (Swivel)

Definition: **SWIVEL.** A chain link that provides for rotary motion between the lengths of chain that it connects. (IALA Dictionary, 8-5-165)

S-201 Geo Feature: Swivel

Primitives: NoGeometry

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
Remote Monitoring System		TE	0, *
Nature of Construction	1: Masonry 2: Concreted	EN	0, 1

	3: Loose Boulders 4: Hard Surfaced 5: Unsurfaced 6: Wooden 7: Metal 8: Glass Reinforced Plastic 9: Painted 10: Framework 11: Latticed 12: Glass 13: Fiberglass 14: Plastic		
Weight		RE	0, 1
Swivel Type		TE	0, 1
9.5.1 Swivel If it is required to encode a swivel, it should be done using the feature Swivel .			

9.6 Counter weight (CounterWeight)

Definition: **COUNTER WEIGHT** A weight that is used to balance another weight. In the context of navigational aids or other structures, a counterweight is often used to stabilize or maintain the equilibrium of a structure, ensuring that it remains upright or balanced. It is typically employed in systems involving moving parts, like cranes, lifts, or buoys.

S-201 Geo Feature: Counter Weight

Primitives: NoGeometry

Real World	Paper Chart Symbol	ECDIS Symbol	
S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
Remote Monitoring System		TE	0, *

Nature of Construction	1: Masonry 2: Concreted 3: Loose Boulders 4: Hard Surfaced 5: Unsurfaced 6: Wooden 7: Metal 8: Glass Reinforced Plastic 9: Painted 10: Framework 11: Latticed 12: Glass 13: Fiberglass 14: Plastic	EN	0, 1
Weight		RE	0, 1
Counter Weight Type		TE	0, 1
9.6.1 Counter Weight If it is required to encode a counter weight, it should be done using the feature Counter Weight .			

10 Recommended Track and Line

10.1 Recommended track (RecommendedTrack)

Definition: **RECOMMENDED TRACK**. A track recommended to all or only certain vessels. (IHO Dictionary – S-32).

S-201 Geo Feature: Recommended track

Primitives: Curve, Surface

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *

Based On Fixed Marks		BO	1, 1
Depth Range Minimum Value		RE	0, 1
Maximal Permitted Draught		RE	0, 1

10.1.1 Recommended tracks

Recommended tracks and fairways usually comprise a number of sections (sometimes termed “legs”) which lead between dangers lying close on both sides of the track or fairway. Tracks commonly include some sections which are leading lines. The distinction between tracks and fairways, in this context, is that tracks have no specified outer limits and fairways do have specified outer limits.

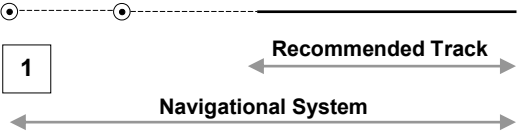
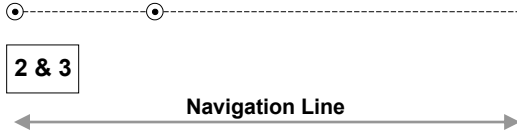
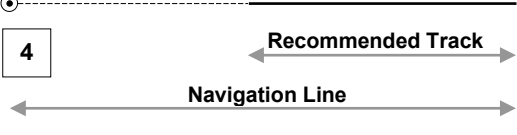
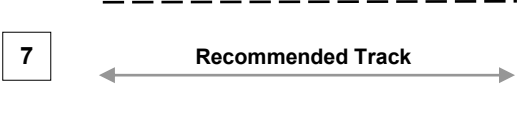
It is important to recognise that it is not the role of cartographers to create “recommended” tracks and other “recommended” routeing measures; such recommendations are made by other authorities. The word “Recommended”, used in connection with recommended tracks and other recommended routeing measures usually implies that it has been recommended by a competent authority (such as a port authority within its port limits or a maritime safety authority) and may be adopted by IMO. Occasionally, the recommendation may be based on advice directly from a competent surveyor or established by precedent.

Recommended tracks include all channels recommended for hydrographic reasons to lead safely between shoal depths. The use of such tracks is generally left to the discretion of the mariner and will depend on the vessel's draught, the state of the tide, adequacy of navigational aids and so on.

If it is required to encode a recommended track, it must be done using the feature **Recommended Track**.

The use of **Navigation Line** and **Recommended Track** is defined in more detail in the following Table, and in the Figure below.

Figure		Navigation Line	Recommended Track	Navigational Aids
1	Recommended track on a leading line	category of navigation line = 3	category of recommended track = 1	at least 2
2	Clearing line on marks in line	category of navigation line = 1	none	at least 2
3	Transit line on marks in line	category of navigation line = 2	none	at least 2
4	Recommended track on a bearing	category of navigation line = 3	category of recommended track = 1	1
5	Clearing line on a bearing	category of navigation line = 1	none	1
6	Transit line on a bearing	category of navigation line = 2	none	1
7	Recommended track not based on fixed marks	none	category of recommended track = 2	none

Two navigational aids 	Two navigational aids 
One navigational aid 	One navigational aid 
No navigational aids 	

Remarks:

- The attribute **depth range minimum value** is used to encode the shallowest depth along the track, where required.
- The attribute **maximum permitted draught** is used to encode the maximum draught permitted on the track, where required.
- The recommended track is that portion of a navigation line that a ship should use for navigation.
- In the case of a two-way recommended track, only one value of orientation is encoded (in the complex attribute **orientation**); the other value can be deduced (i.e. the value in **orientation (orientation value)** + 180 degrees). The value of orientation encoded on **orientation (orientation value)** should be the value of the bearing from seaward. If it is not possible to define a seaward direction, the value that is less than 180° should be used.
- When the traffic flow along a recommended track is one way (attribute **traffic** = 1, 2 or 3), the resultant direction of the line (accounting for the direction of digitising and any subsequent reversal of the line) associated with the **Recommended Track** must be the same as the direction of the traffic flow, in order to ensure the correct representation in the ECDIS of the direction to be followed.

Distinction: Fairway; navigation line; recommended route centreline; recommended traffic lane part.

10.2 Navigation Line (NavigationLine)

Definition: **NAVIGATION LINE.** A straight line extending towards an area of navigational interest and generally generated by two navigational aids or one navigational aid and a bearing. (Service Hydrographique et Oceanographique de la Marine, France).

S-201 Geo Feature: Navigation line

Primitives: Curve

Real World

Paper Chart Symbol

ECDIS Symbol

S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
ID Code		TE	0, 1
Information		C	0, *
File Locator		(S) TE	0, 1
File Reference		(S) TE	0, 1
Headline		(S) TE	0, 1
Language		(S) TE	1, 1
Text		(S) TE	0, 1
Feature Name		C	0, *
Display Name		(S) BO	0, 1
Language		(S) TE	0, 1
Name		(S) TE	1, 1
Scale Minimum		IN	0, 1
Source Date		DA	0, 1
Source		TE	0, 1
Pictorial Representation		TE	0, 1
Inspection Frequency		TE	0, 1
Inspection Requirements		TE	0, 1
AtoN Maintenance Record		TE	0, 1
Installation Date		DA	0, 1
Fixed Date Range		C	0, 1
Date End		(S) TD	0, 1
Date Start		(S) TD	0, 1
Periodic Date Range		C	0, 1
Date End		(S) TD	1, 1
Date Start		(S) TD	1, 1
Seasonal Action Required		TE	0, *
Category of Navigation Line	1: Clearing Line 2: Transit Line	EN	1, 1

3: Leading Line Bearing a Recommended Track

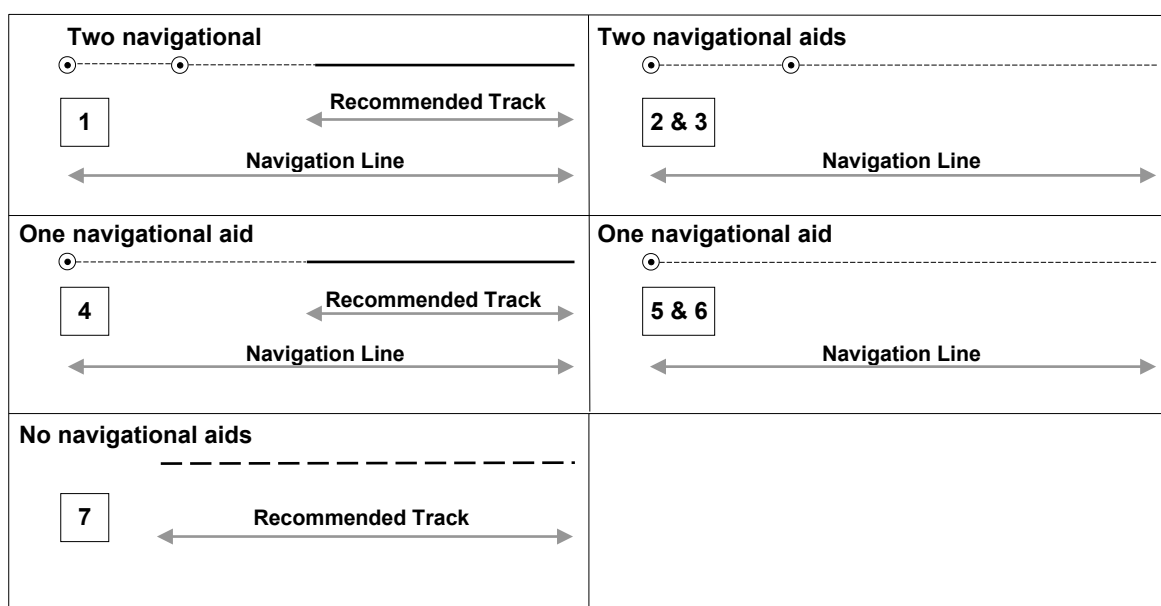
10.2.1 Navigation lines

Clearing Lines are important in rocky areas where dangers are not guarded by buoys and where sailing vessels (which are not always able to keep to a direct track) and other small craft may navigate close inshore. Transits marking isolated dangers are based on beacons or other marks which are erected on shore to indicate (approximately, unless there are two pairs of beacons) the position of an isolated danger. Leading lines based on beacons or lights must be encoded where the maximum display scale for the ENC data permits. Leading lines based on natural features should be encoded on the largest maximum display scale ENC data where they appear to be useful, particularly if other navigational aids seem inadequate.

If it is required to encode a navigation line, it must be done using the feature **Navigation Line**.

The use of **Navigation Line** and **Recommended Track** is defined in more detail in the following Table, and in the Figure below:

Figure		Navigation Line	Recommended Track	Navigational Aids
1	Recommended track on a leading line	category of navigation line = 3	category of recommended track = 1	at least 2
2	Clearing line on marks in line	category of navigation line = 1	none	at least 2
3	Transit line on marks in line	category of navigation line = 2	none	at least 2
4	Recommended track on a bearing	category of navigation line = 3	category of recommended track = 1	1
5	Clearing line on a bearing	category of navigation line = 1	none	1
6	Transit line on a bearing	category of navigation line = 2	none	1
7	Recommended track not based on fixed marks	none	category of recommended track = 2	none



Remarks:

- The value populated for the mandatory complex attribute **orientation** must be the value of the bearing from seaward.
- The extent of the navigation line depends on the visibility of the navigational aids.
- The recommended track is that portion of a navigation line that a ship should use for navigation.

10.2.2 Measured distances

If the track to be followed is on a leading line or a bearing, it must be encoded in the way described in the Table and Figure above (cases 1 or 4). If the track is not on a leading line or bearing, it must be encoded only as a **Navigation Line** feature with the attribute **category of navigation line** being set to an empty (null) value. In either case, if it is required to encode the measured distance, it must be done using the attribute **information** (e.g. *Measured distance = 1450 metres*).

If it is required to encode the transit lines, they must be done using **Navigation Line** features, with **category of navigation line** = 2 (transit line).

If it is required to encode the beacons, they must be done using **Beacon Special Purpose/General** features, with attribute **category of special purpose mark** = 17 (measured distance mark).

On occasions, one or more of the transits used for the measured distance may incorporate an existing landmark as the front or rear mark. In this case, if **Landmark** is encoded, **category of special purpose mark** = 17 must also be populated.

Where the entire measured distance system exists within a single dataset, each transit line with its beacons must be aggregated into a collection feature **Aggregation**. These two aggregation features and the track to be followed must be aggregated into another **Aggregation** feature.

Remarks:

- All features comprising a measured distance must have the same value populated for the attribute SCAMIN.

Distinction: Recommended route; recommended track.

11 Collection Objects

11.1 Aton Association

<u>Definition:</u> ATON ASSOCIATION. Used to identify an association between two or more objects.			
<u>S-201 Collection Feature:</u> Aton Association			
<u>Primitives:</u> None			
<i>Real World</i>	<i>Paper Chart Symbol</i>	<i>ECDIS Symbol</i>	
S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
Category of Association	1 : channel markings 2 : danger markings	CL (OE)	1,1
Danger marks The association object is used to show the relationship between the navigation lines and tracks and the associated dangers. Other associations If it is necessary to code AtonAssociation objects other than channel markings or danger markings, they must be encoded as “other: ...” where the ellipses are replaced by a short label conforming to S-100 clause 3-6.7 indicating the type of association. <u>Remarks:</u> No remarks <u>Distinction:</u> Aton Aggregation.			

11.2 Aton Aggregation

<u>Definition:</u> ATON AGGREGATION. Used to identify an aggregation of two or more objects.			
<u>S-201 Collection Feature:</u> Aton Aggregation			
<u>Primitives:</u> None			
<i>Real World</i>	<i>Paper Chart Symbol</i>	<i>ECDIS Symbol</i>	
S-201 Attribute	Allowable Encoding Value	Type	Multiplicity
Category of Aggregation	1 : leading line 2 : range system 3: measured distance	CL (OE)	1,1

Range system

To encode a range system the features Navigation Line, Recommended Track and the lights or other navigational aids making up the range system must be aggregated using the collection feature Aggregation.

Leading line (clearing line or transit line)

A leading line is distinct from a range system in that it does not have a section of the navigation line designated a recommended track.

Measured distance

To encode a measured distance the features Navigation Line and fixed marks (BeaconSpecialPurposeGeneral or Landmark) must be aggregated using the collection feature Aggregation.

Other aggregations

If it is necessary to code other types of **AtonAggregation** objects, they must be encoded as “other: ...” where the ellipses are replaced by a short label conforming to S-100 clause 3-6.7 indicating the type of association.

Remarks:

No remarks

Distinction: Association.